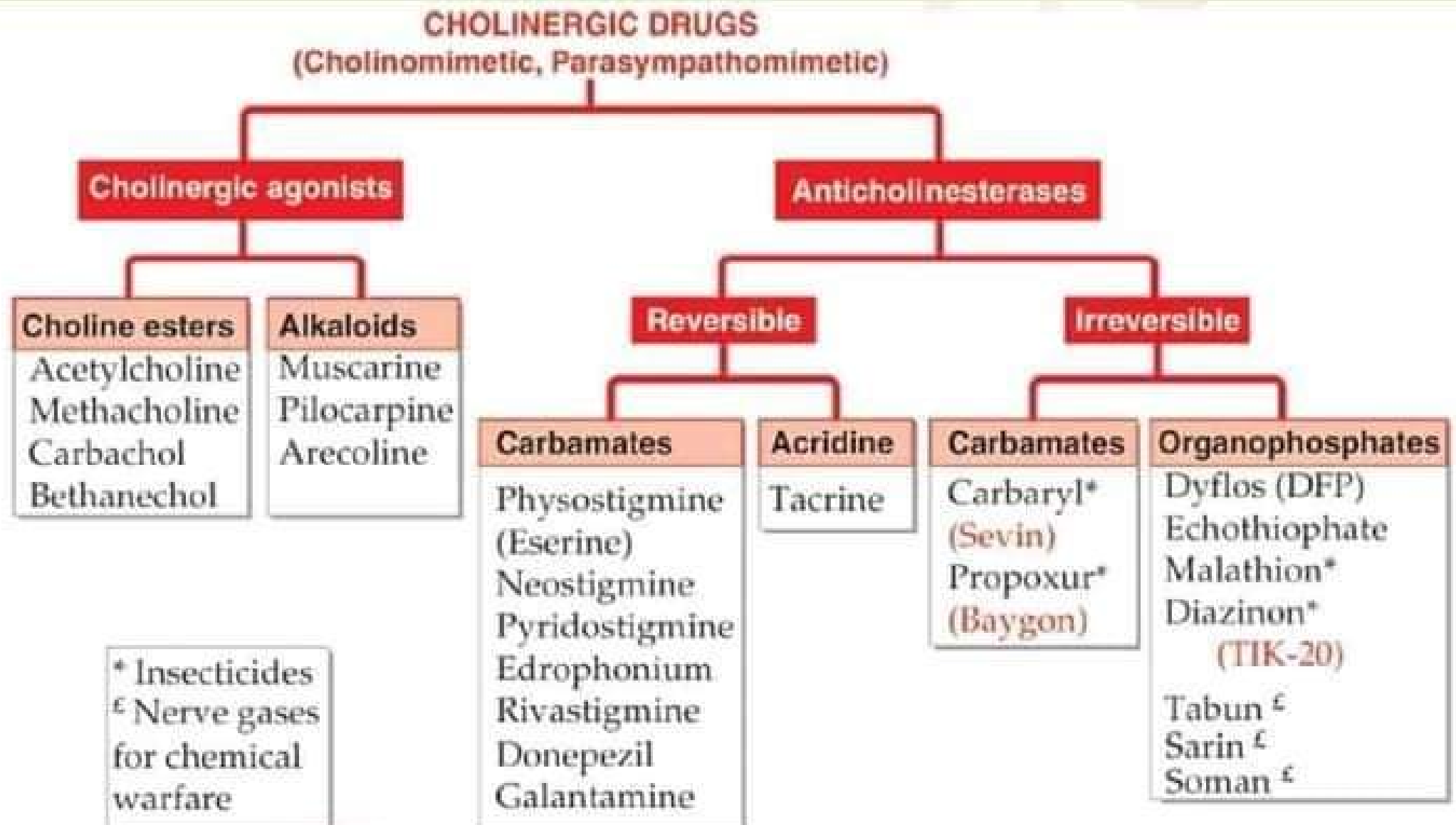


Classification of Drugs

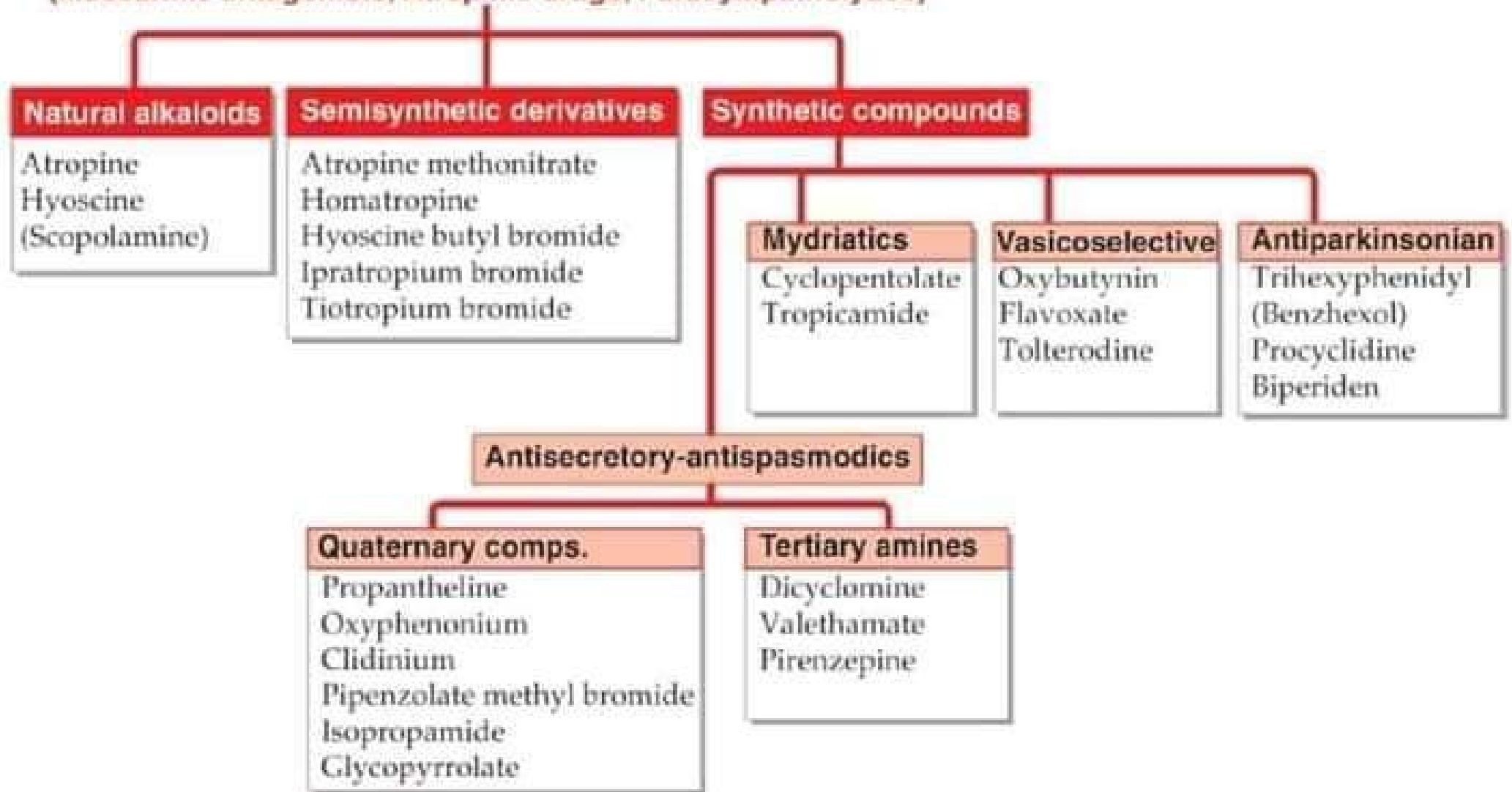
Drug Acting on Autonomic Nervous system



Anticholinergic Drugs

ANTICHOLINERGIC DRUGS

(Muscarinic antagonists, Atropinic drugs, Parasympatholytics)



Ganglionic Stimulants and Block Agents

GANGLIONIC STIMULANTS

Selective nicotinic agonists

Nicotine (small dose)
Lobeline
Dimethyl phenyl piperazinium (DMPP)
Tetramethyl ammonium (TMA)
Varenicline

Nonselective/muscarinic agonists

Acetylcholine
Carbachol
Pilocarpine
Anticholinesterases
MCN 343-A

GANGLIONIC BLOCKING AGENTS

Competitive blockers

Persistent depolarizing blockers

Nicotine (large dose)
Anticholinesterases (large dose)

Quaternary ammonium comps.

Hexamethonium
Pentolinium

Amines

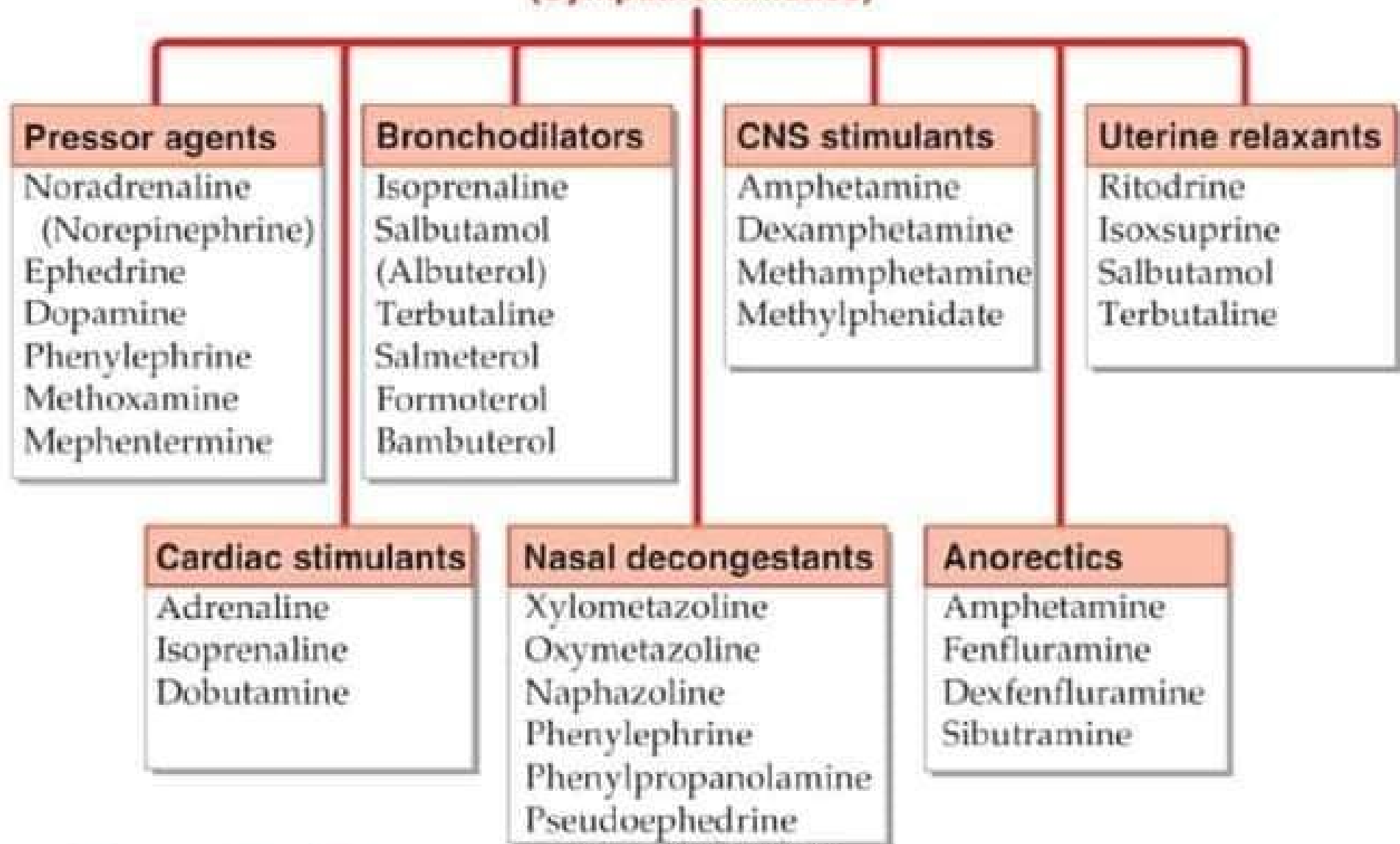
Mecamylamine
Pempidine

Monosulfonium comp.

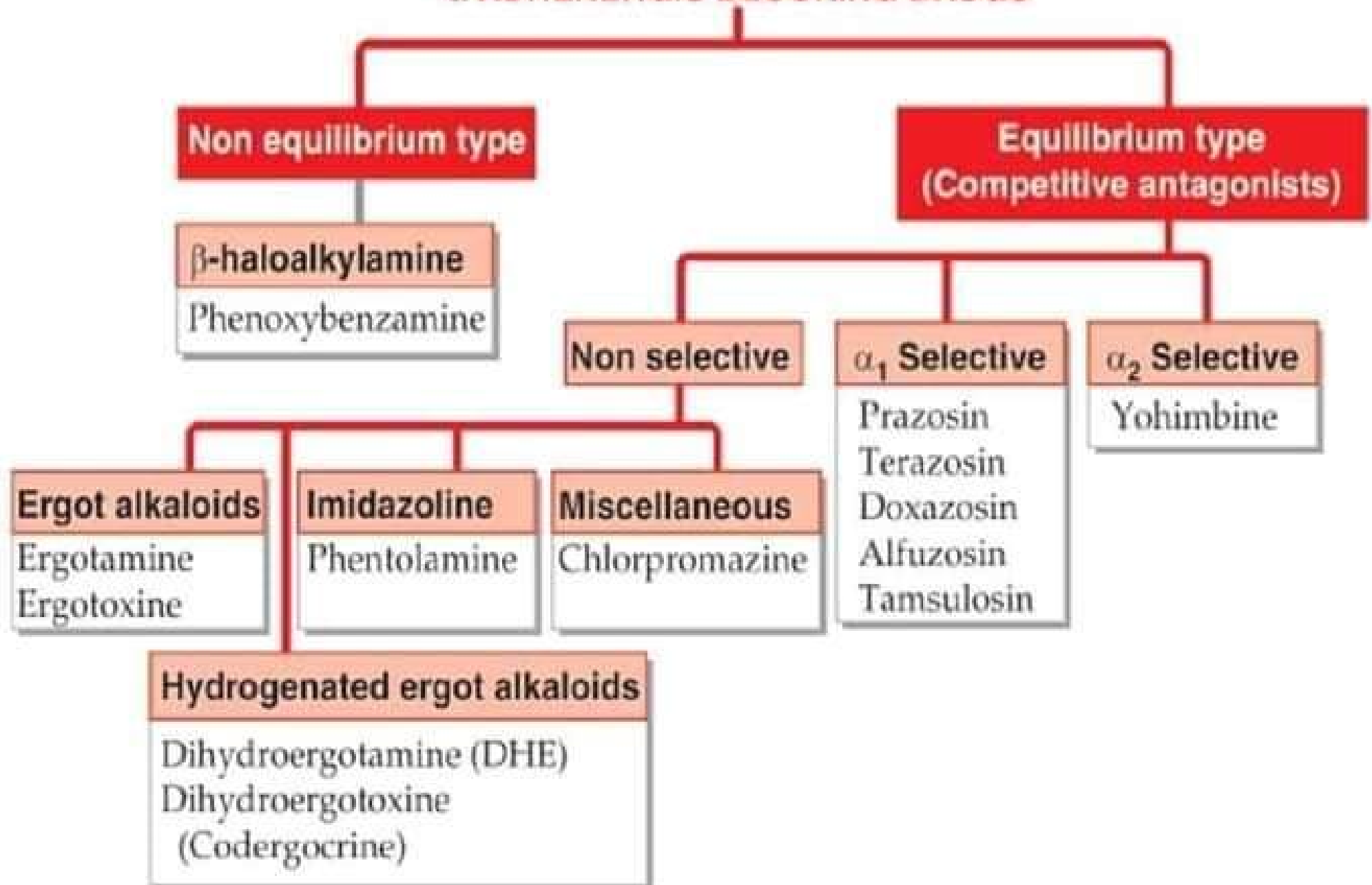
Trimethaphan-camforsulfonate

Adrenergic Drugs

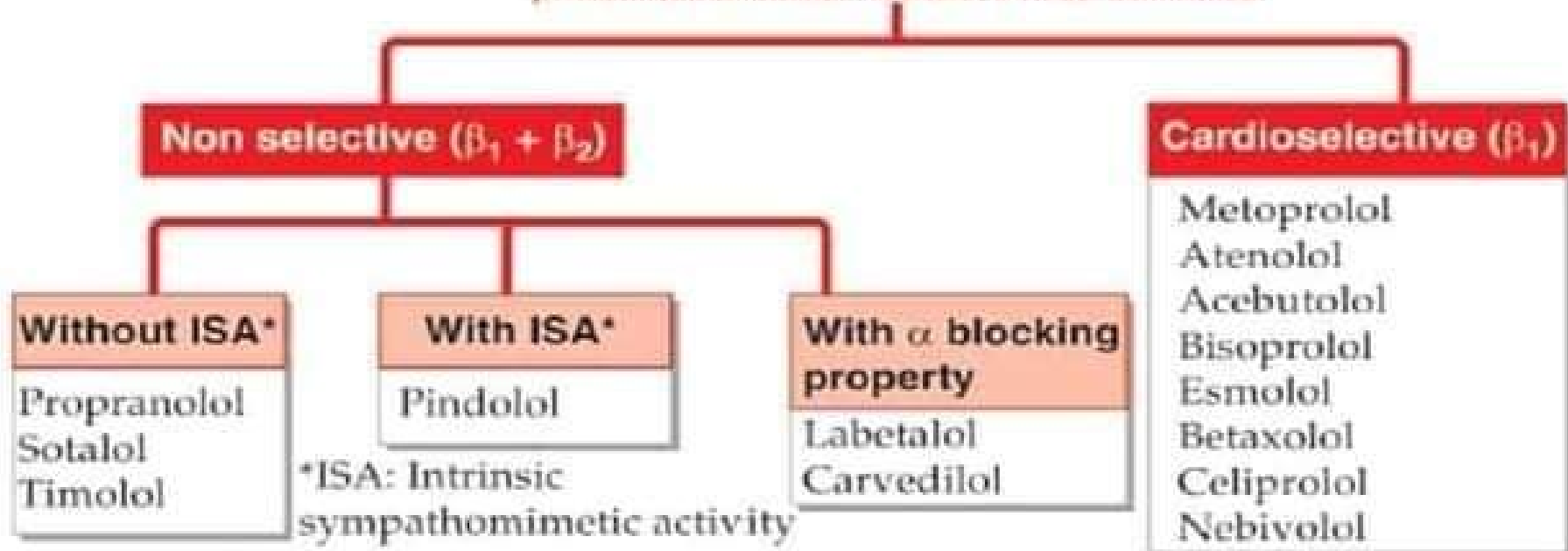
ADRENERGIC DRUGS (Sympathomimetics)



α ADRENERGIC BLOCKING DRUGS

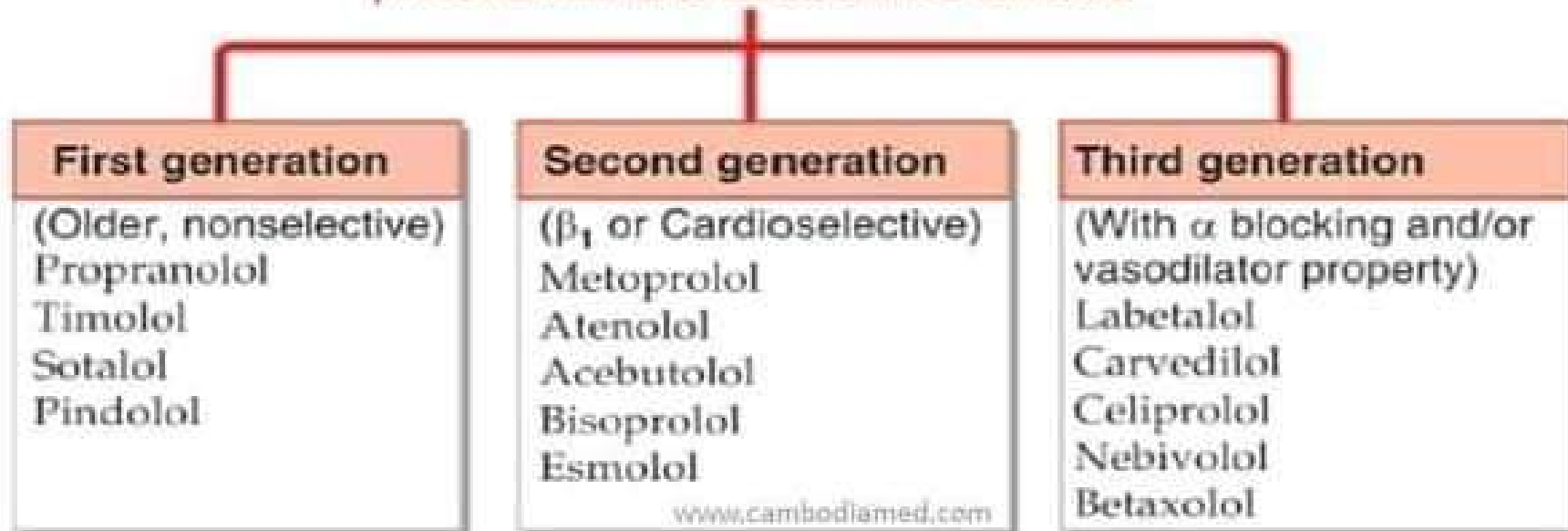


β ADRENERGIC BLOCKING DRUGS

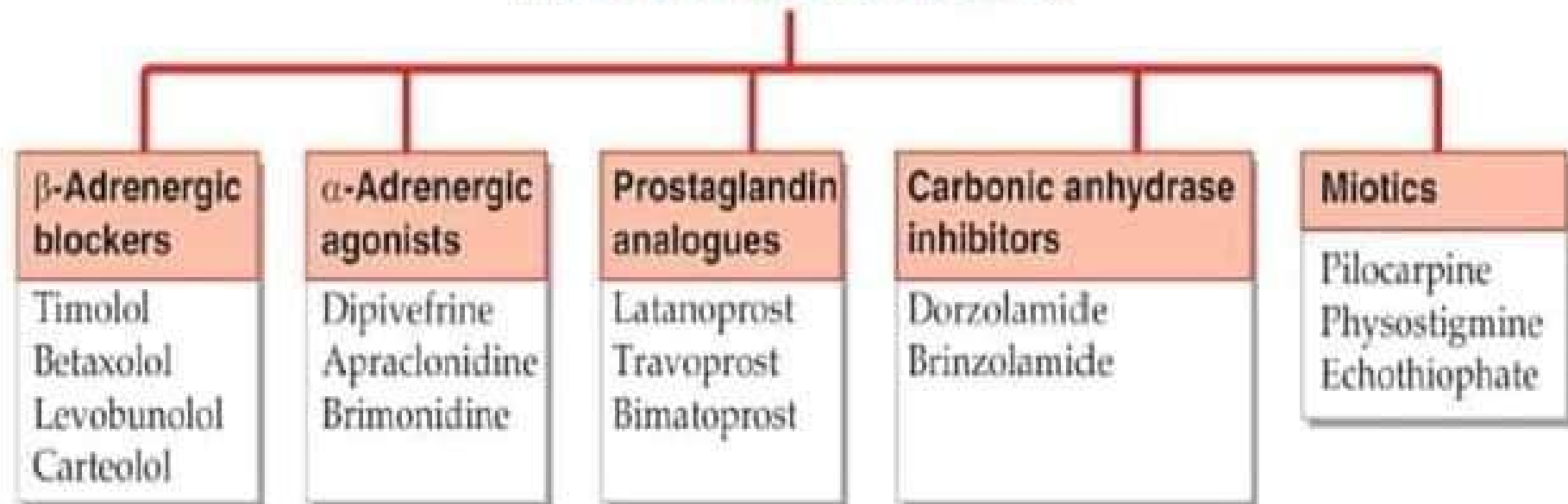


GENERATIONWISE CLASSIFICATION

β ADRENERGIC BLOCKING DRUGS



TOPICAL DRUGS FOR GLAUCOMA



Preparations

1. Timolol: Start with 0.25% eye drops BD, change to 0.5% drops in case of inadequate response. 0.5% OD as gel forming solution. **GLUCOMOL, OCUPRES, IOTIM, LOPRES** 0.25% and 0.5% eye drops. **TIMOLAST** 0.5% gel forming eye drops (long acting).
Timolol 0.5% + Latanoprost 0.005%: **LAPROST PLUS, LATOCHEK-T** eye drops.
2. Betaxolol: 0.5% topically in eye BD; **OPTIPRES, IOBET, OCUBETA** 0.5% eye drops.
3. Levobunolol: 0.5% topically in eye OD; **BETAGAN** 0.5% ophthalmic solution.
4. Dipivefrine: 0.1% topically in eye BD; **PROPINE** 0.1% eye drops.
5. Apraclonidine: 0.5–1.0% topically in eye; **ALFADROPS-DS** 1% eye drops.
6. Brimonidine: 0.2% topically in eye TDS;
ALPHAGAN-P, BRIMODIN-P 0.15% eye drops; **IOBRIM** 0.2% eye drops.

Autacoid and Related Drugs

HISTAMINERGIC AGONISTS

Nonselective agonists (H₁ + H₂ + H₃)

Histamine
Betahistine

Selective H₁ agonists

2-Methyl histamine
2-Pyridyl ethylamine
2-Thizolyl ethylamine

Selective H₂ agonists

Dimaprit
Impromidine

Selective H₃ agonists

(R) α -Methylhistamine
Imetit

H₁ ANTAGONISTS (Conventional antihistaminics)

Highly sedative

Diphenhydramine
Dimenhydrinate
Promethazine
Hydroxyzine

Moderately sedative

Pheniramine
Cyproheptadine
Meclozine
Cinnarizine

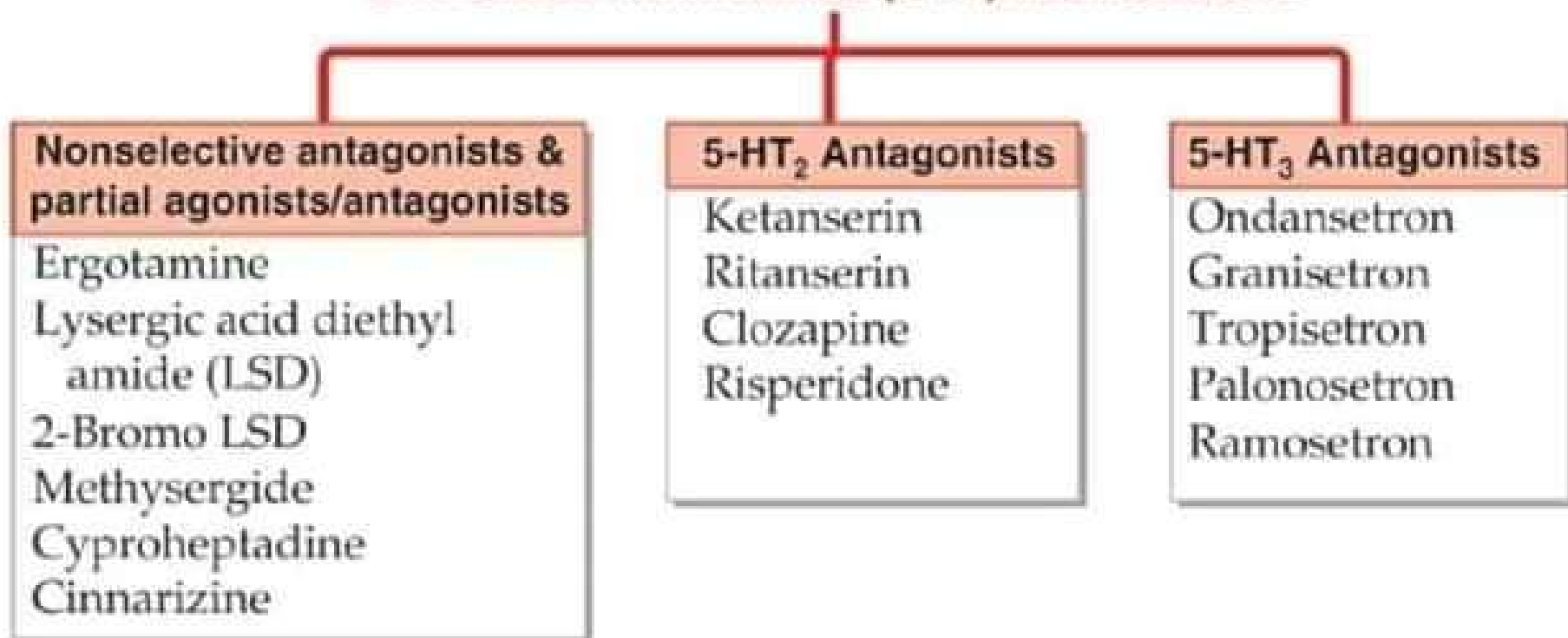
Mildly sedative

Chlorpheniramine
Dexchlorpheniramine
Triprolidine
Clemastine

Second generation (nonsedating) anti- histaminics

Fexofenadine
Loratadine
Desloratadine
Cetirizine
Levocetirizine
Azelastine
Mizolastine
Ebastine

5-HYDROXY TRYPTAMINE (5-HT) ANTAGONISTS



DRUGS FOR MIGRAINE

To treat attacks

Simple analgesics

Paracetamol
Aspirin
Metamizol
Codeine

Antiemetics

Metoclopramide
Domperidone
Prochlorperazine
Promethazine

NSAIDs

Ibuprofen
Naproxen
Diclofenac
Mephenamic acid

Selective 5-HT_{1D/1B} agonists

Sumatriptan
Rizatriptan
Naratriptan
Zolmitriptan
Frovatriptan

Ergot alkaloids

Ergotamine
Dihydroergotamine
(DHE)

To prevent attacks

Propranolol
(other β blockers)
Amitriptyline
(other TCAs)
Flunarizine
Valproic acid
Gabapentin
Topiramate
Cyproheptadine

PROSTAGLANDINS (PGs)

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graph TD; A[PROSTAGLANDINS (PGs)] --> B[Natural PGs]; A --> C[Prostaglandin analogues];
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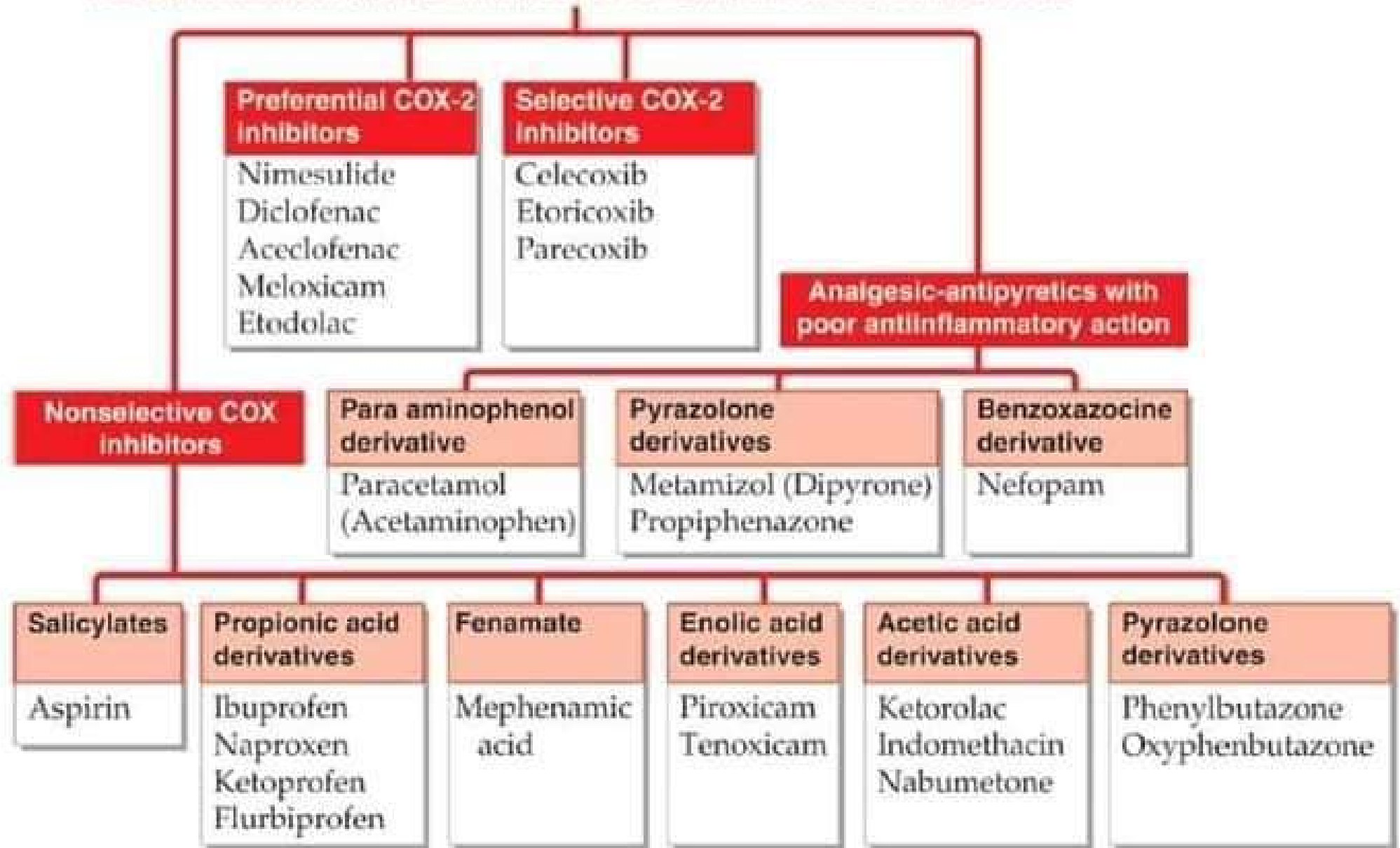
Natural PGs

Dinoprostone (PGE_2)
Gemeprost
Dinoprost ($\text{PGF}_{2\alpha}$)
Alprostadiol (PGE_1)
Prostacyclin (PGI_2)
(Epoprostenol)

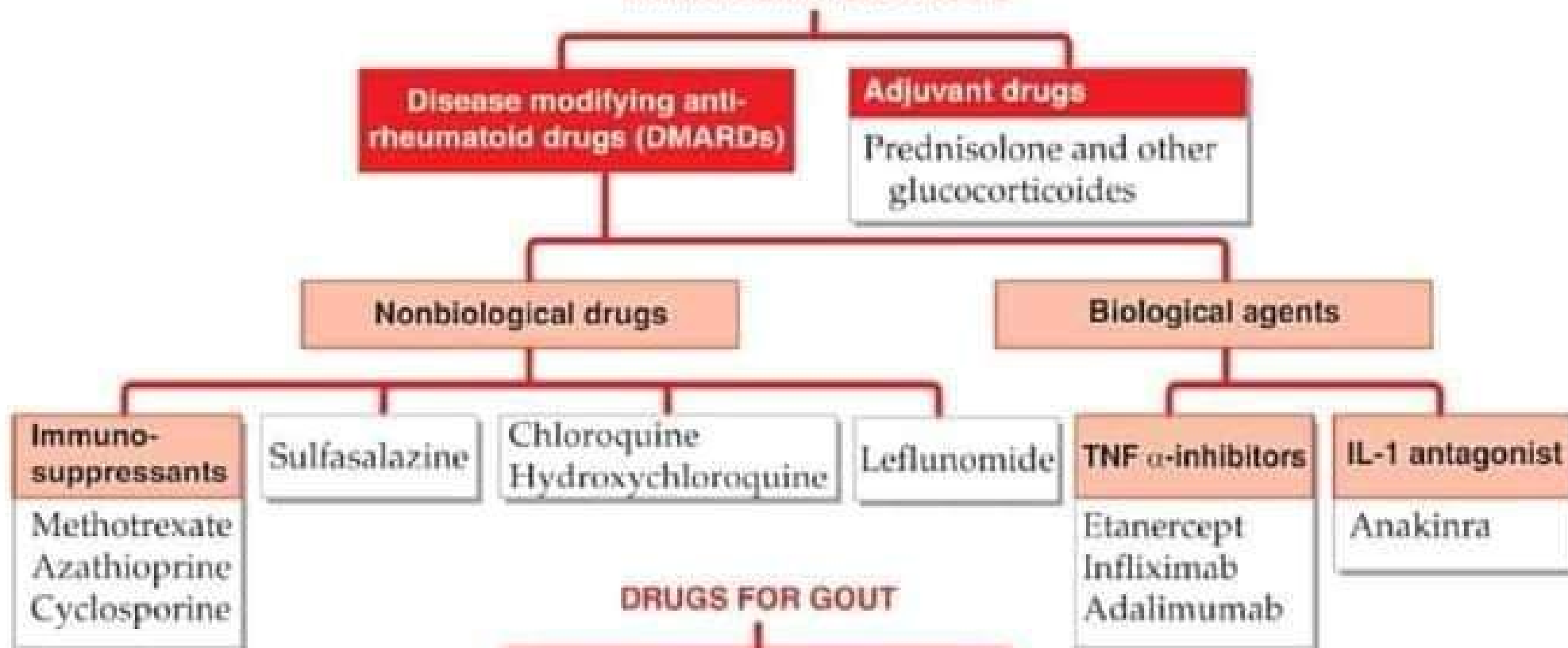
Prostaglandin analogues

Carboprost (15-methyl $\text{PGF}_{2\alpha}$)
Misoprostol (methyl PGE_1 ester)
Latanoprost (PGE_2 analogue)
Travoprost
Bimatoprost

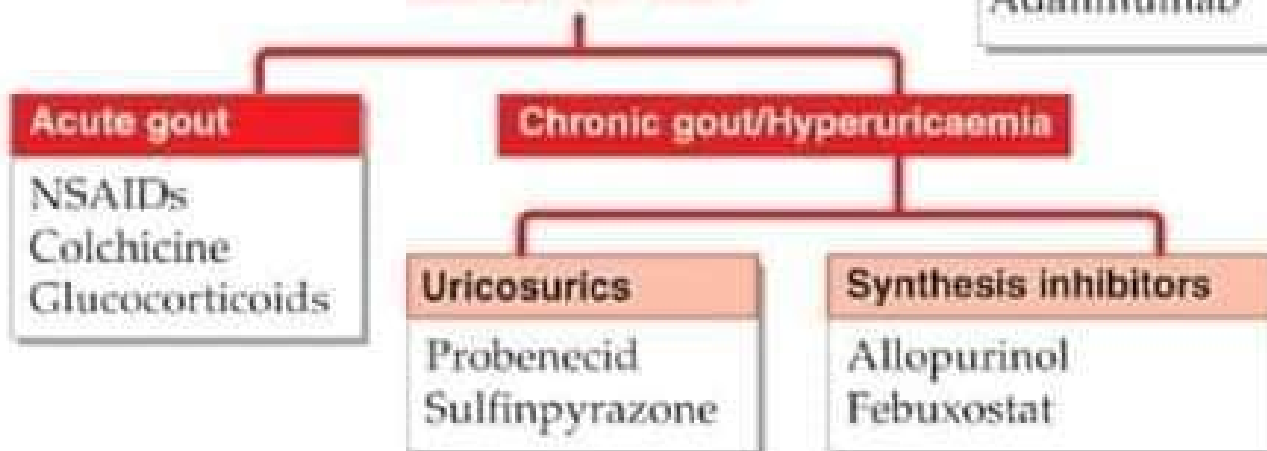
NONSTEROIDAL ANTIINFLAMMATORY DRUGS/ANTIPYRETIC-ANALGESICS



ANTIRHEUMATOID DRUGS

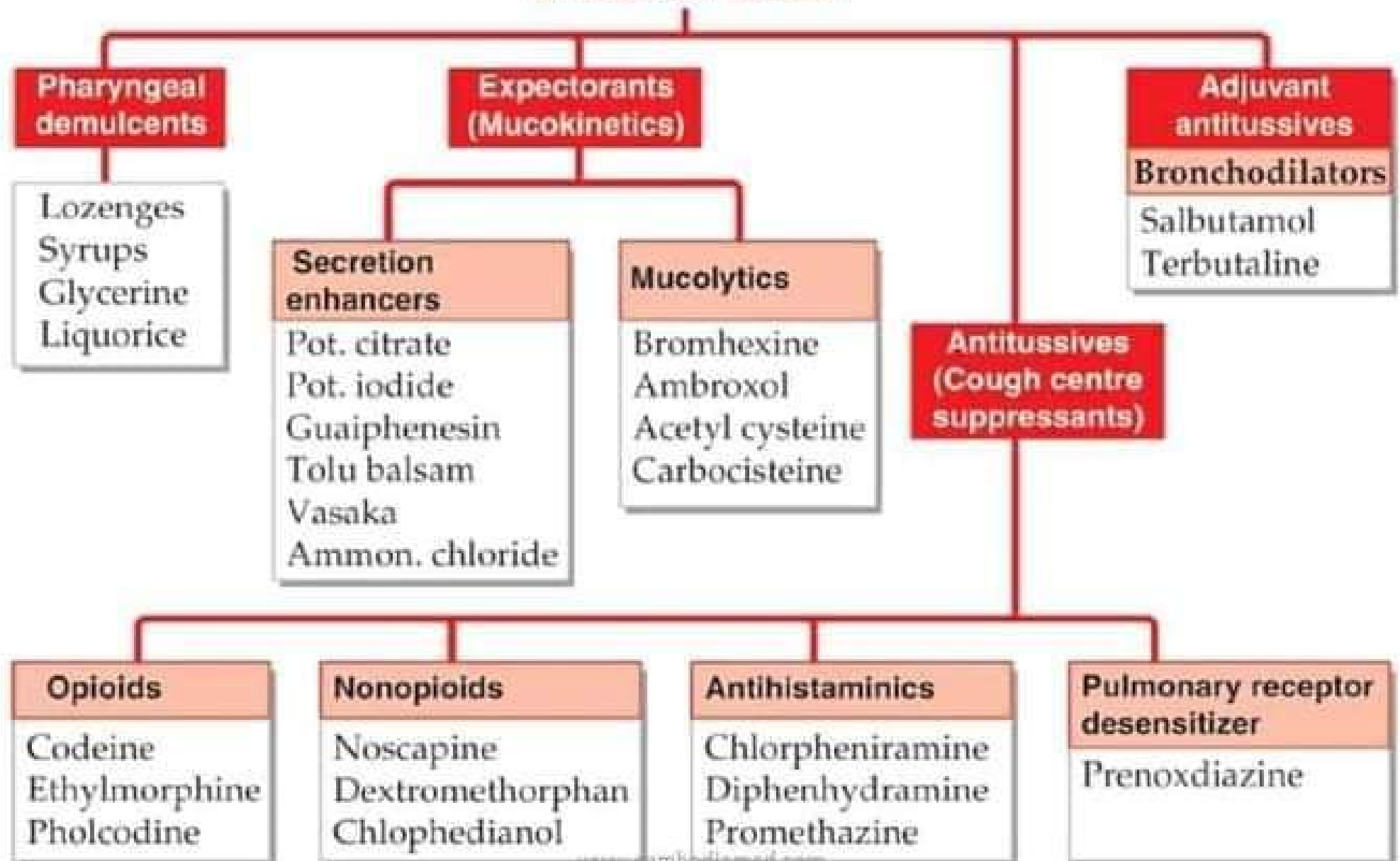


DRUGS FOR GOUT

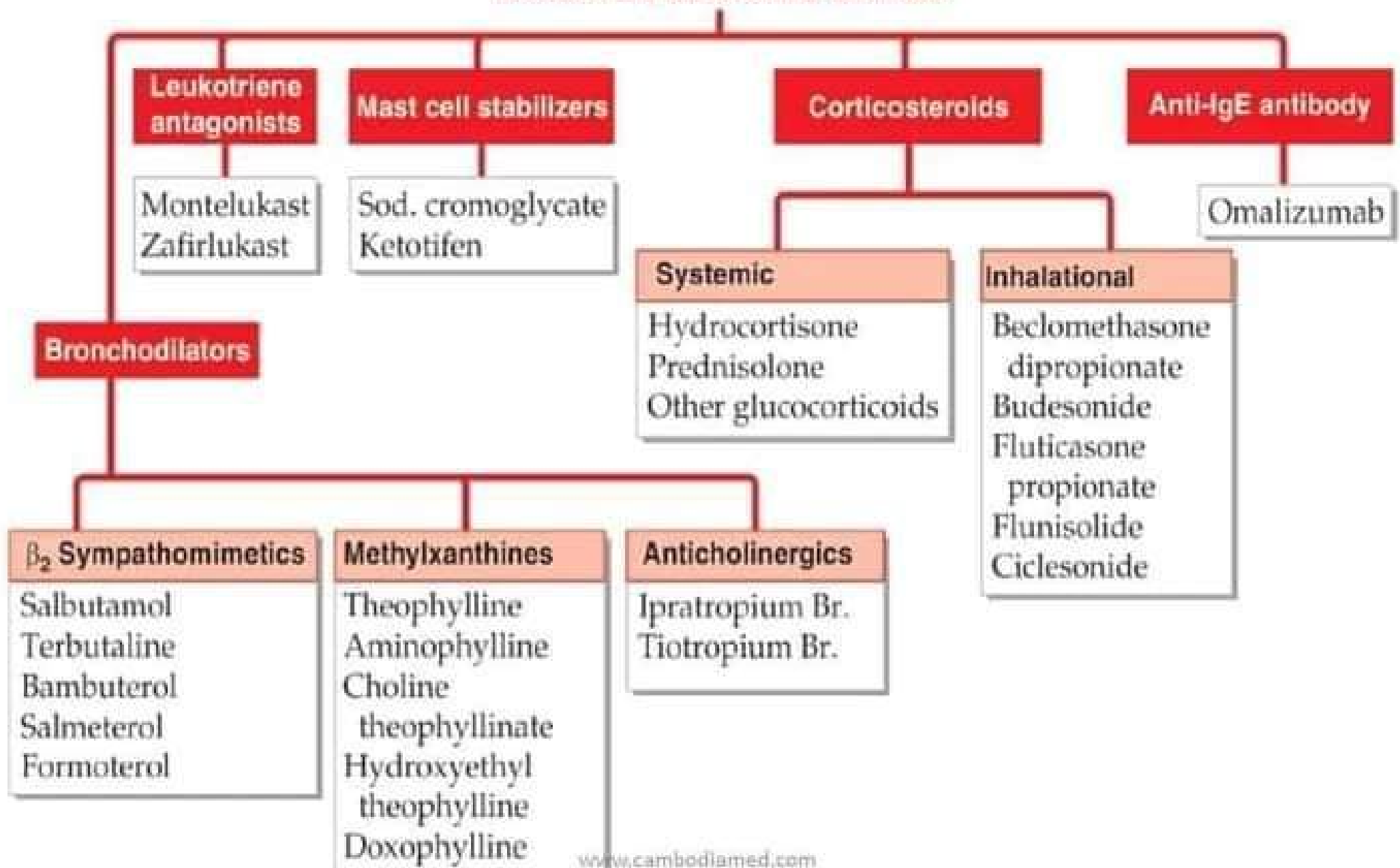


Drugs for Respiratory Disorder

DRUGS FOR COUGH



DRUGS FOR BRONCHIAL ASTHMA



Hormones and Related Drugs

ANTERIOR PITUITARY HORMONES

From acidophil cells

Growth hormone (GH)
Prolactin

From basophil cells

Thyroid stimulating hormone (TSH)
Adrenocorticotrophic hormone (ACTH)
Follicle stimulating hormone (FSH)
Luteinizing hormone (LH)

DRUGS ALTERING ANTERIOR PITUITARY HORMONE SECRETION

Inhibit GH release

Somatostatin
Octreotide
Lanreotide

Inhibit prolactin release

Bromocriptine
Cabergoline
Apomorphine

Enhance prolactin release

Chlorpromazine
(other neuroleptics)
Metoclopramide
Reserpine

Inhibit gonadotropin (Gn) release

Superactive GnRH agonists

(Initially enhance Gn release)
Nafarelin
Triptorelin
Goserelin
Leuprolide

www.cambridgemed.com

GnRH antagonists

Ganirelix
Cetrorelix

Thyroid hormone

Thyroxine (T_4)

Triiodothyronine (T_3 , Liothyronine)

THYROID INHIBITORS

Inhibit hormone synthesis

(Antithyroid drugs)

Propylthiouracil

Methimazole

Carbimazole

Inhibit iodide trapping

(Ionic inhibitors)

Thiocyanates ($-SCN$)

Perchlorates ($-ClO_4$)

Nitrates ($-NO_3$)

Inhibit hormone release

Iodine

Iodides of Na, K

Organic iodide

Destroy thyroid tissue

Radioactive iodine ^{131}I

INSULINS

```
graph TD; INSULINS --> Rapid_acting[Rapid acting]; INSULINS --> Short_acting[Short acting]; INSULINS --> Intermediate_acting[Intermediate acting]; INSULINS --> Long_acting[Long acting]; Rapid_acting --> RA["(Insulin analogues)  
Insulin lispro  
Insulin aspart  
Insulin glulisine"]; Short_acting --> SA["Regular (soluble)  
insulin"]; Intermediate_acting --> IA["Insulin zinc suspension  
(Lente insulin)  
Neutral protamine hagedorn  
(NPH) or Isophane insulin"]; Long_acting --> LA["(Insulin analogues)  
Insulin glargine  
Insulin detemir"];
```

Rapid acting

(Insulin analogues)
Insulin lispro
Insulin aspart
Insulin glulisine

Short acting

Regular (soluble)
insulin

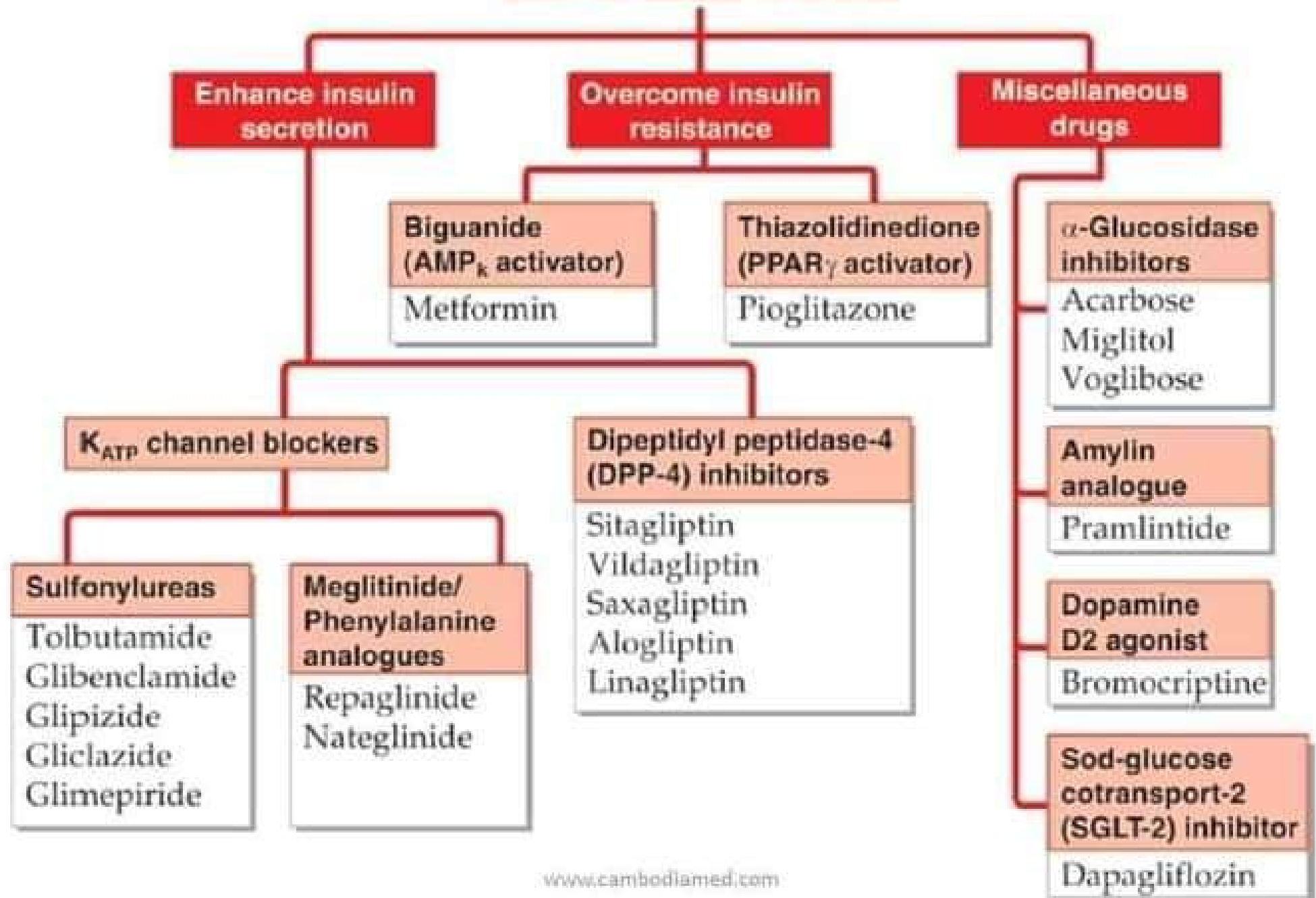
Intermediate acting

Insulin zinc suspension
(Lente insulin)
Neutral protamine hagedorn
(NPH) or Isophane insulin

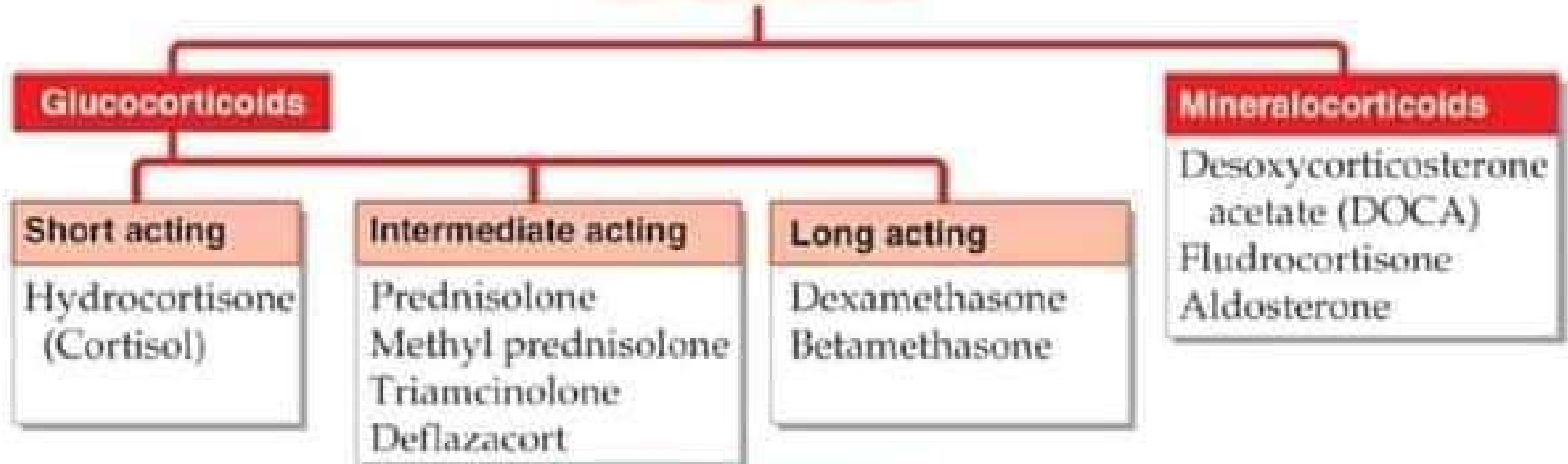
Long acting

(Insulin analogues)
Insulin glargine
Insulin detemir

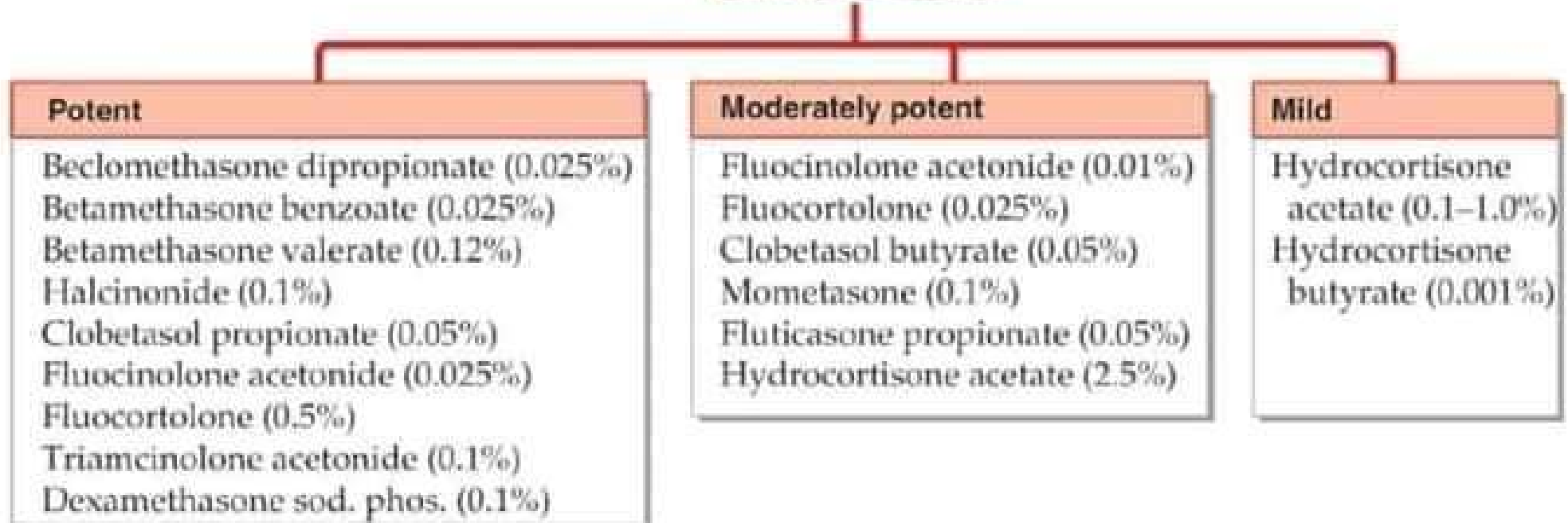
ORAL ANTIDIABETIC DRUGS



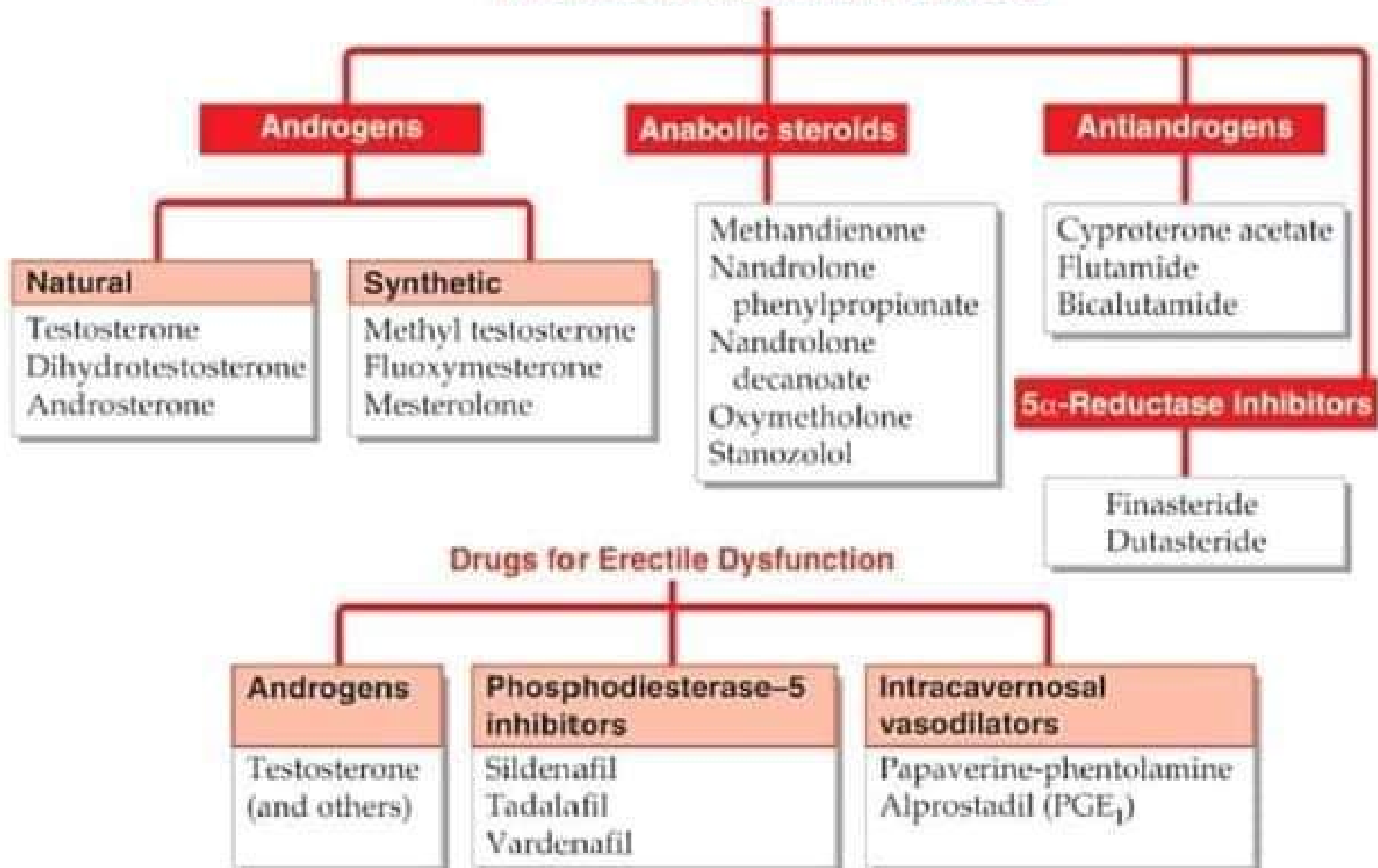
CORTICOSTEROIDS



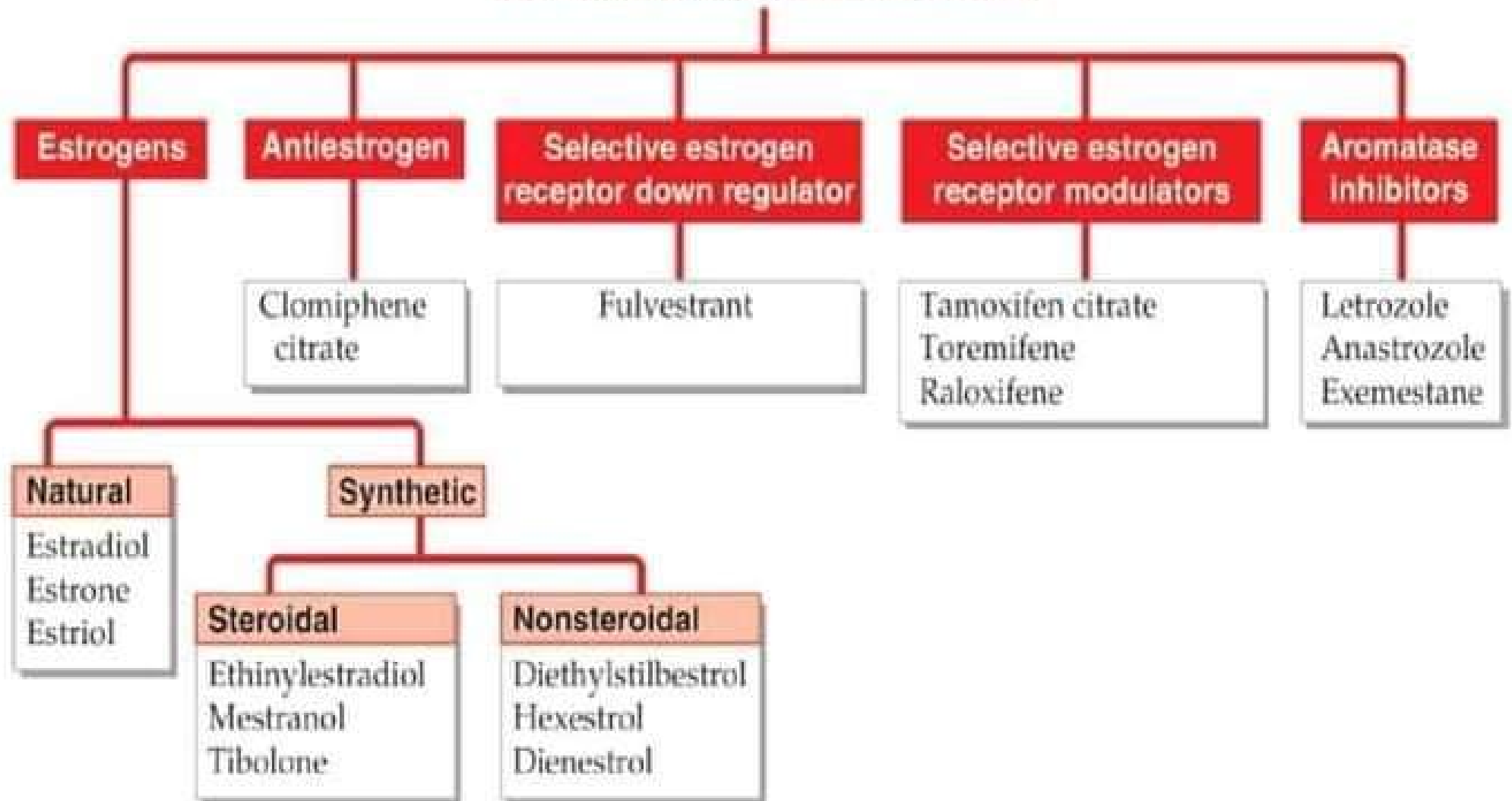
TOPICAL STEROIDS



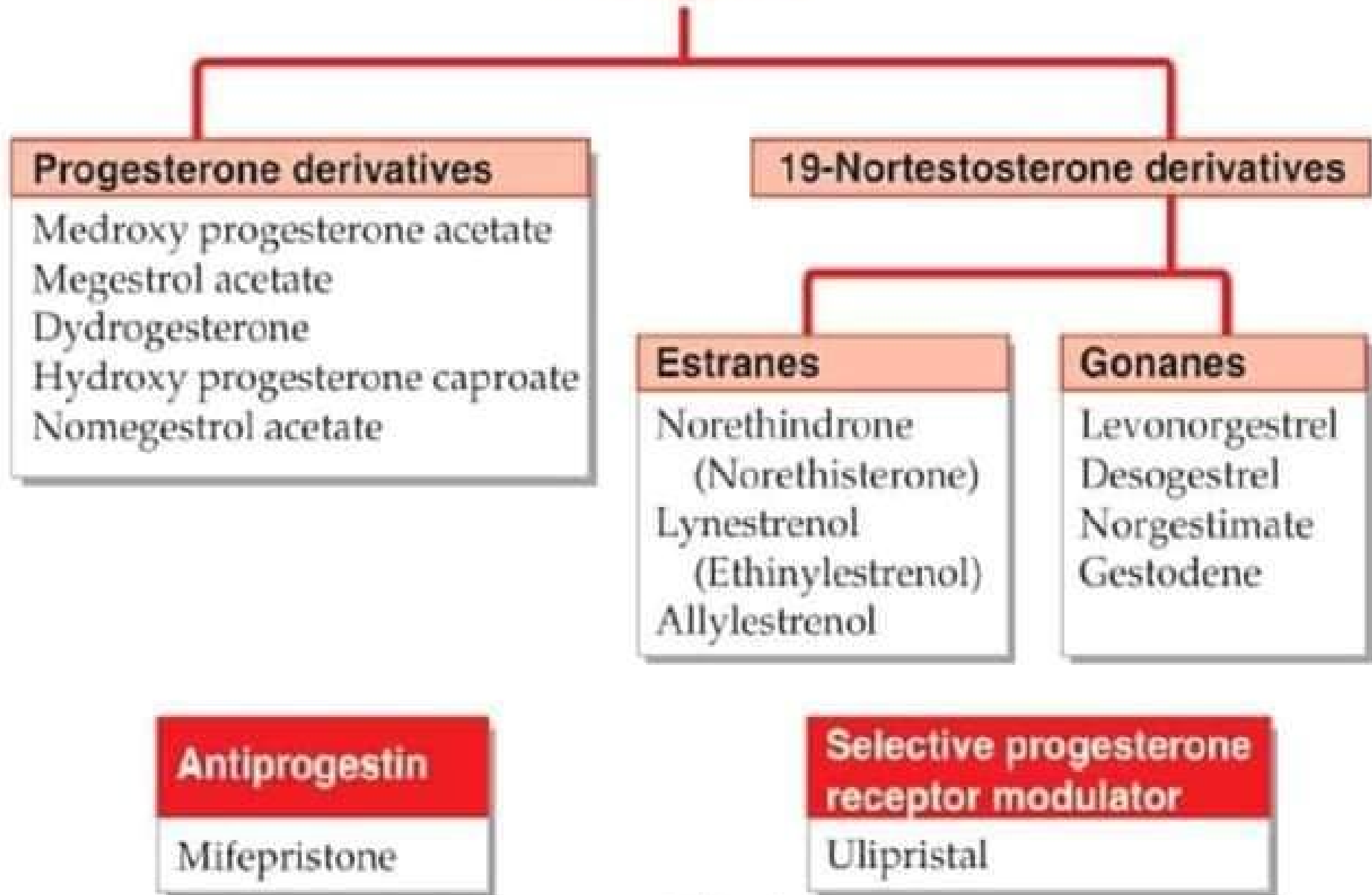
ANDROGENS AND RELATED DRUGS



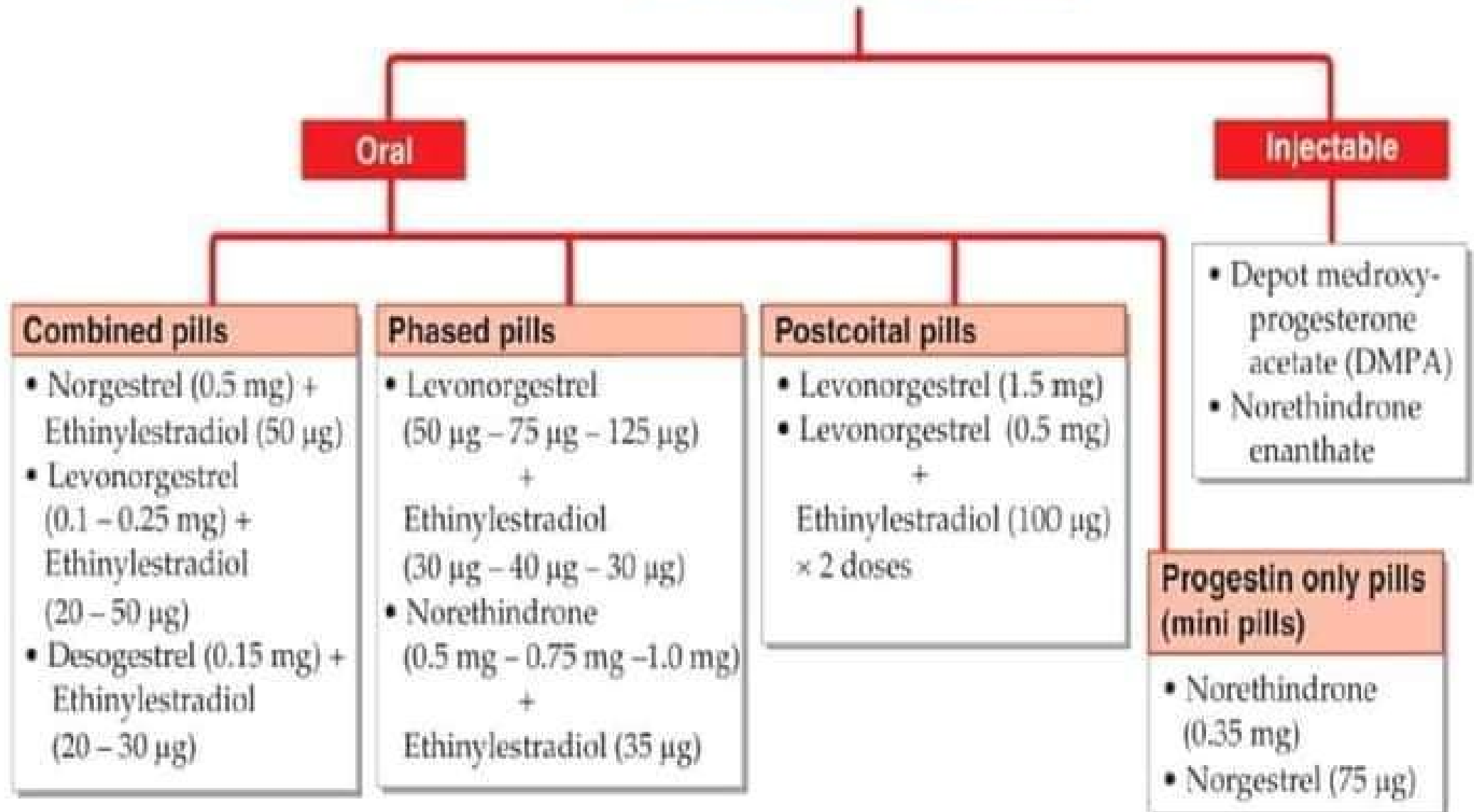
ESTROGENS AND RELATED DRUGS



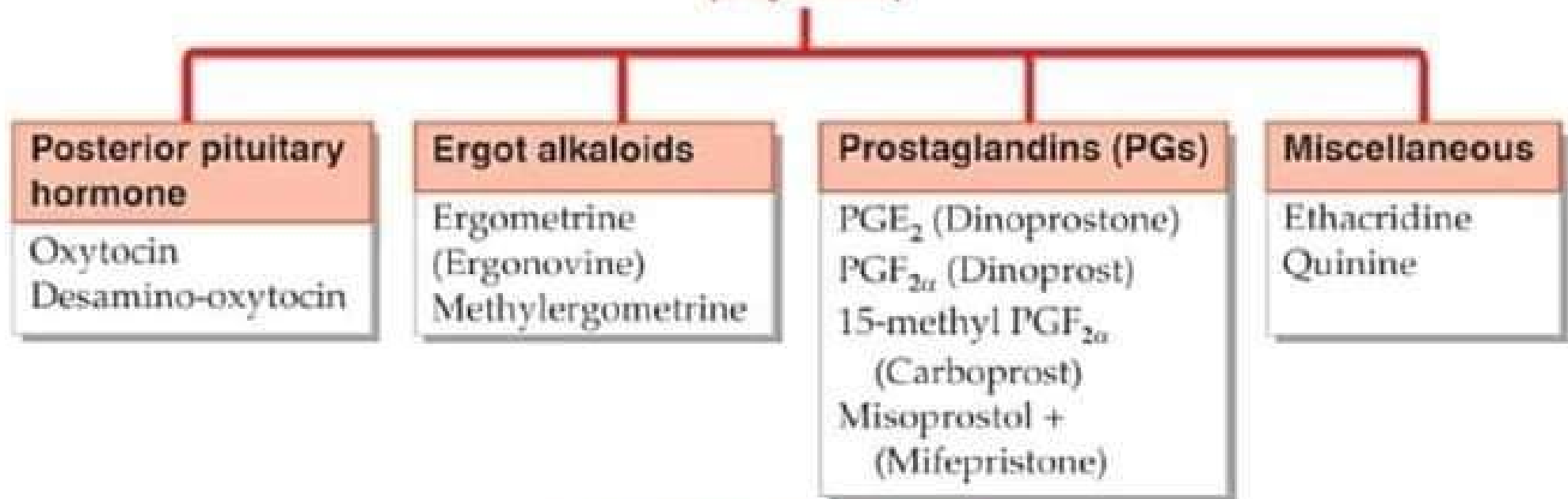
PROGESTINS



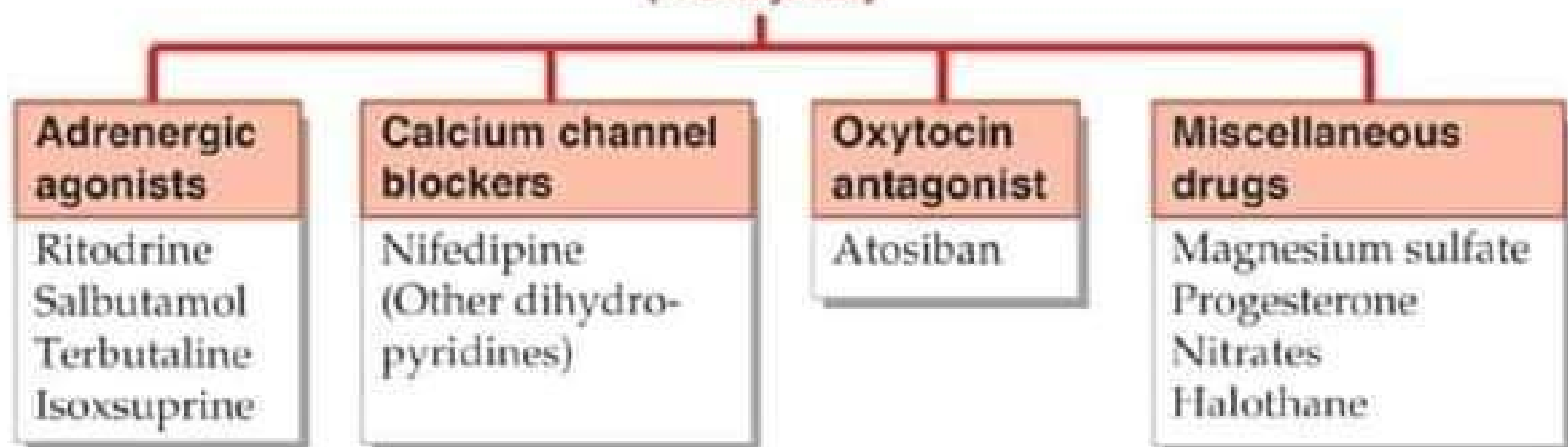
HORMONAL CONTRACEPTIVES



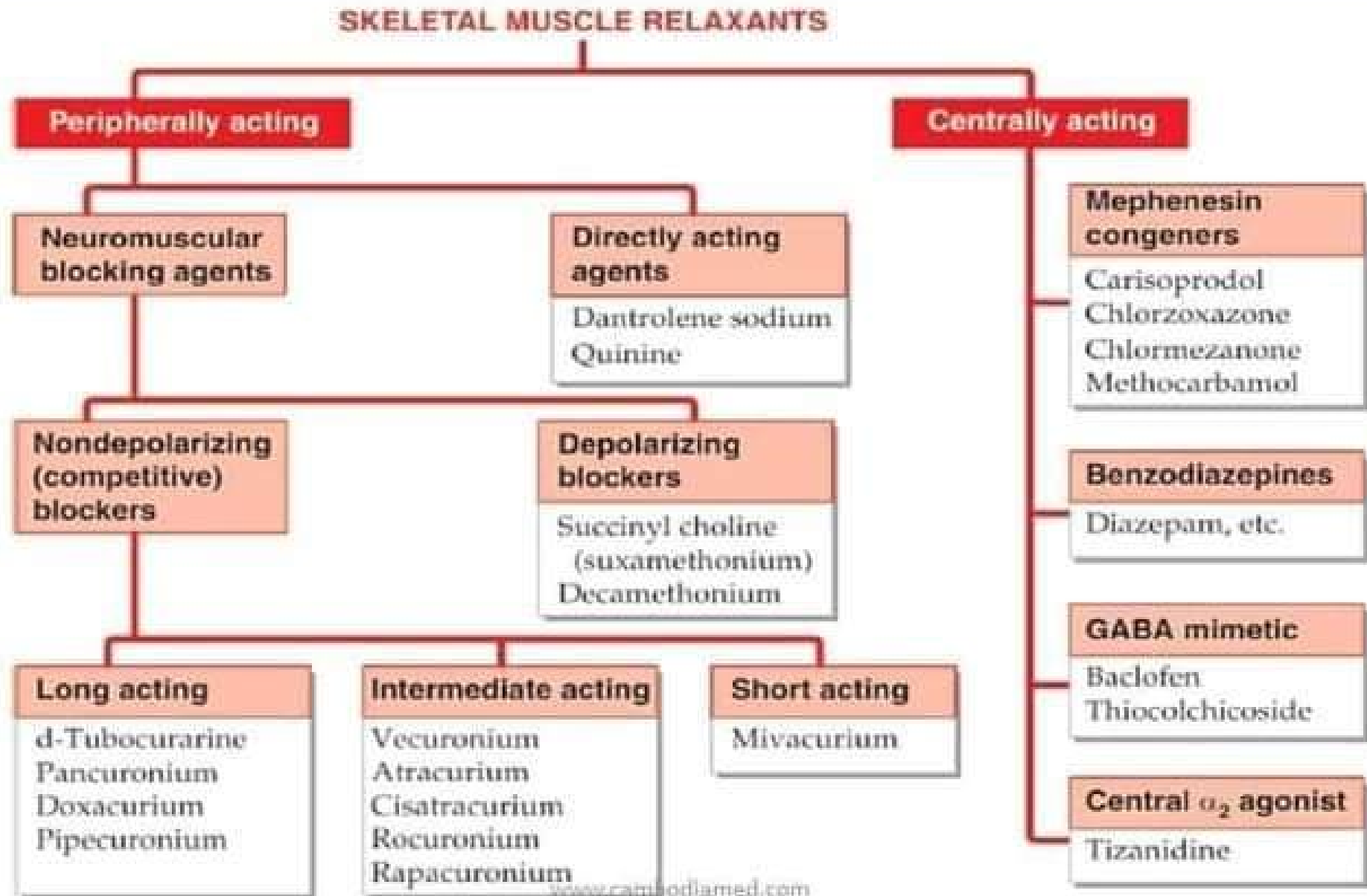
UTERINE STIMULANTS (Oxytocics)



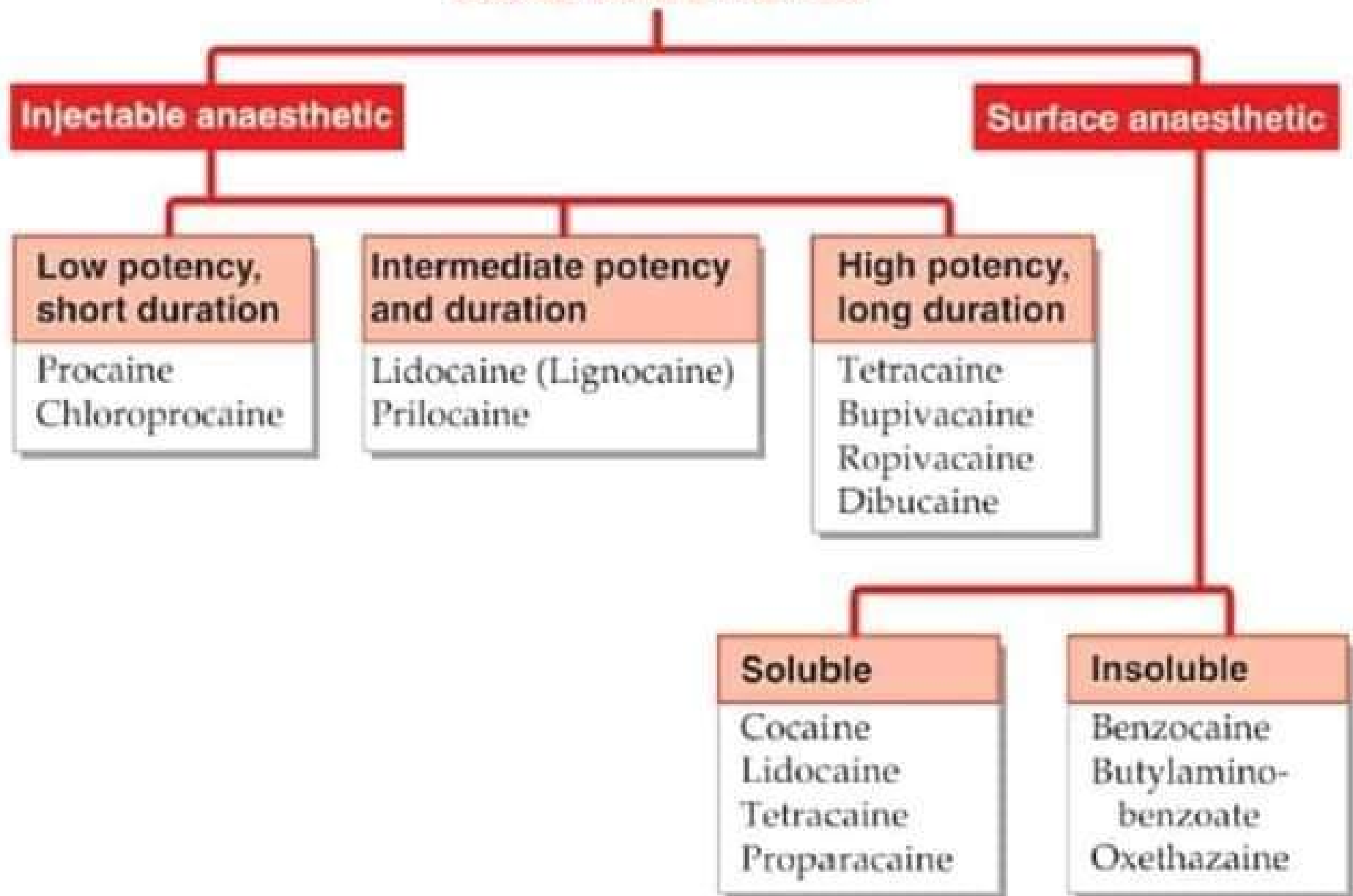
UTERINE RELAXANTS (Tocolytics)



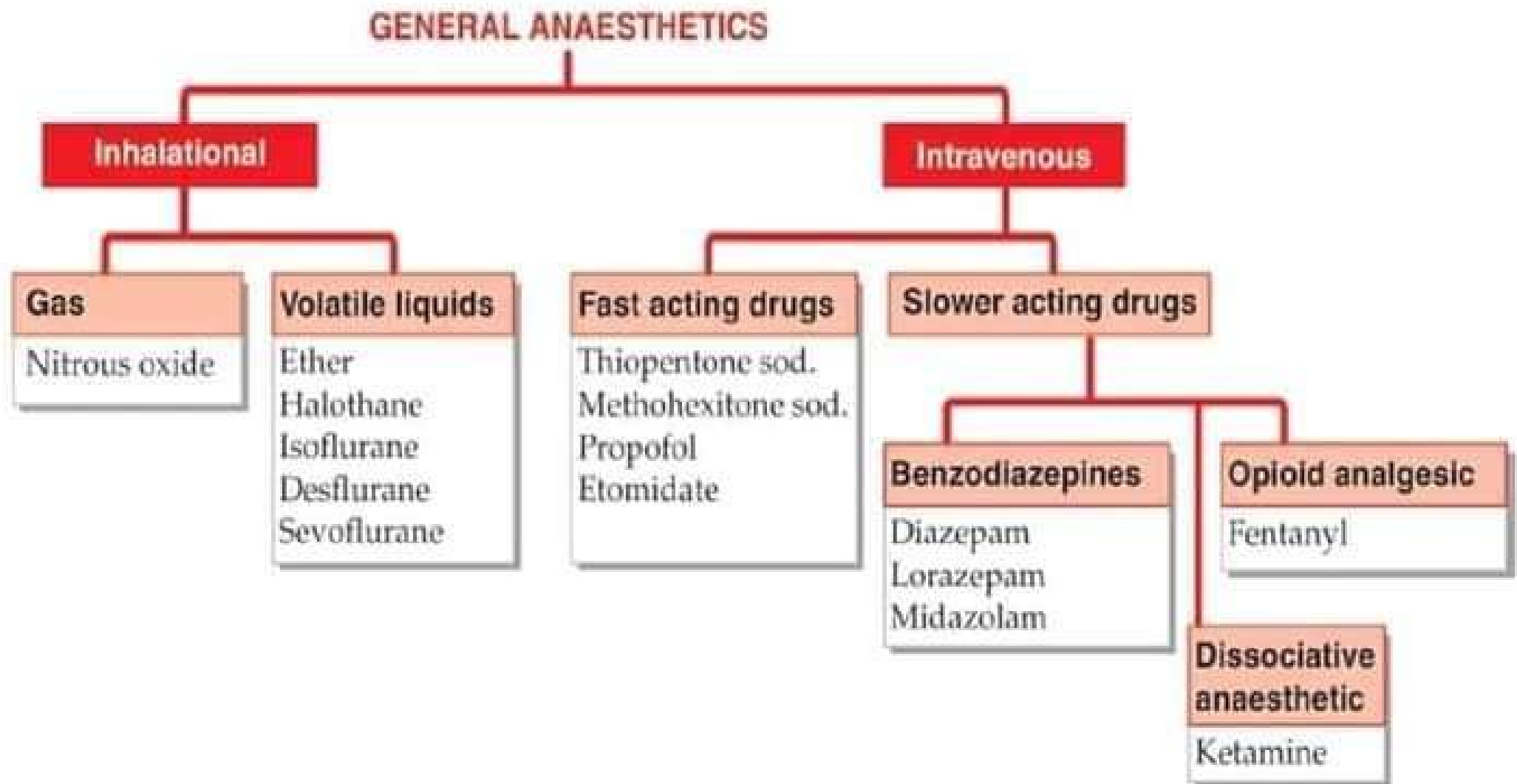
Drugs Acting on Peripheral (somatic) Nervous System



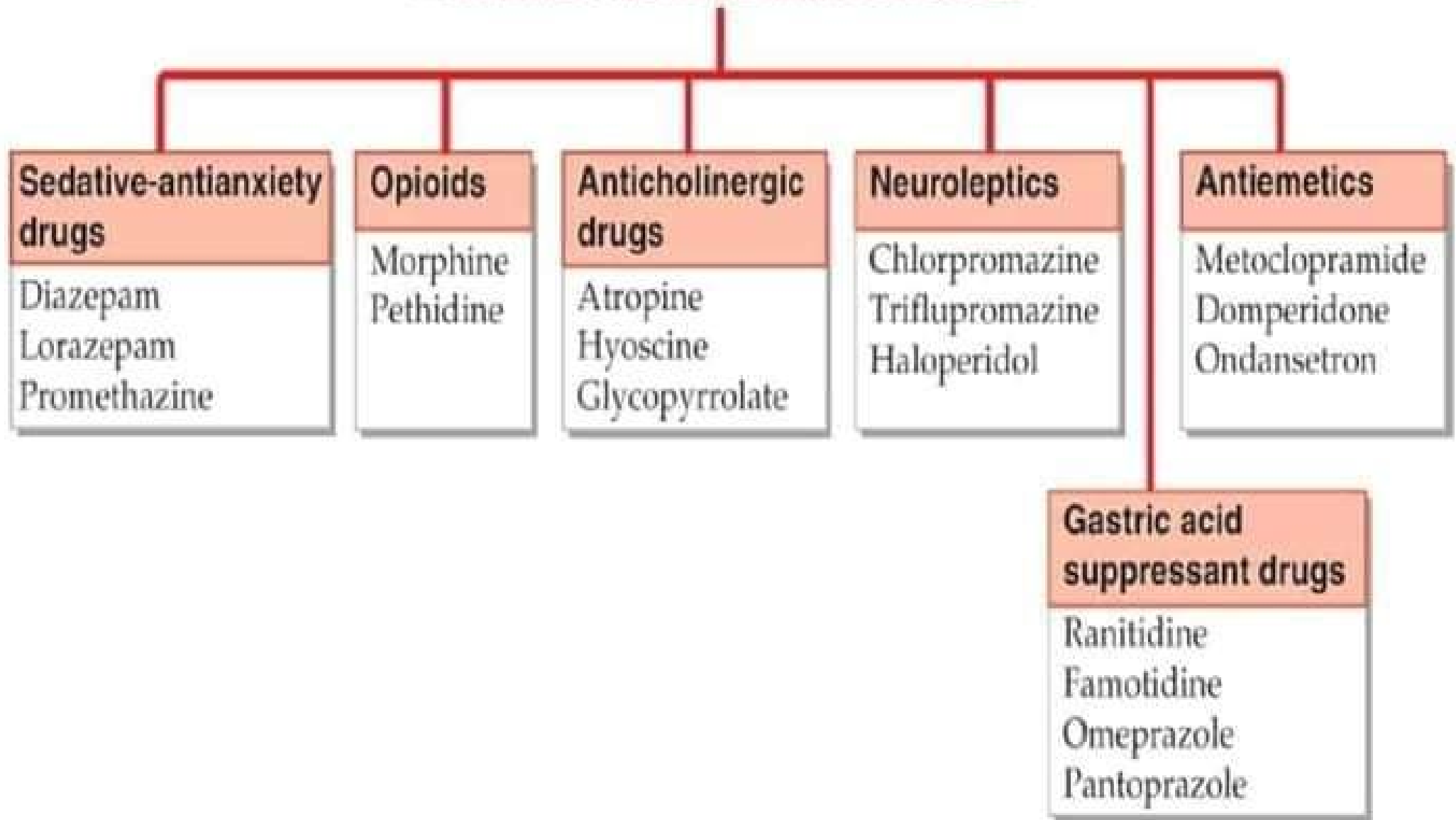
LOCAL ANAESTHETICS



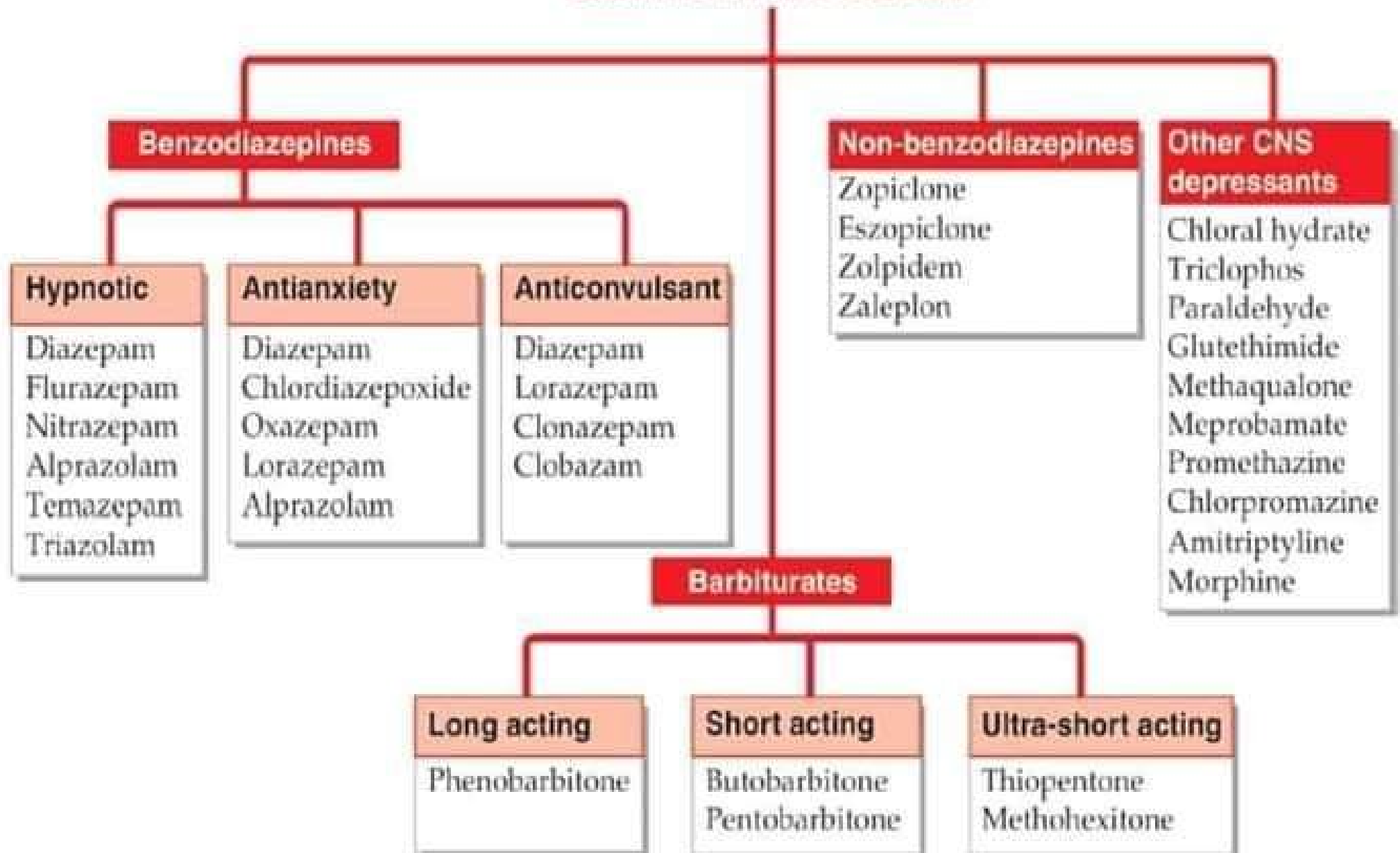
Drugs Acting on Central Nervous System



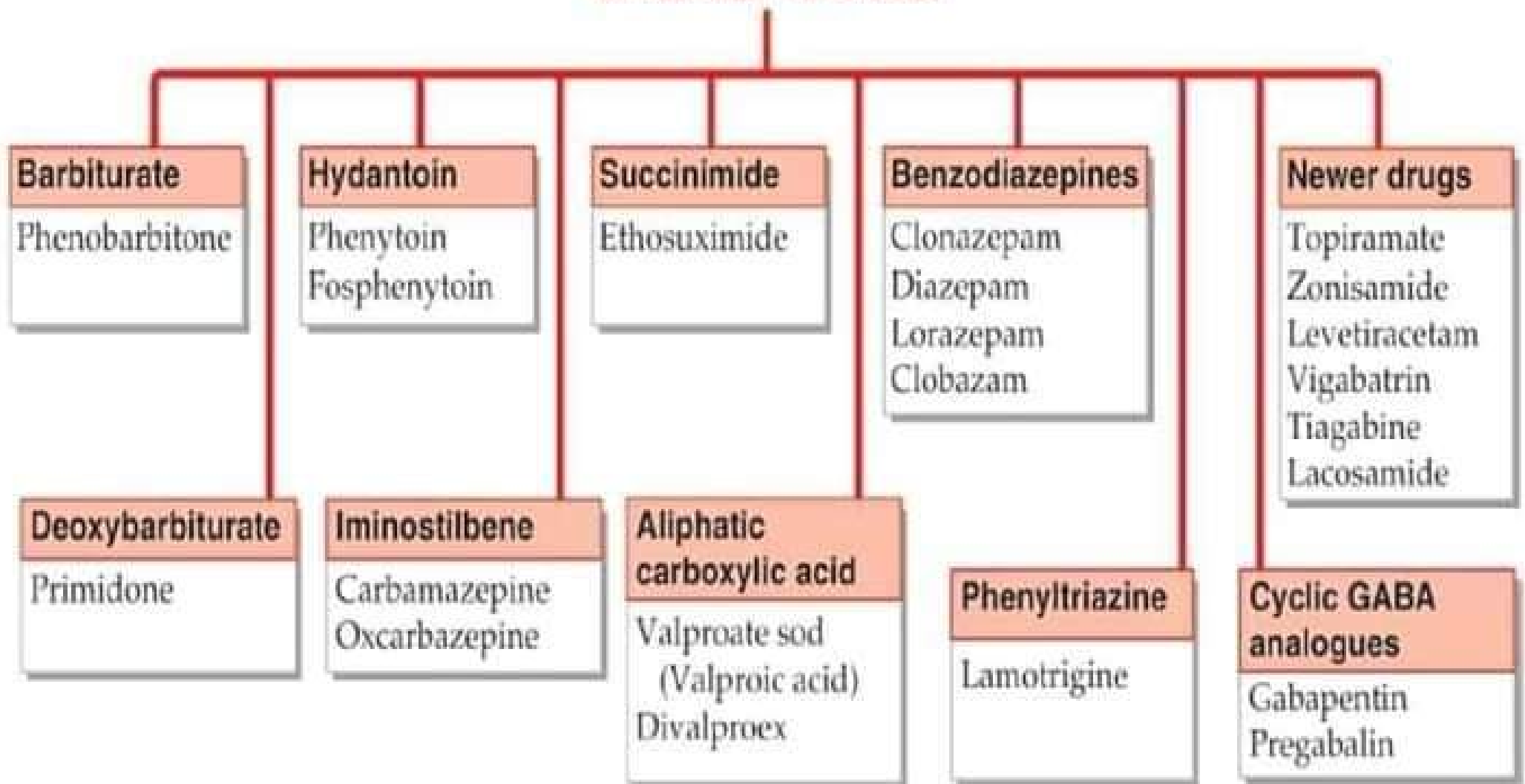
PREANAESTHETIC MEDICATION DRUGS



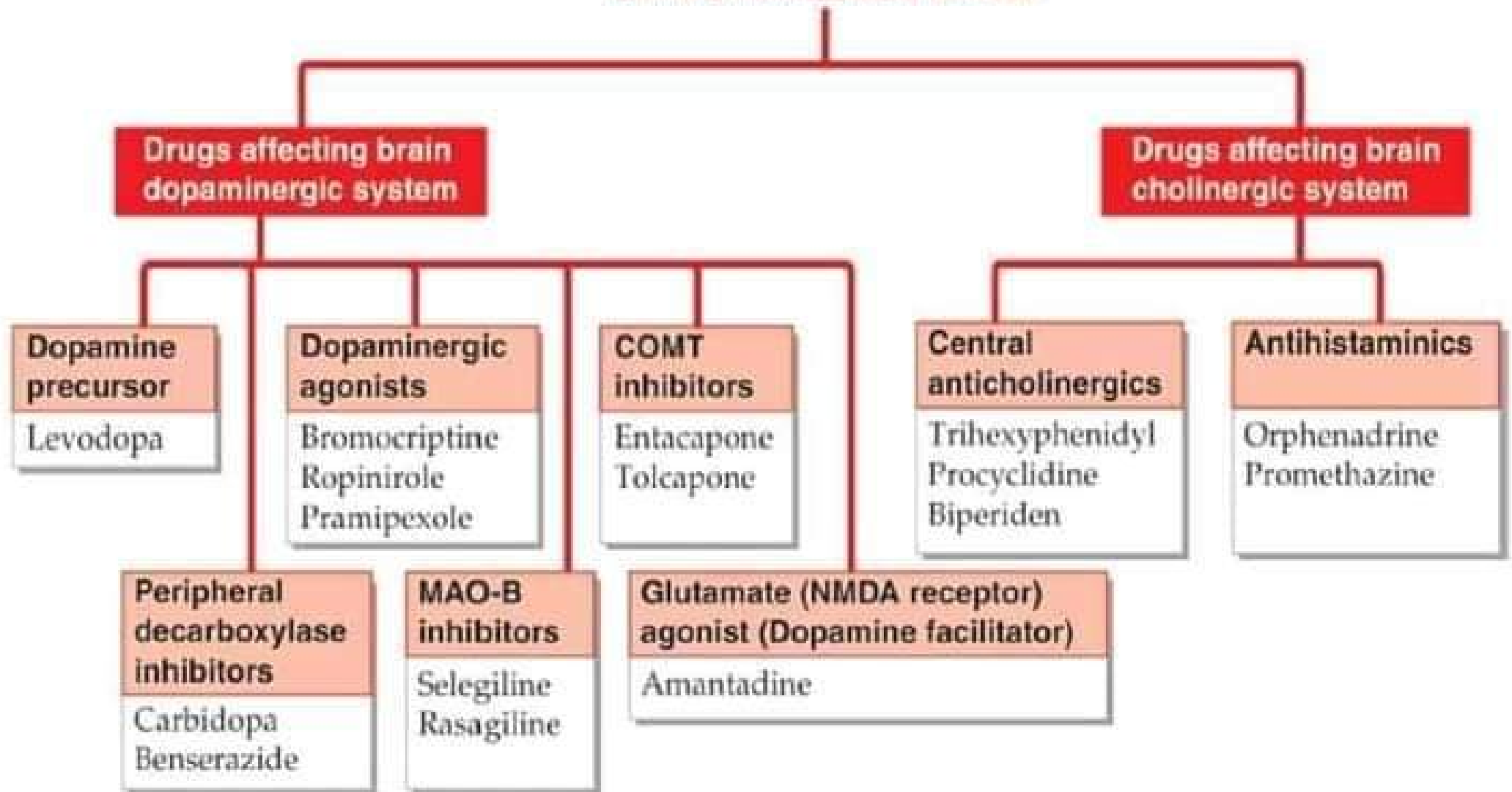
SEDATIVE-HYPNOTIC DRUGS



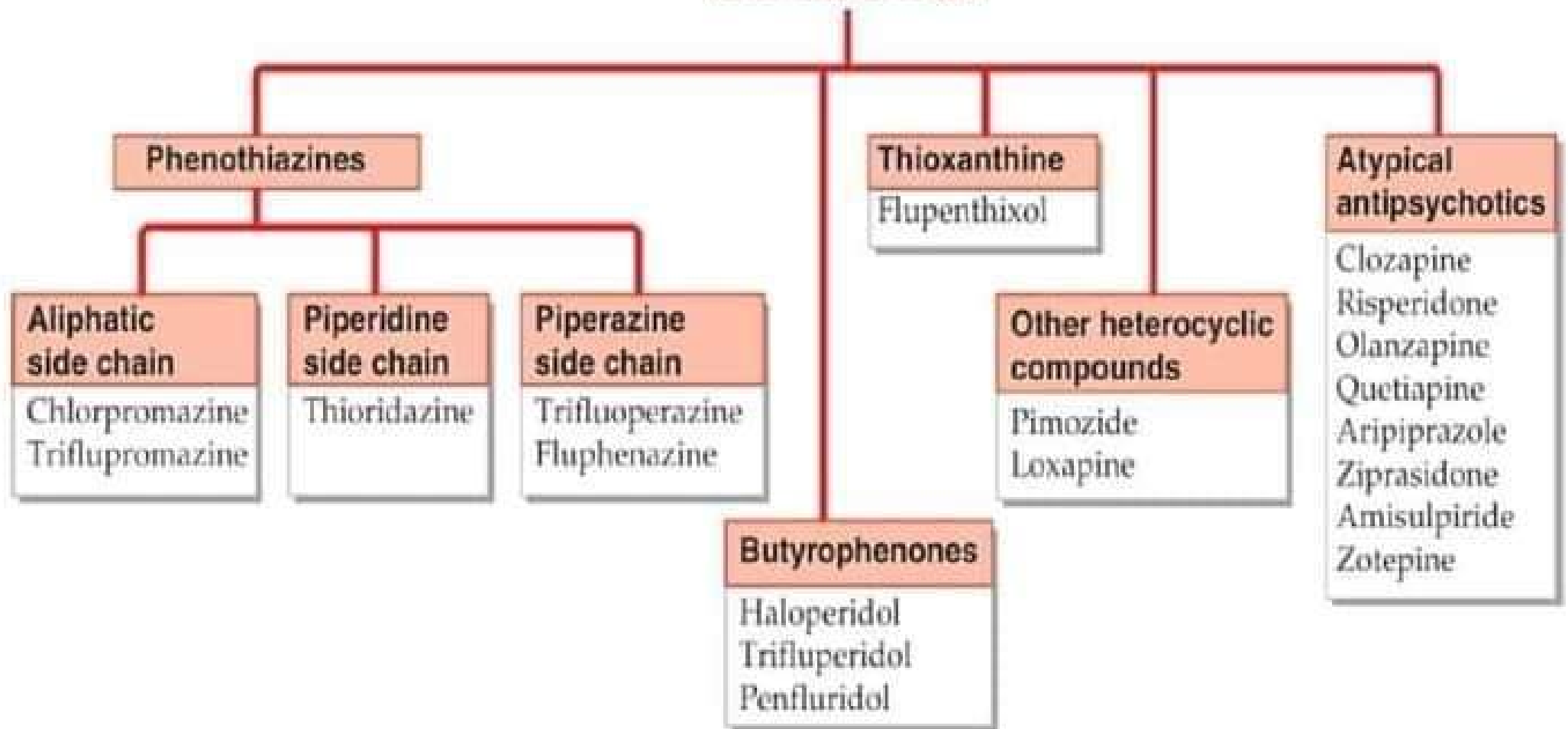
ANTIEPILEPTIC DRUGS



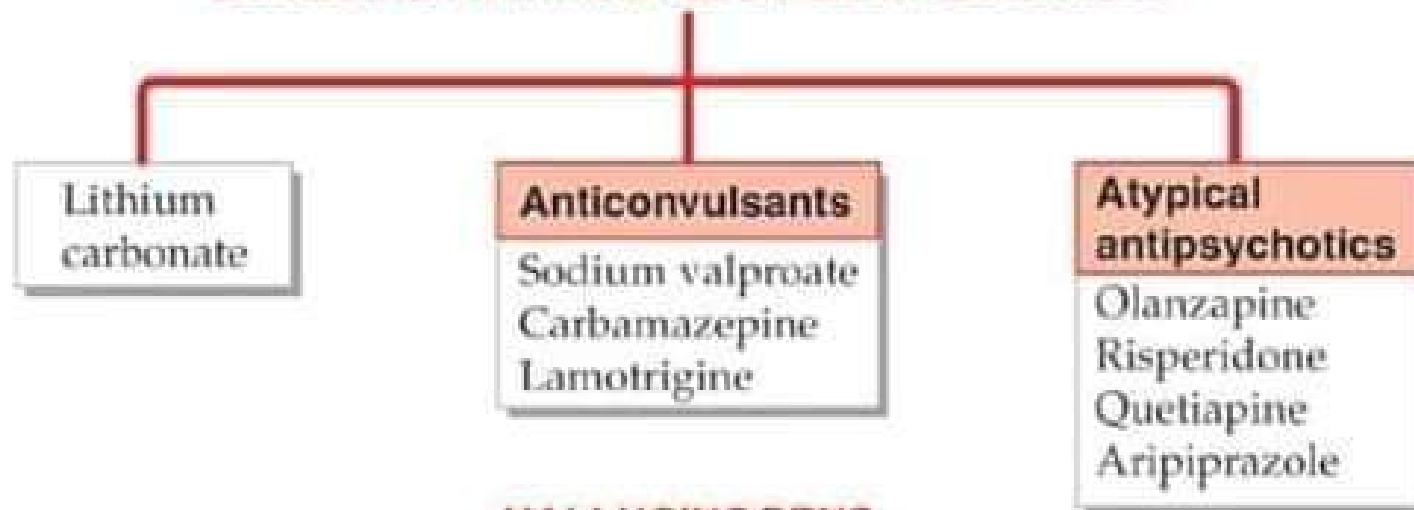
ANTIPARKINSONIAN DRUGS



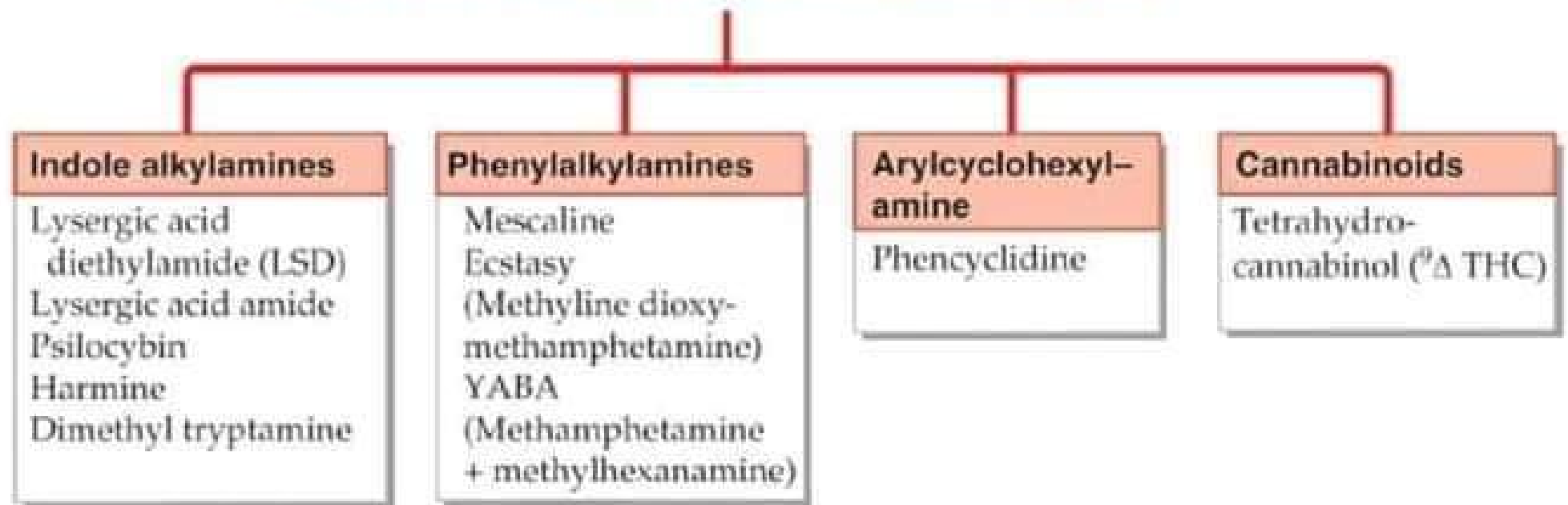
ANTIPSYCHOTIC DRUGS (Neuroleptic drugs)



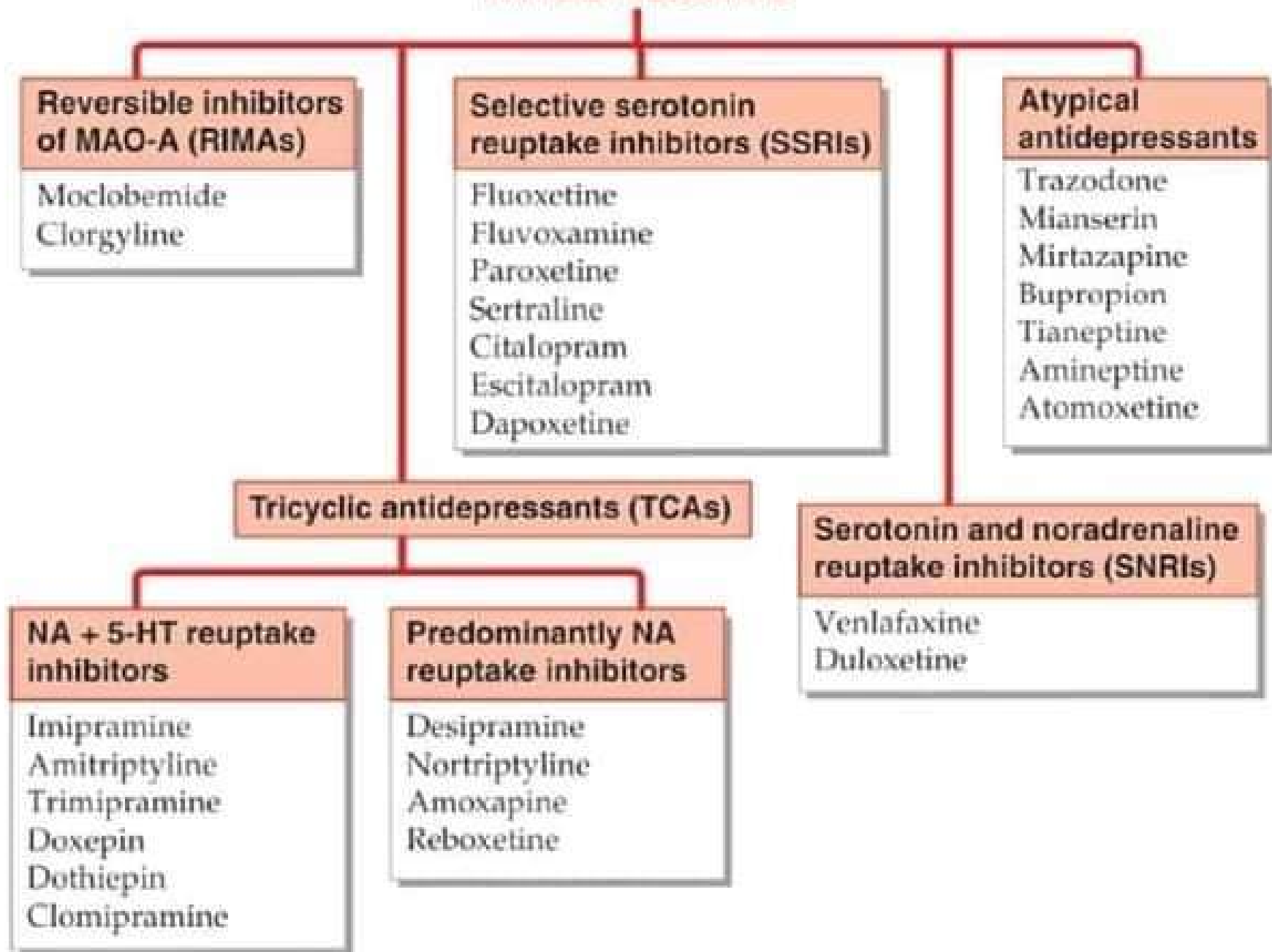
DRUGS FOR MANIA AND BIPOLAR DISORDER



HALLUCINOGENS (Psychotomimetics, Psychedelics, Psychodysleptics)



ANTIDEPRESSANTS



ANTIANXIETY DRUGS

```
graph TD; A[ANTIANXIETY DRUGS] --> B[Benzodiazepines]; A --> C[Azapirones]; A --> D[Sedative-antihistaminic]; A --> E["β-adrenergic blocker"]; B --> B1[Diazepam]; B --> B2[Oxazepam]; B --> B3[Lorazepam]; B --> B4[Chlordiazepoxide]; B --> B5[Alprazolam]; C --> C1[Buspirone]; C --> C2[Gepirone]; C --> C3[Ispapirone]; D --> D1[Hydroxyzine]; E --> E1[Propranolol];
```

Benzodiazepines

Diazepam
Oxazepam
Lorazepam
Chlordiazepoxide
Alprazolam

Azapirones

Buspirone
Gepirone
Ispapirone

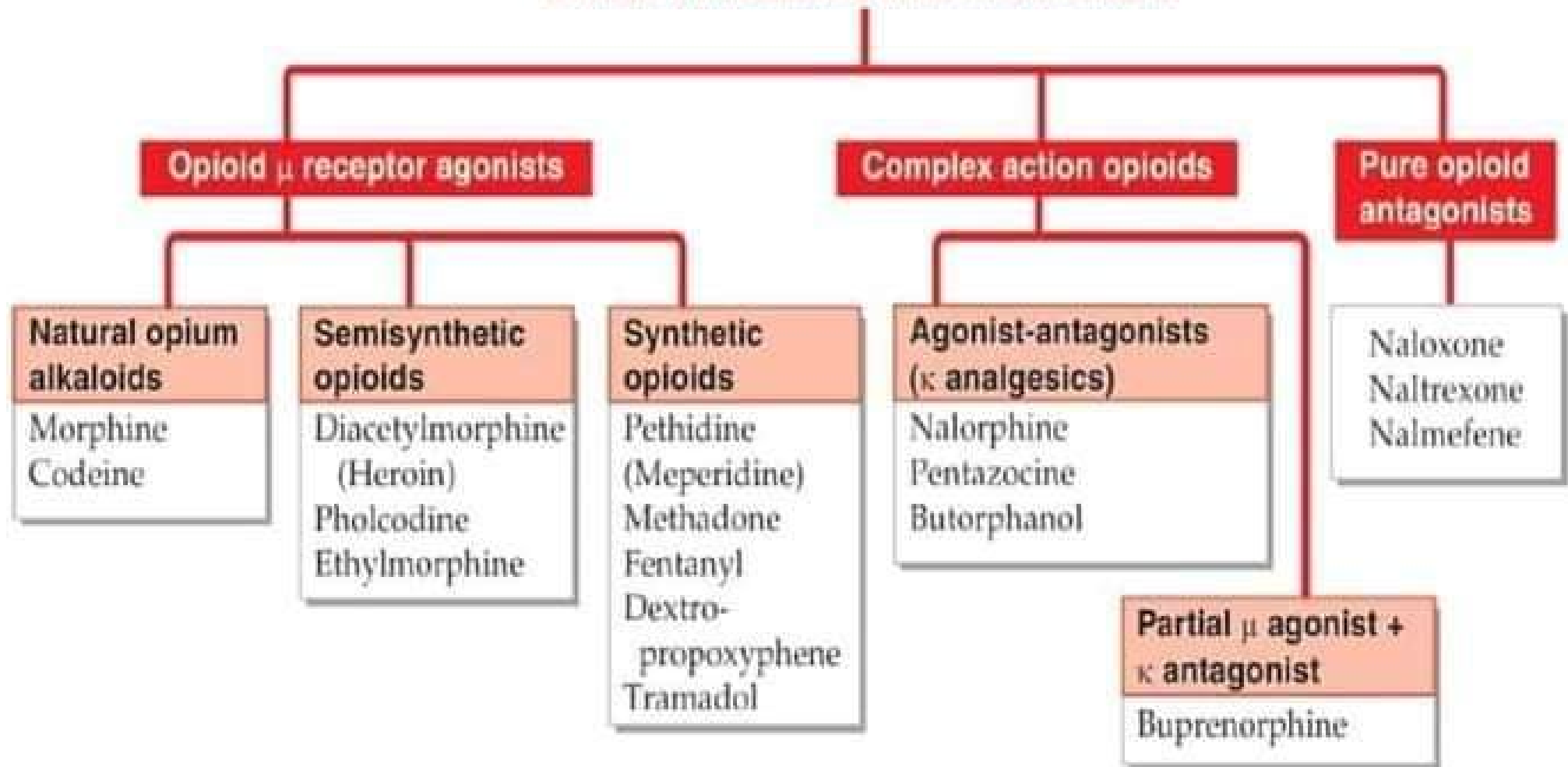
Sedative- antihistaminic

Hydroxyzine

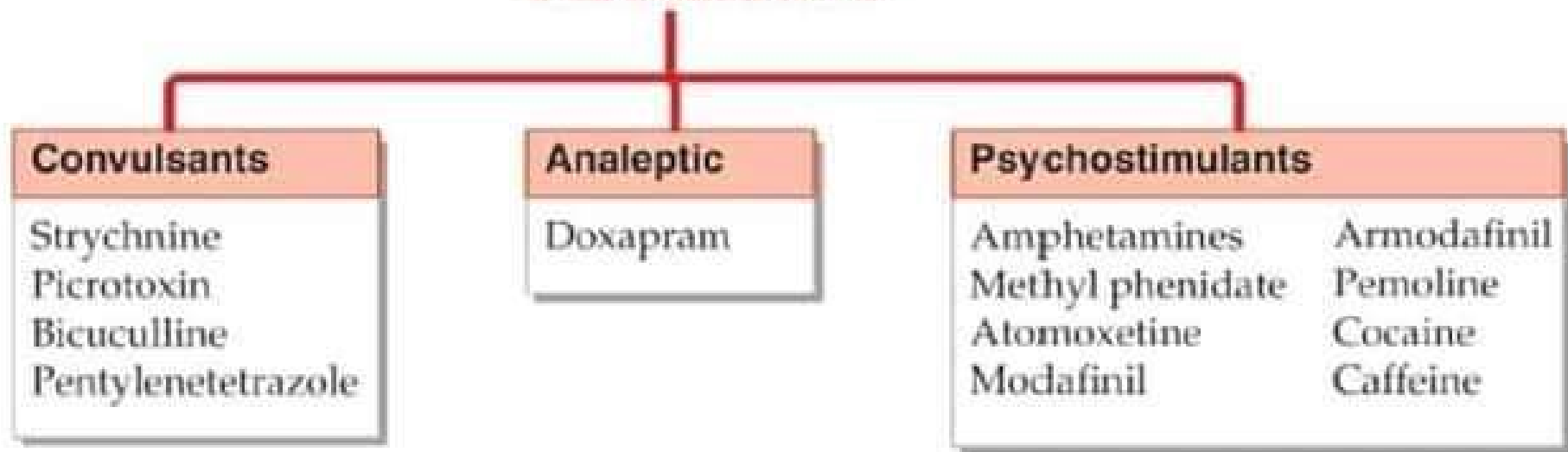
β-adrenergic blocker

Propranolol

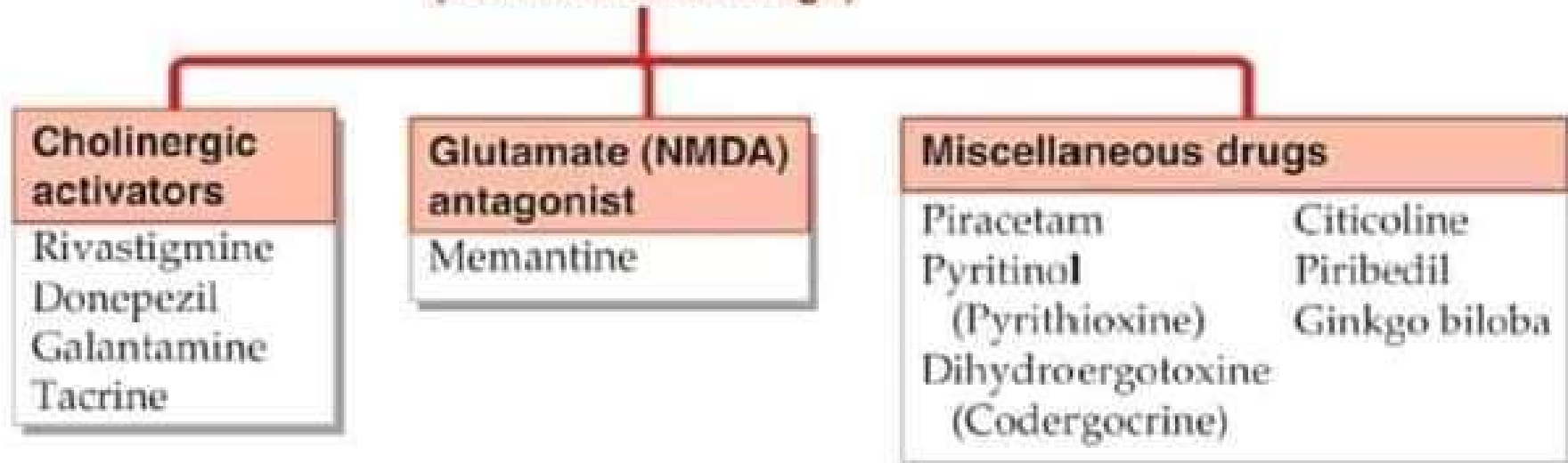
OPIOID ANALGESICS AND ANTAGONISTS



CNS STIMULANTS

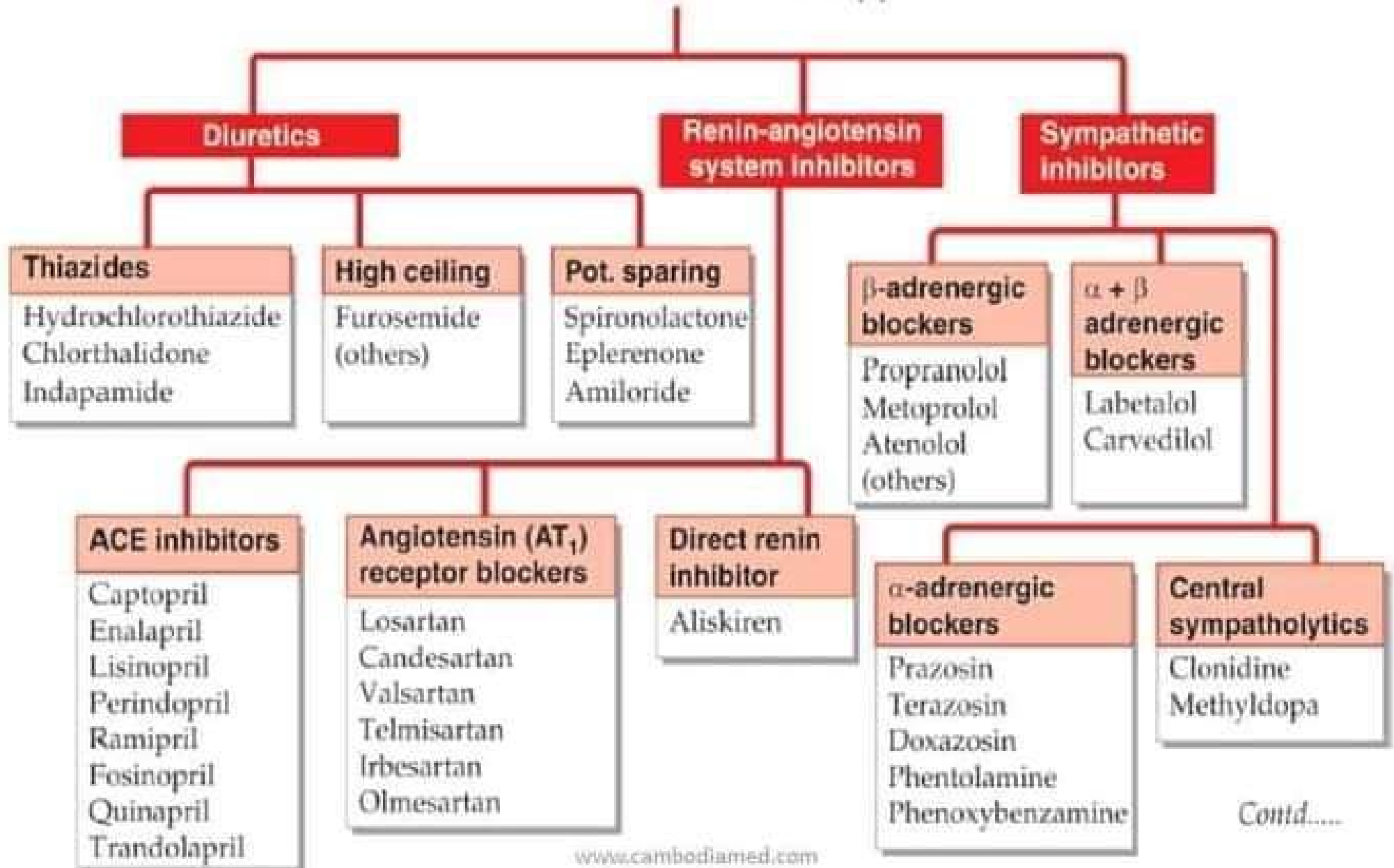


COGNITION ENHANCERS (Cerebroactive drugs)



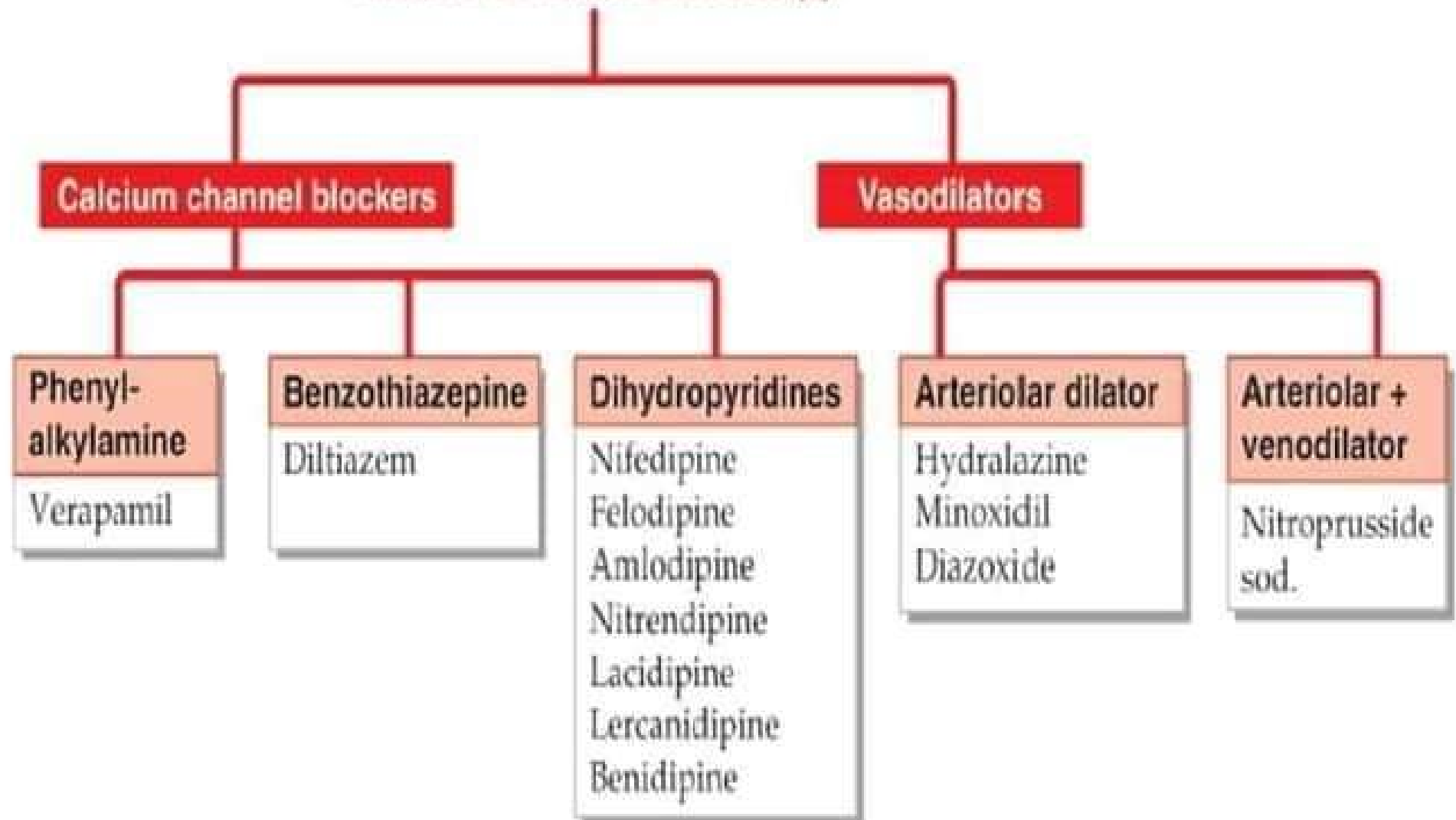
Cardiovascular Drugs

ANTIHYPERTENSIVE DRUGS (1)

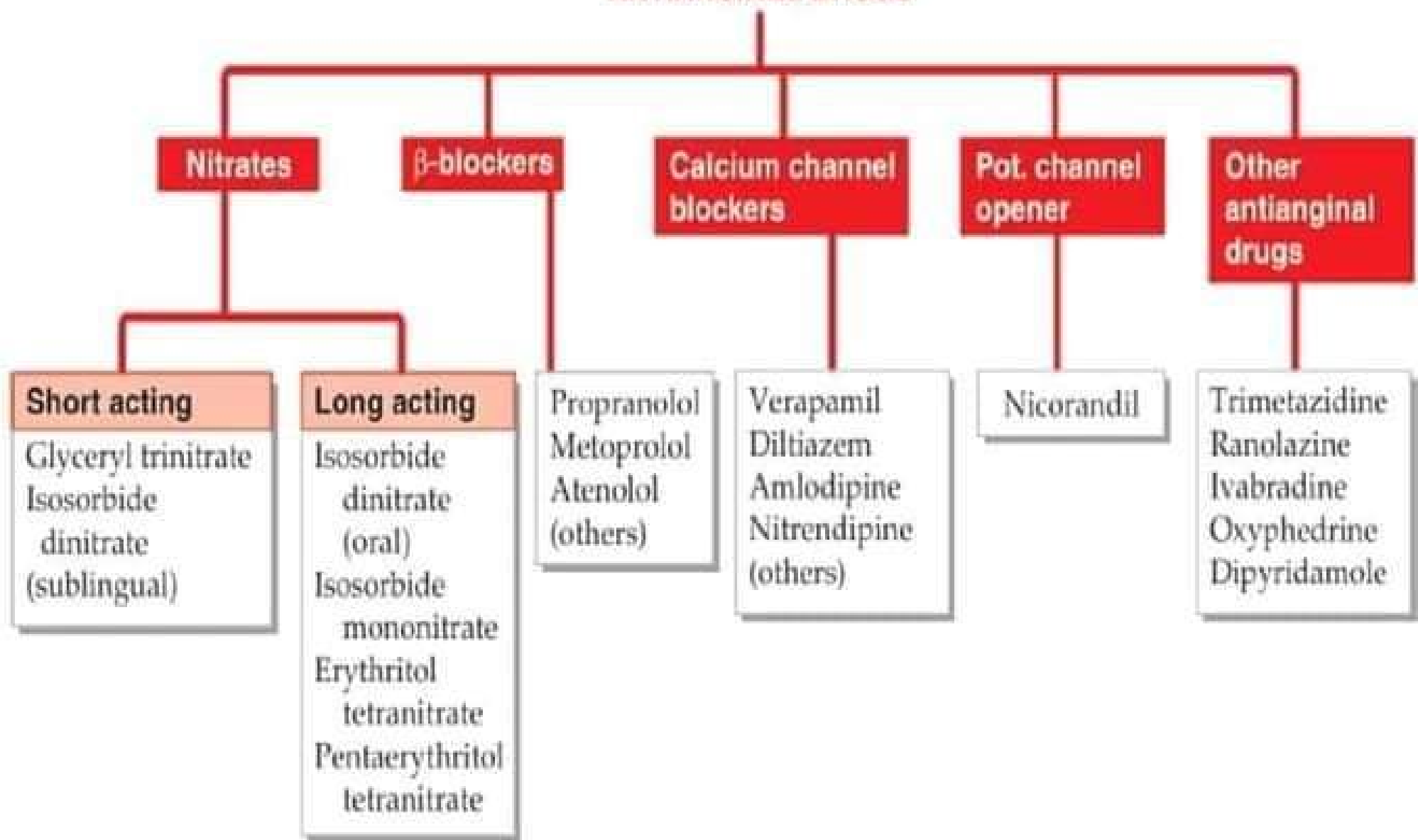


Contd....

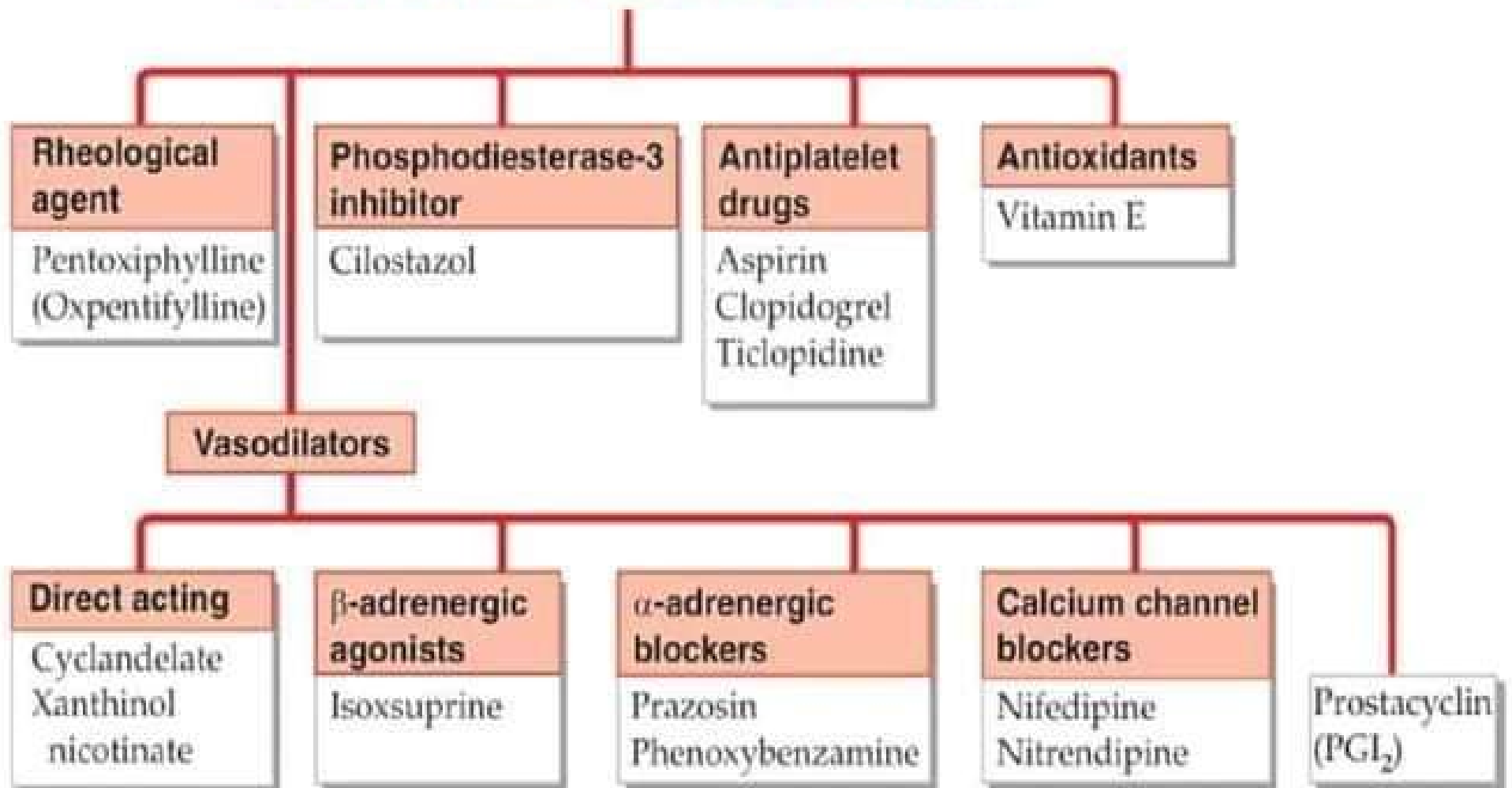
ANTIHYPERTENSIVE DRUGS (2)



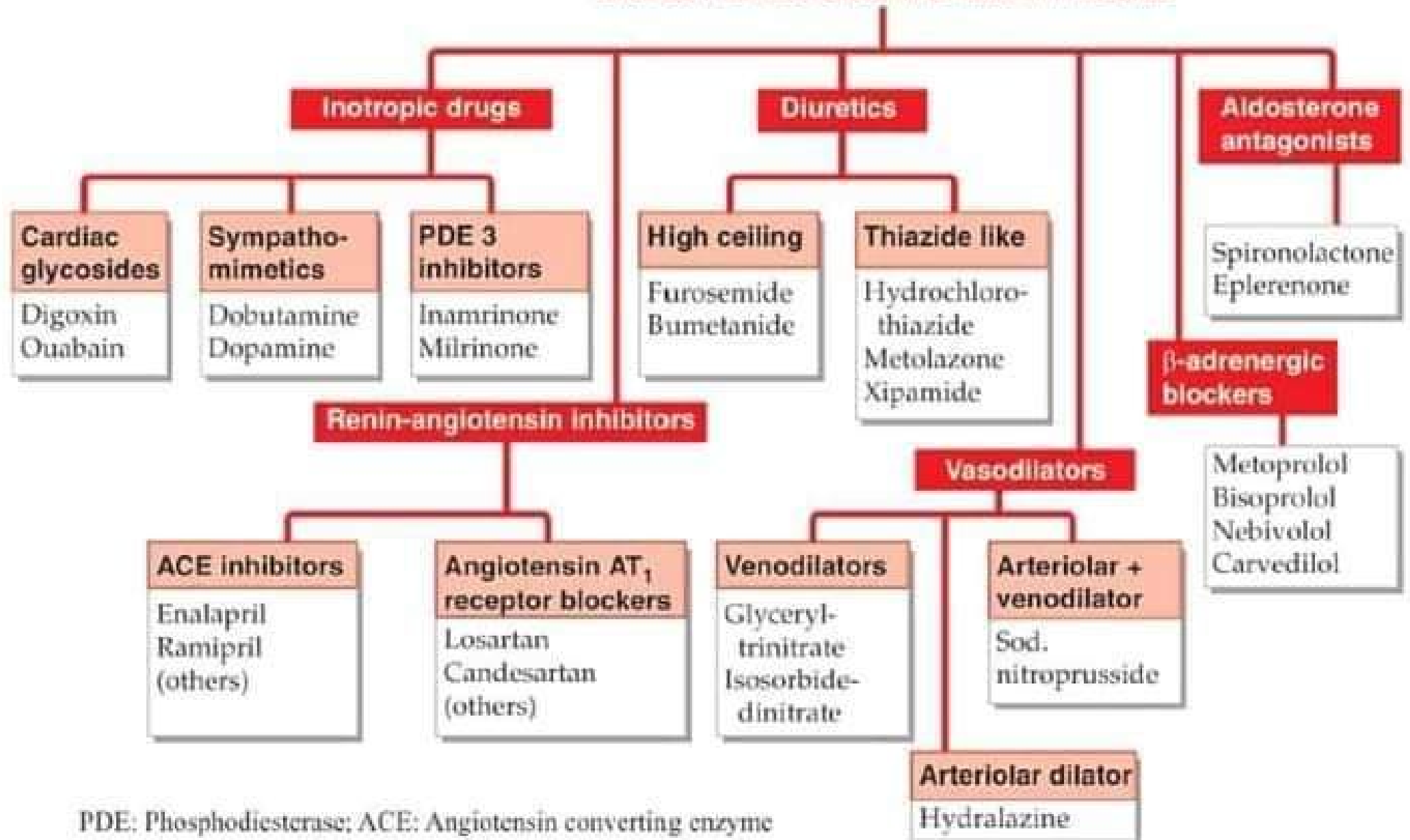
ANTIANGINAL DRUGS



DRUGS FOR PERIPHERAL VASCULAR DISEASES

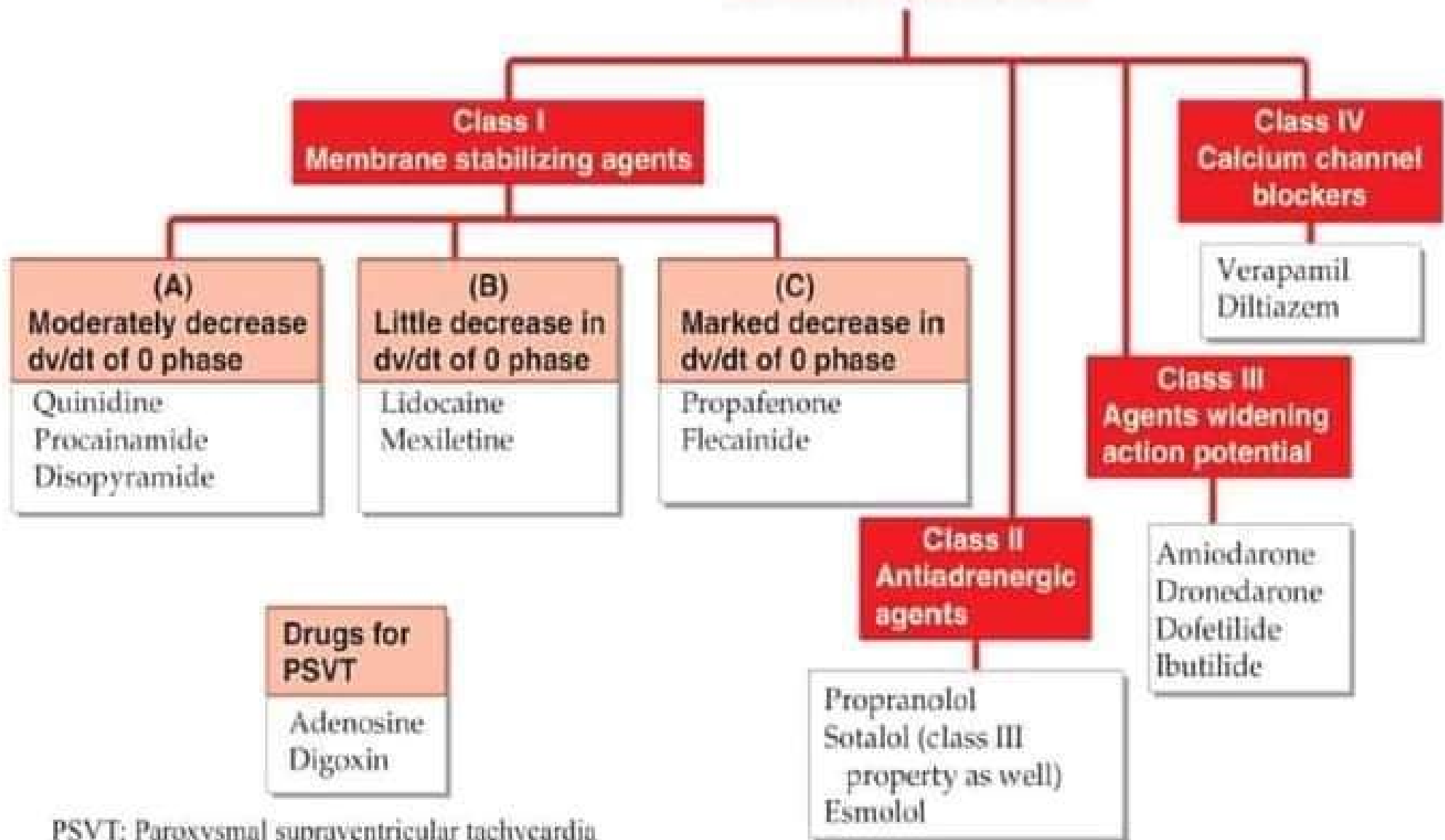


DRUGS FOR CONGESTIVE HEART FAILURE



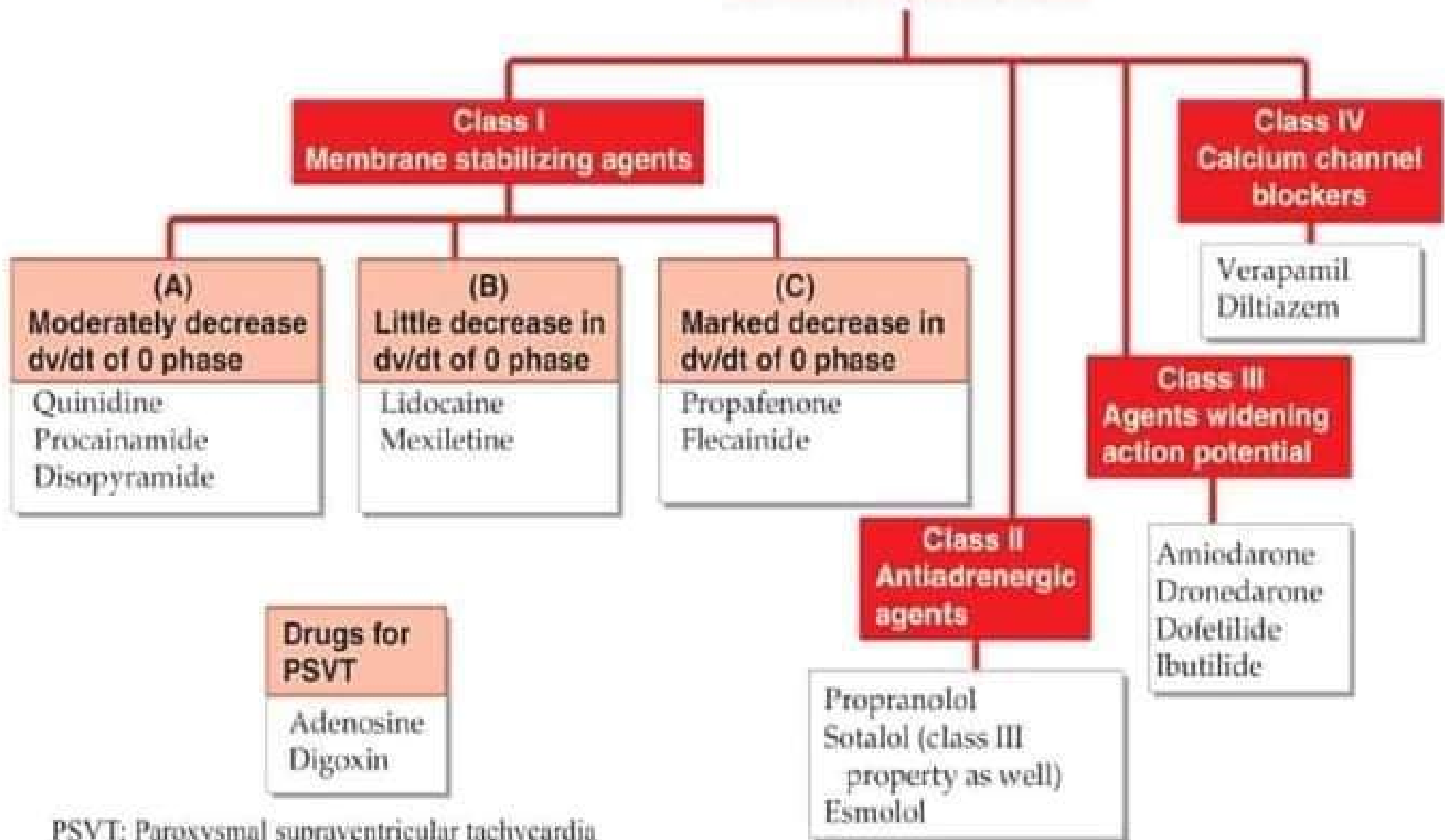
PDE: Phosphodiesterase; ACE: Angiotensin converting enzyme

ANTIARRHYTHMIC DRUGS



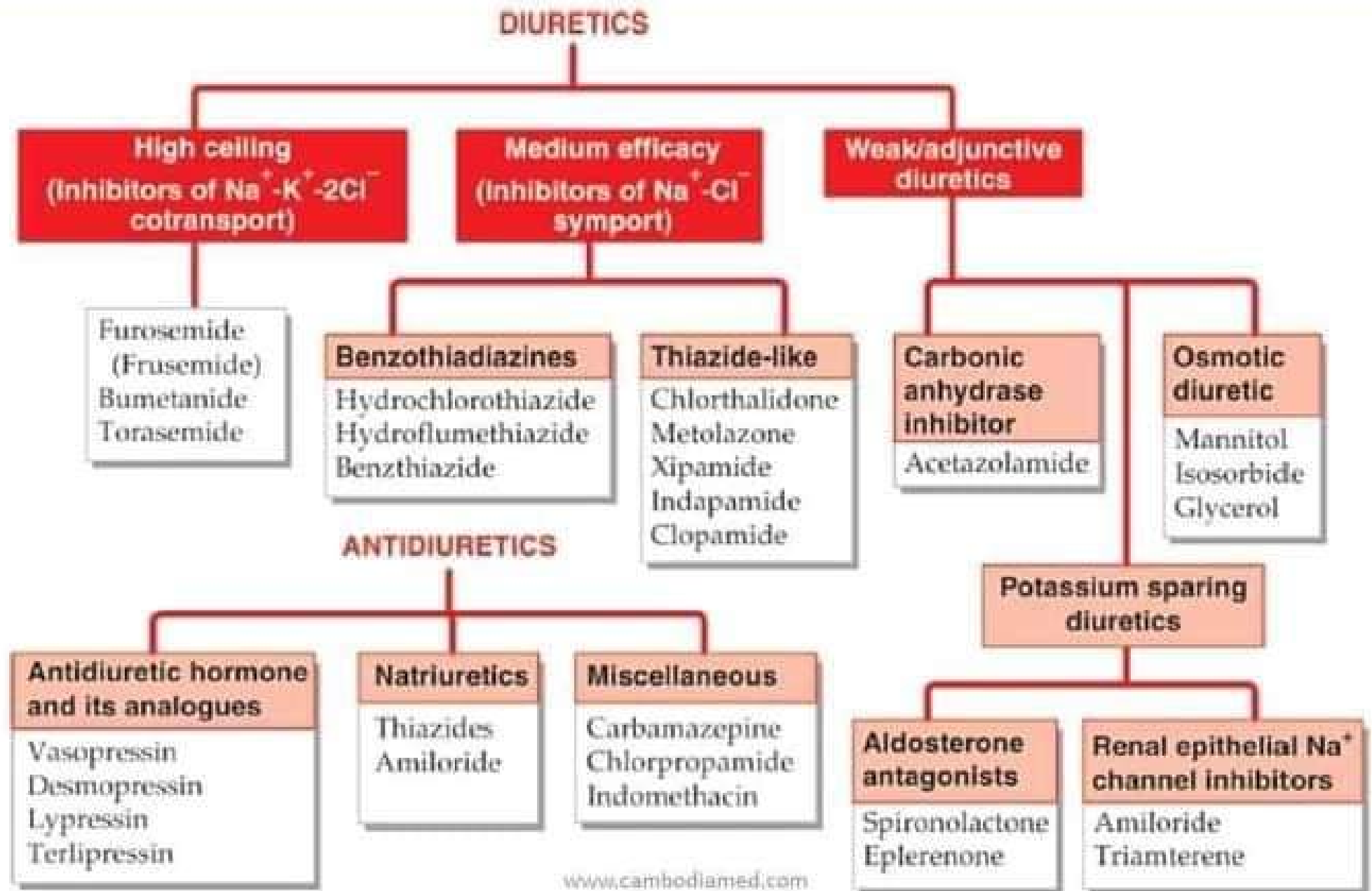
PSVT: Paroxysmal supraventricular tachycardia

ANTIARRHYTHMIC DRUGS

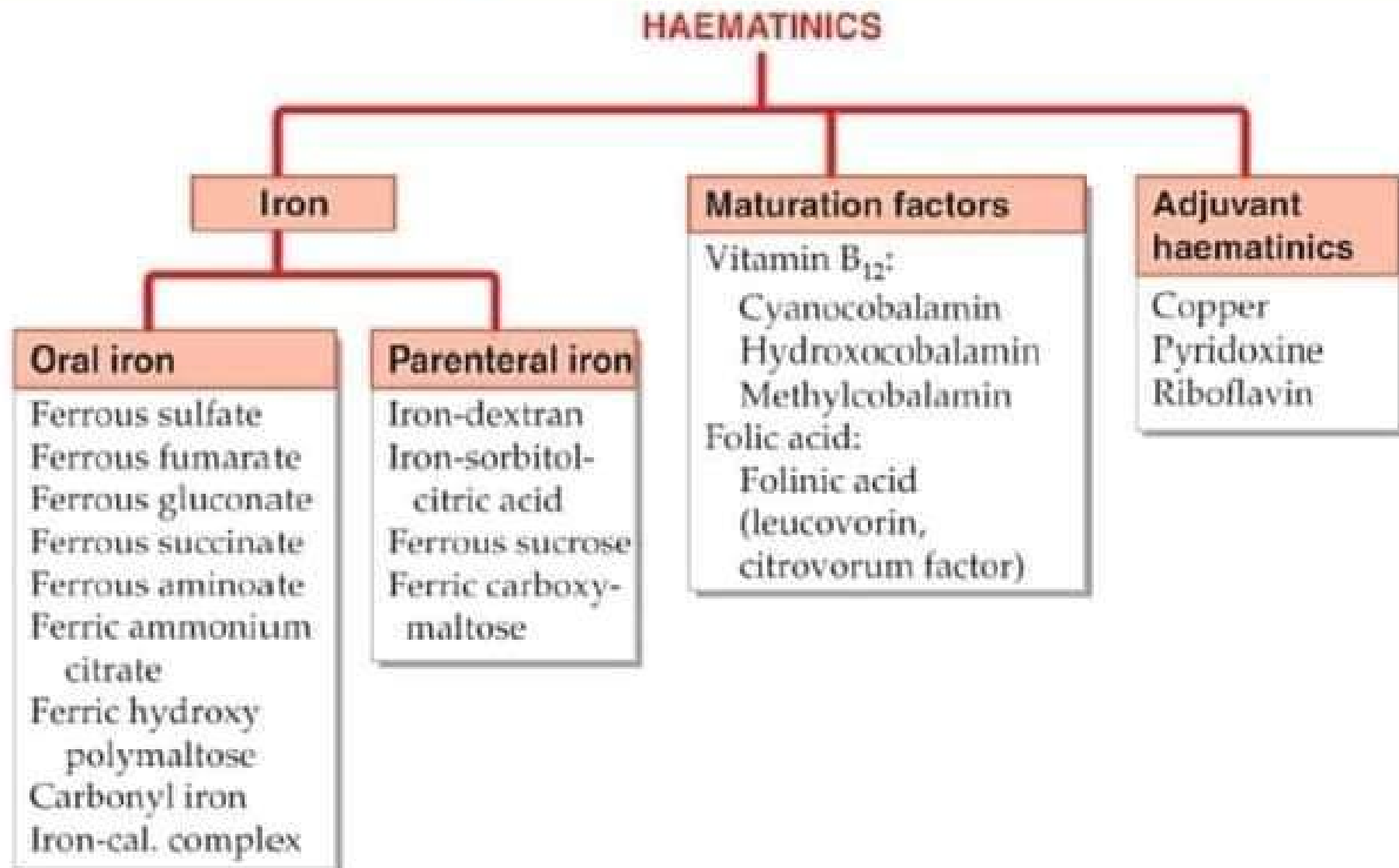


PSVT: Paroxysmal supraventricular tachycardia

Drugs Acting on Kidney



Drugs Affecting Blood



COAGULANTS

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graph TD; COAGULANTS --> Vitamin_K[Vitamin K]; COAGULANTS --> Miscellaneous[Miscellaneous];
```

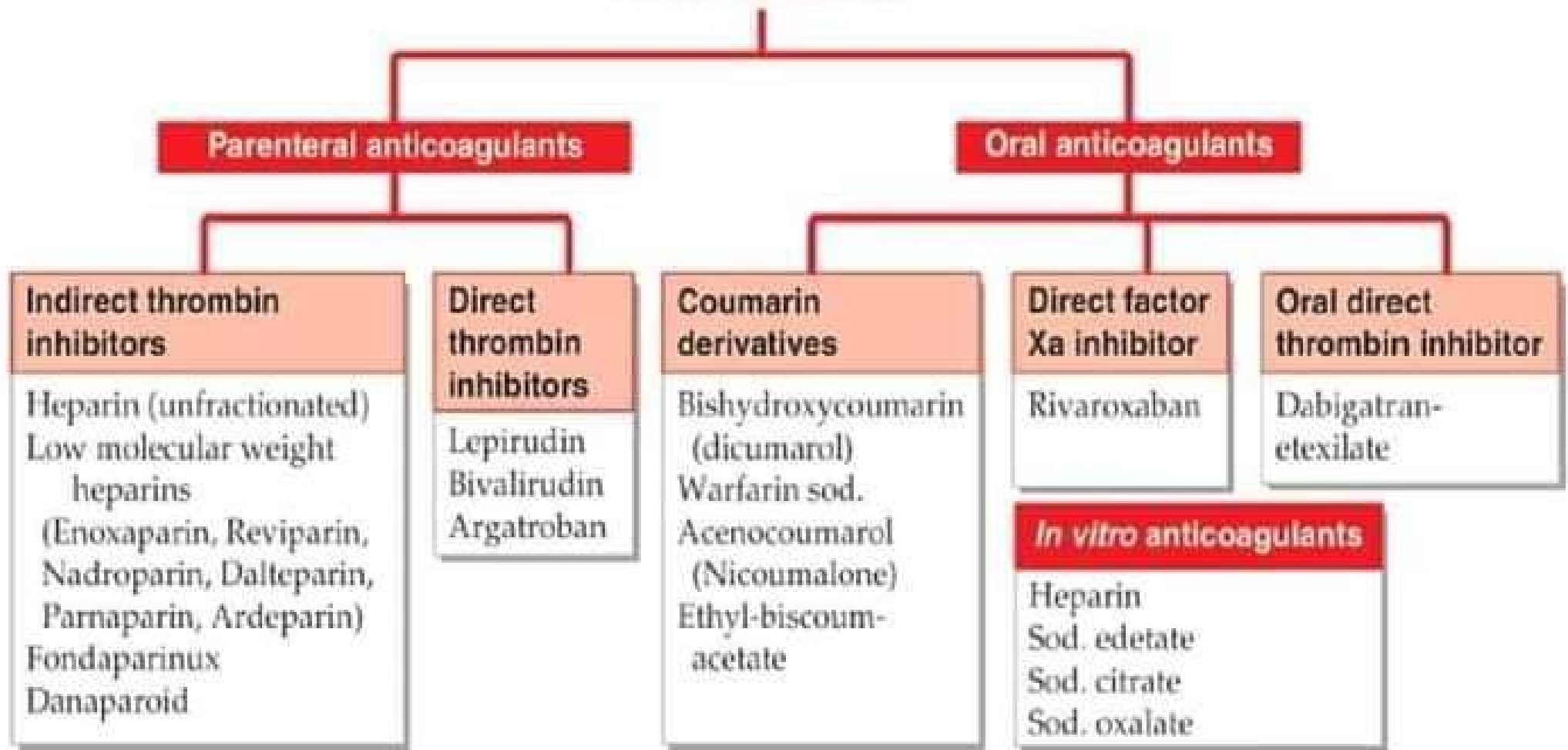
Vitamin K

K₁: Phytonadione
(Phylloquinone)
K₃: Menadione
Acetomenaphthone
Menadione sod.
bisulfite
Menadione sod.
diphosphate

Miscellaneous

Fibrinogen (human)
Antihaemophilic factor
Desmopressin
Adrenochrome mono-
semicarbazone
Ethamsylate
Rutin

ANTICOAGULANTS



Fibrinolytics (Thrombolytics)

Streptokinase
Urokinase
Alteplase (rt-PA)
Reteplase
Tenecteplase

Antifibrinolytics (Antithrombolytics)

Epsilon amino-caproic acid
(EA CA)
Tranexamic acid

ANTIPLATELET DRUGS (Antithrombotic drugs)

Thromboxane synthesis inhibitor

Aspirin

Platelet cAMP enhancer

Dipyridamole

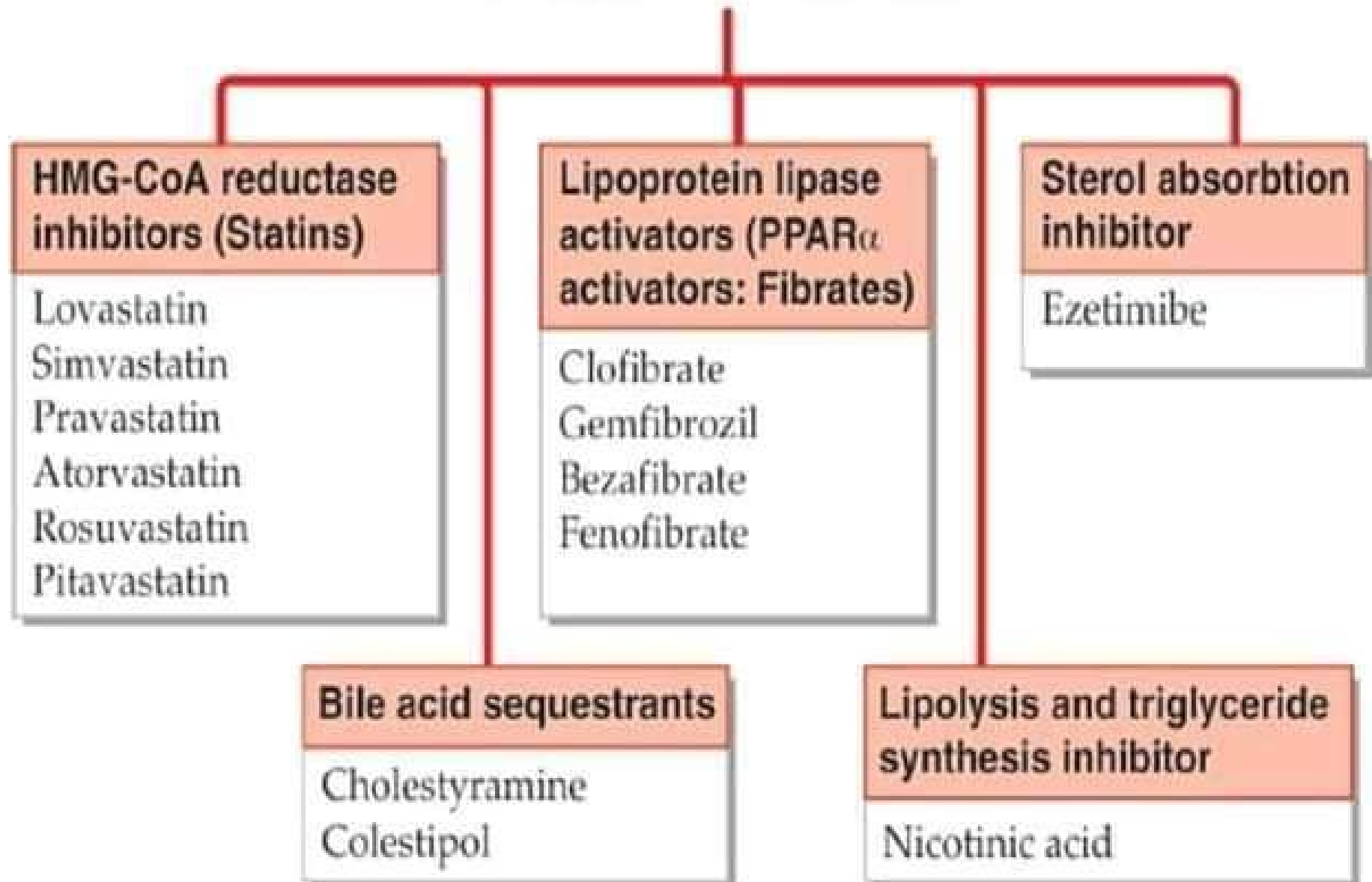
P2Y₁₂ receptor blockers

Ticlopidine
Clopidogrel
Prasugrel

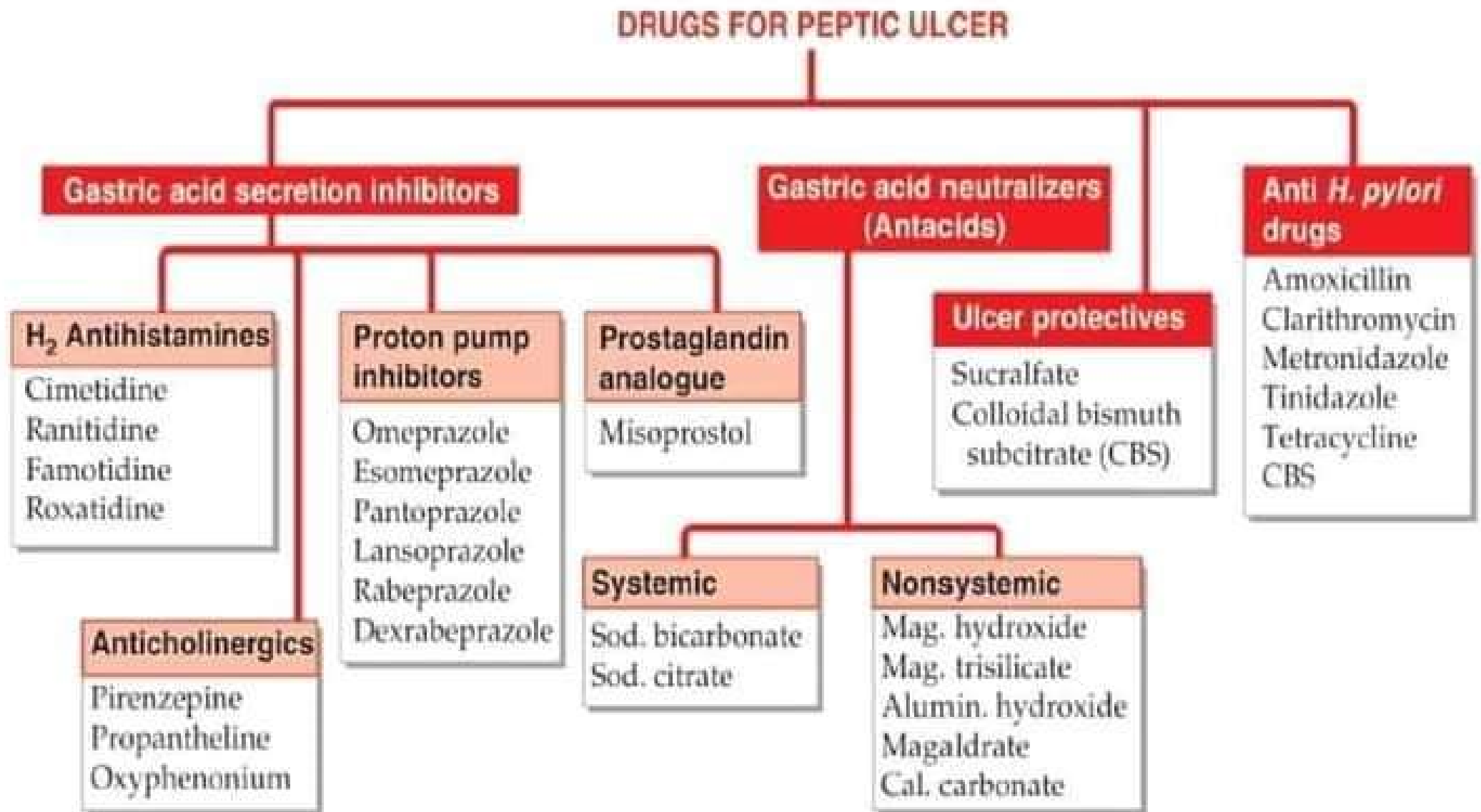
GP II_b/III_a antagonists

Abciximab
Eptifibatide
Tirofiban

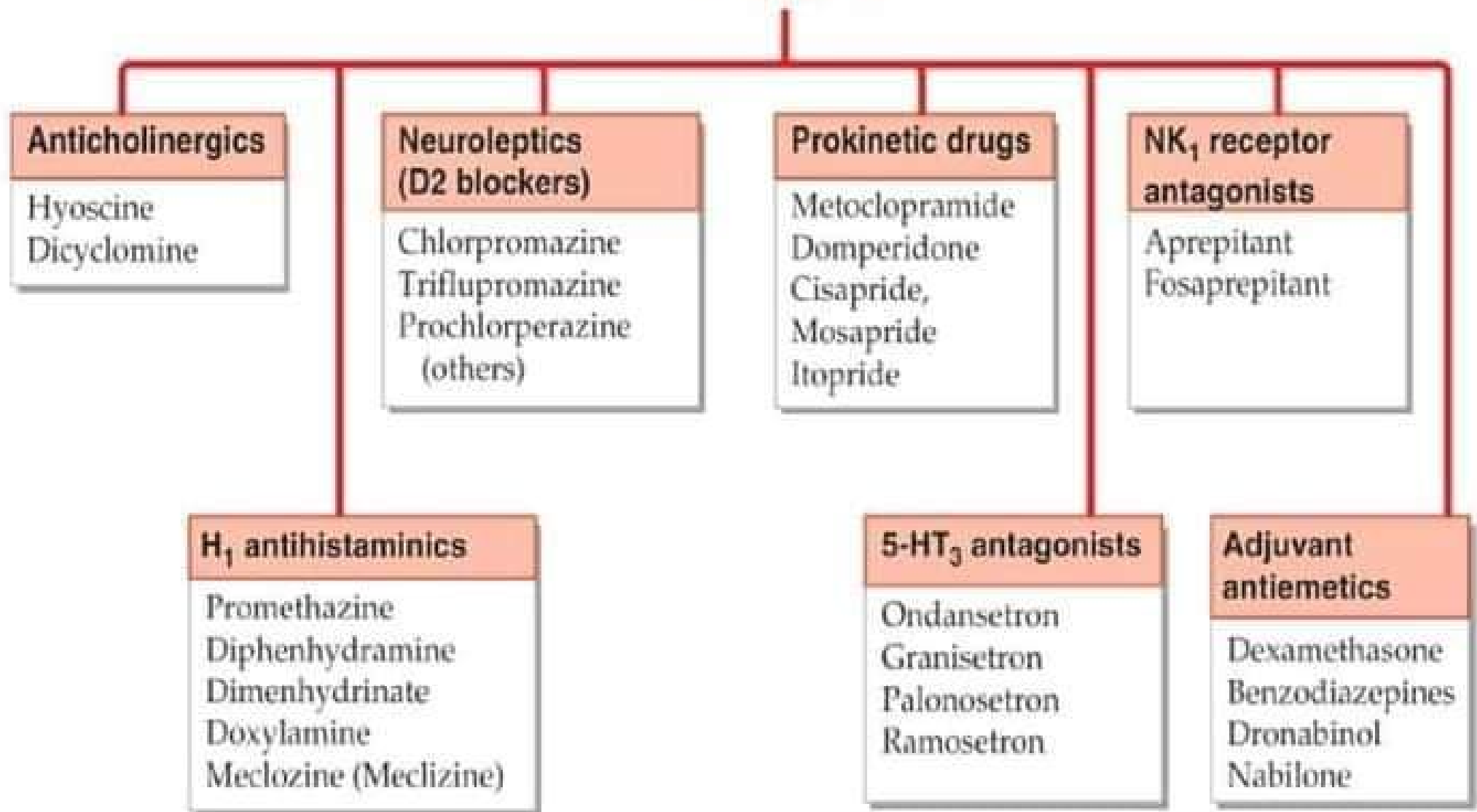
HYPOLIPIDAEMIC DRUGS



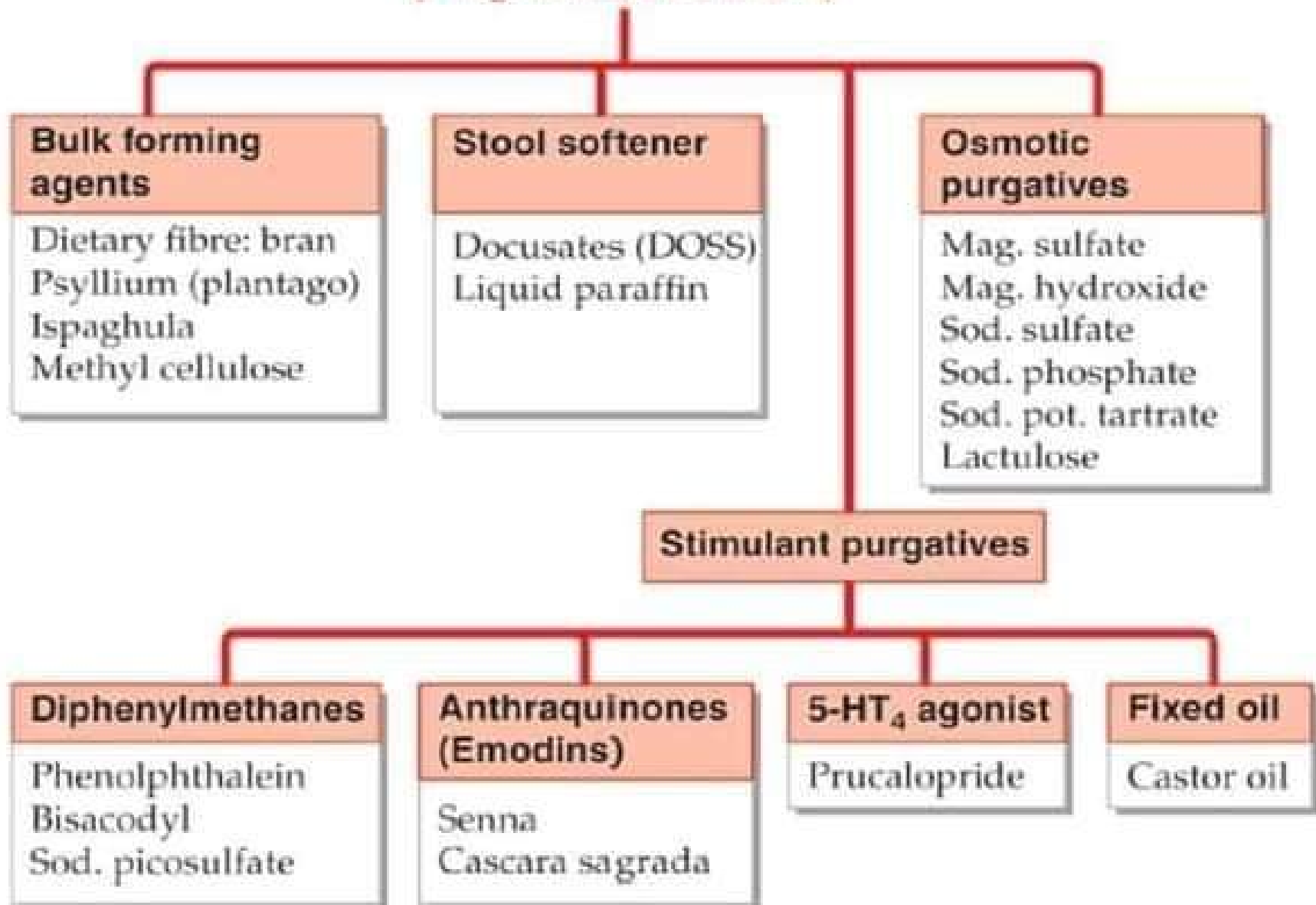
Gastrointestinal Drugs



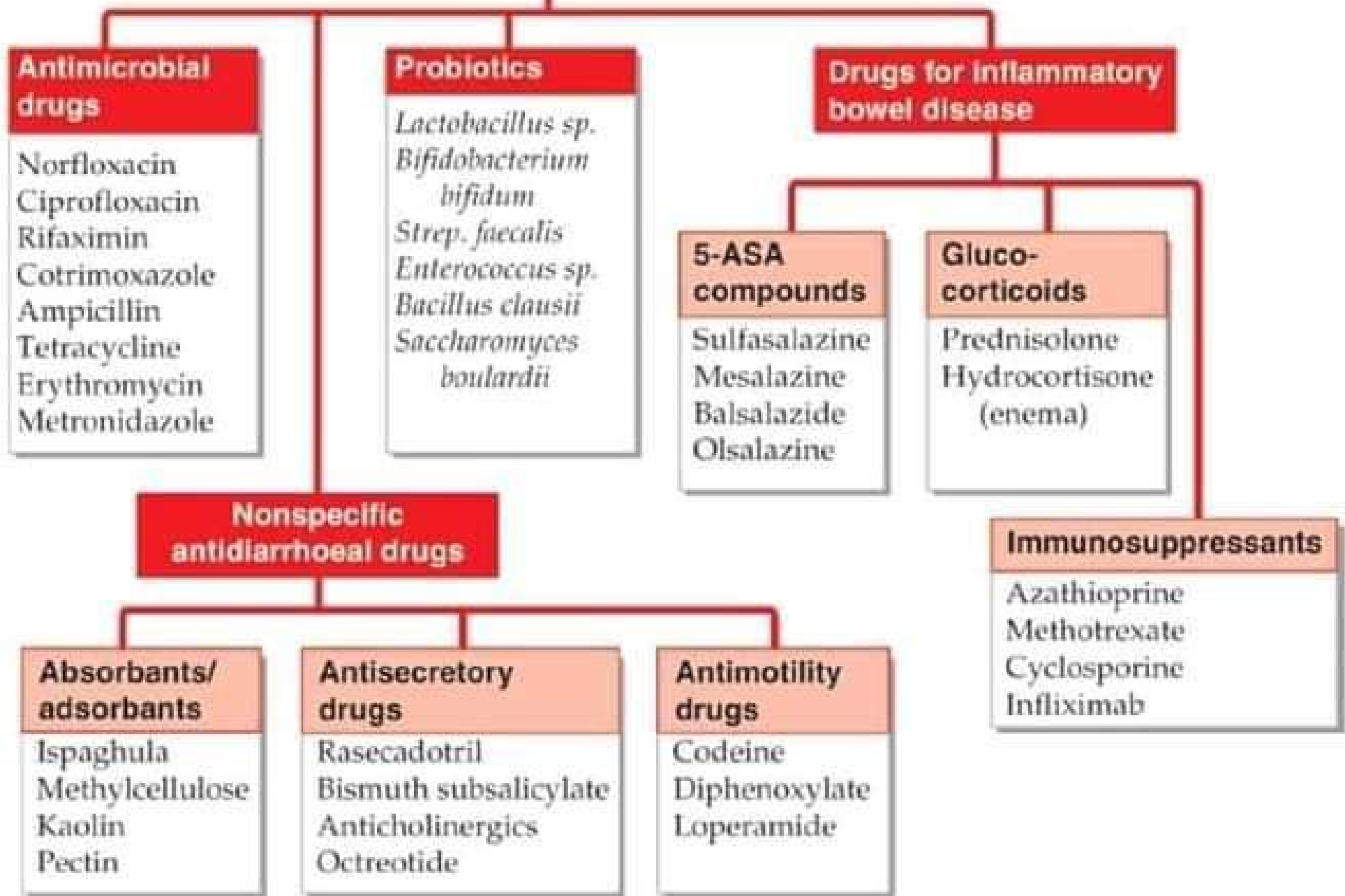
ANTIEMETICS



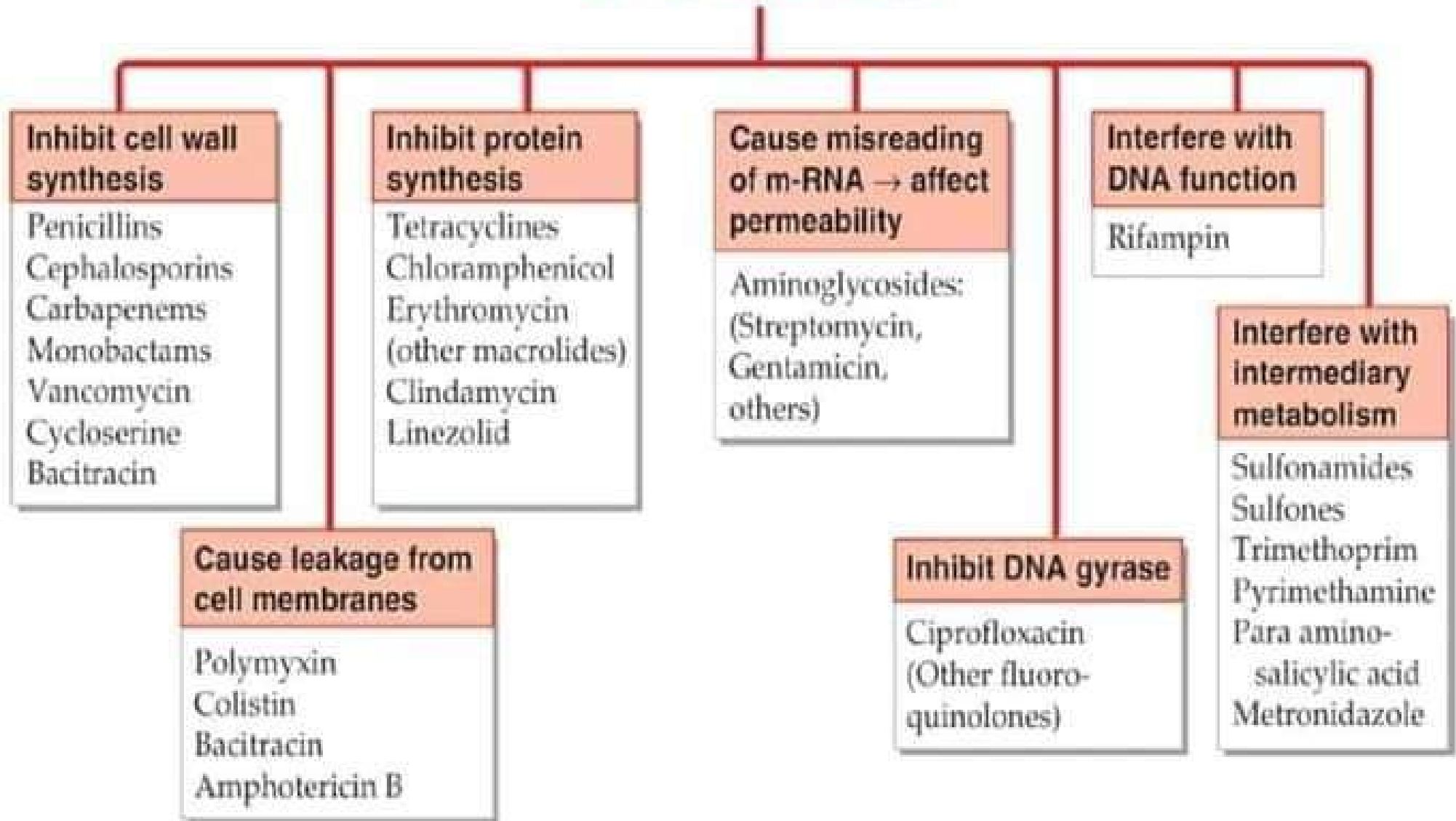
LAXATIVES (Purgatives, Cathartics)



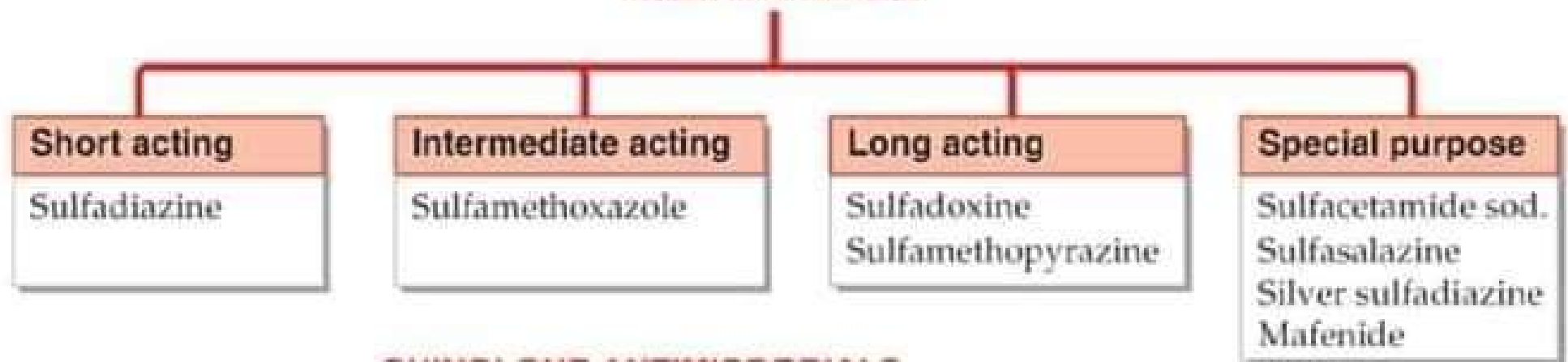
DRUGS FOR DIARRHOEA



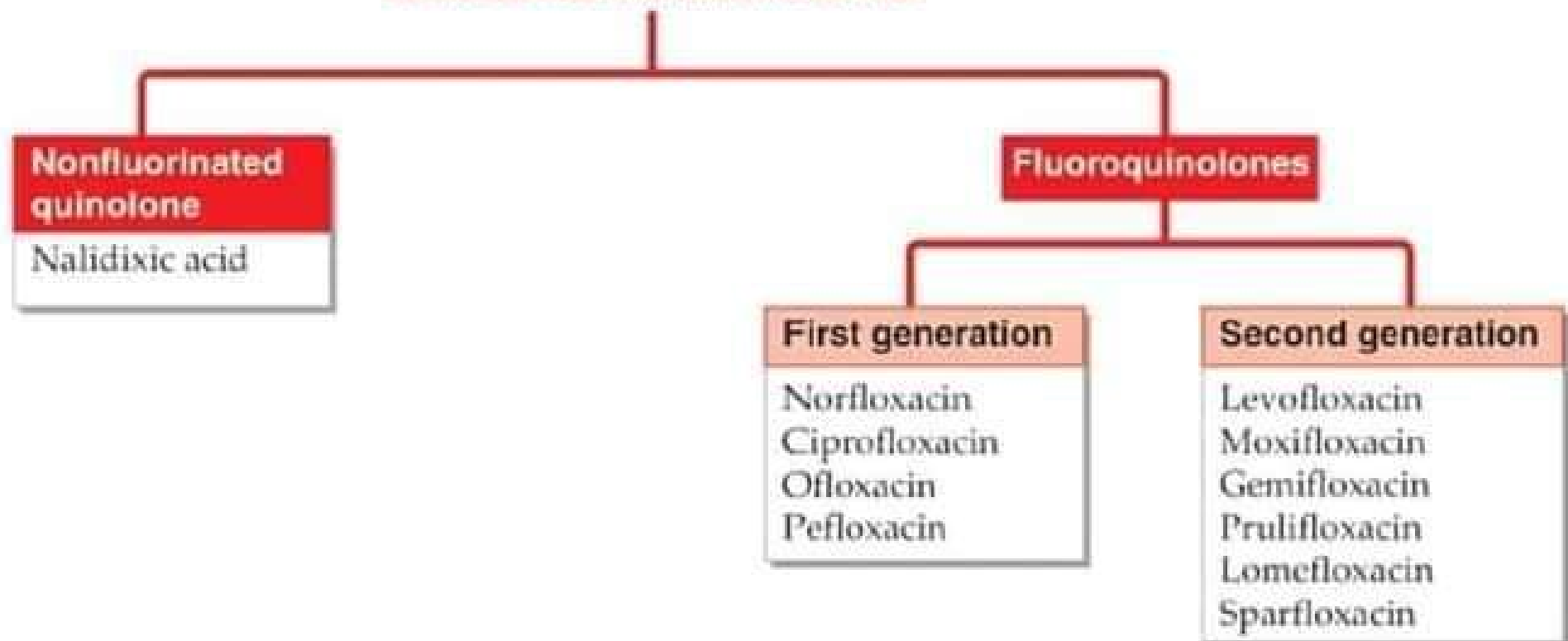
ANTIBACTERIAL DRUGS



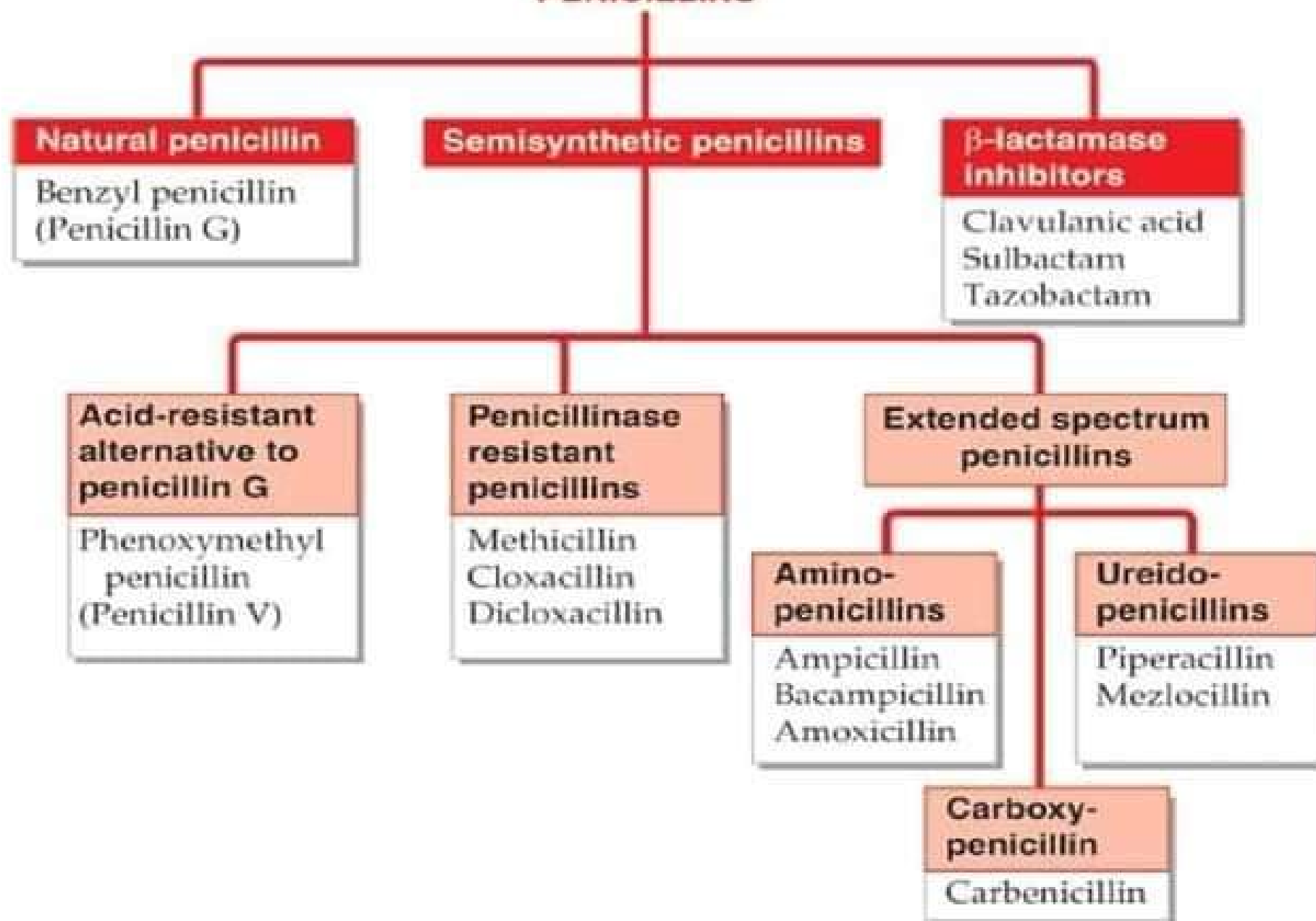
SULFONAMIDES



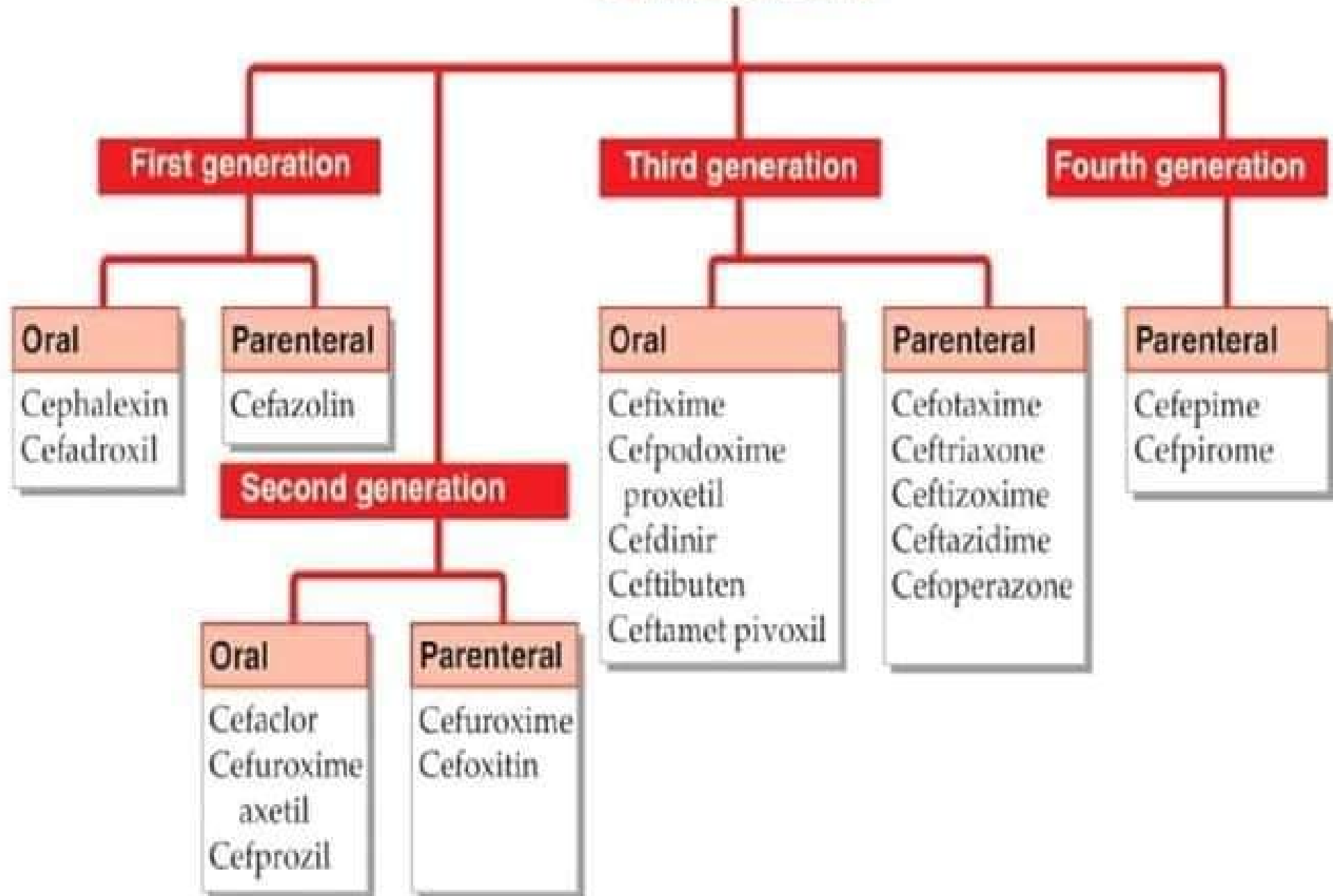
QUINOLONE ANTIMICROBIALS



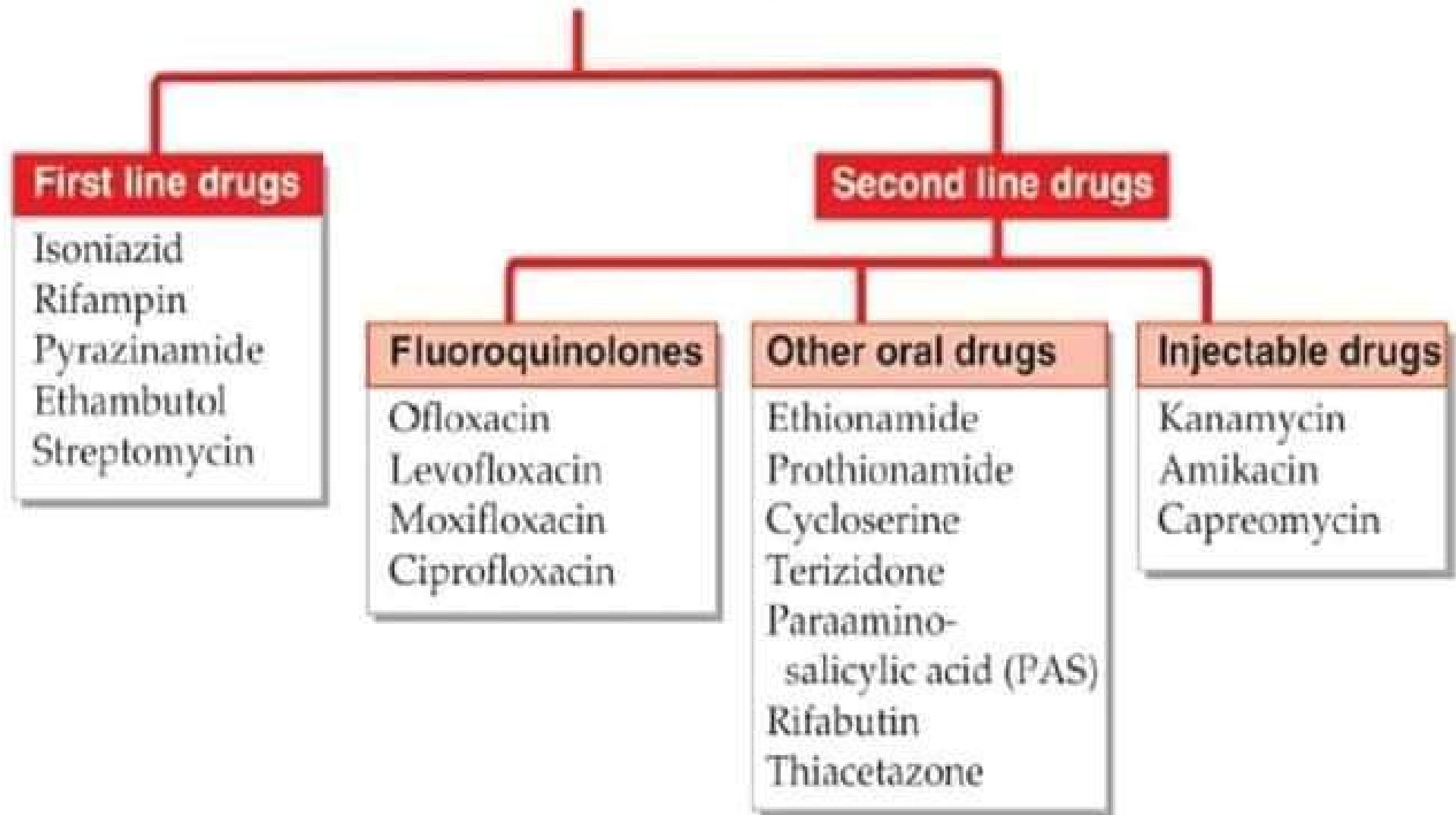
PENICILLINS



CEPHALOSPORINS

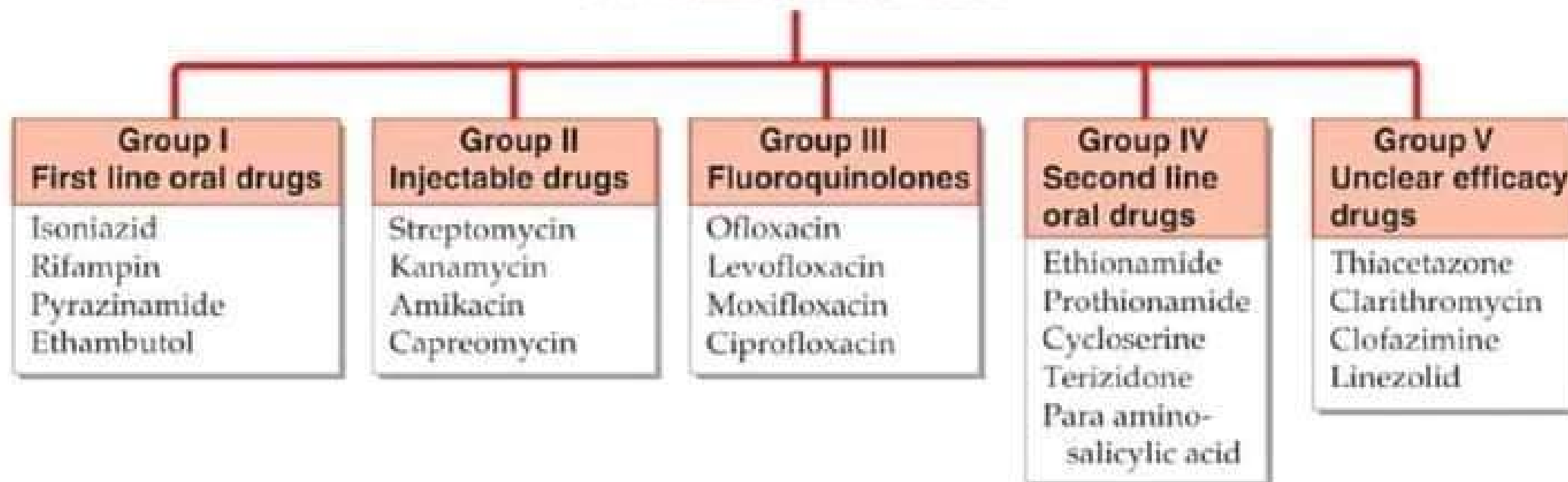


ANTITUBERCULAR DRUGS

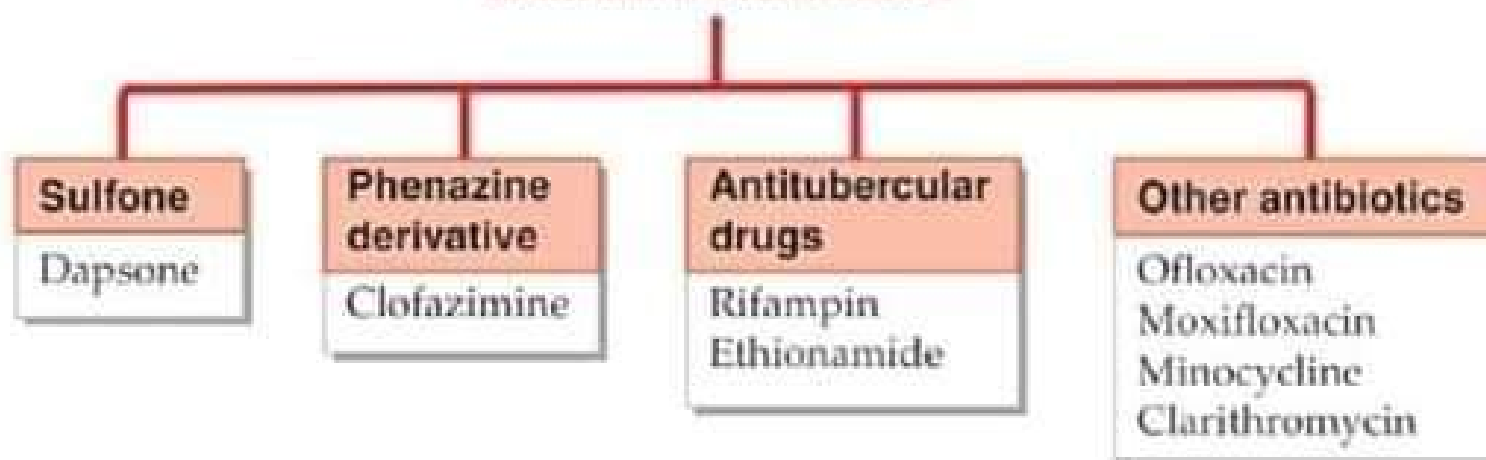


ALTERNATIVE CLASSIFICATION

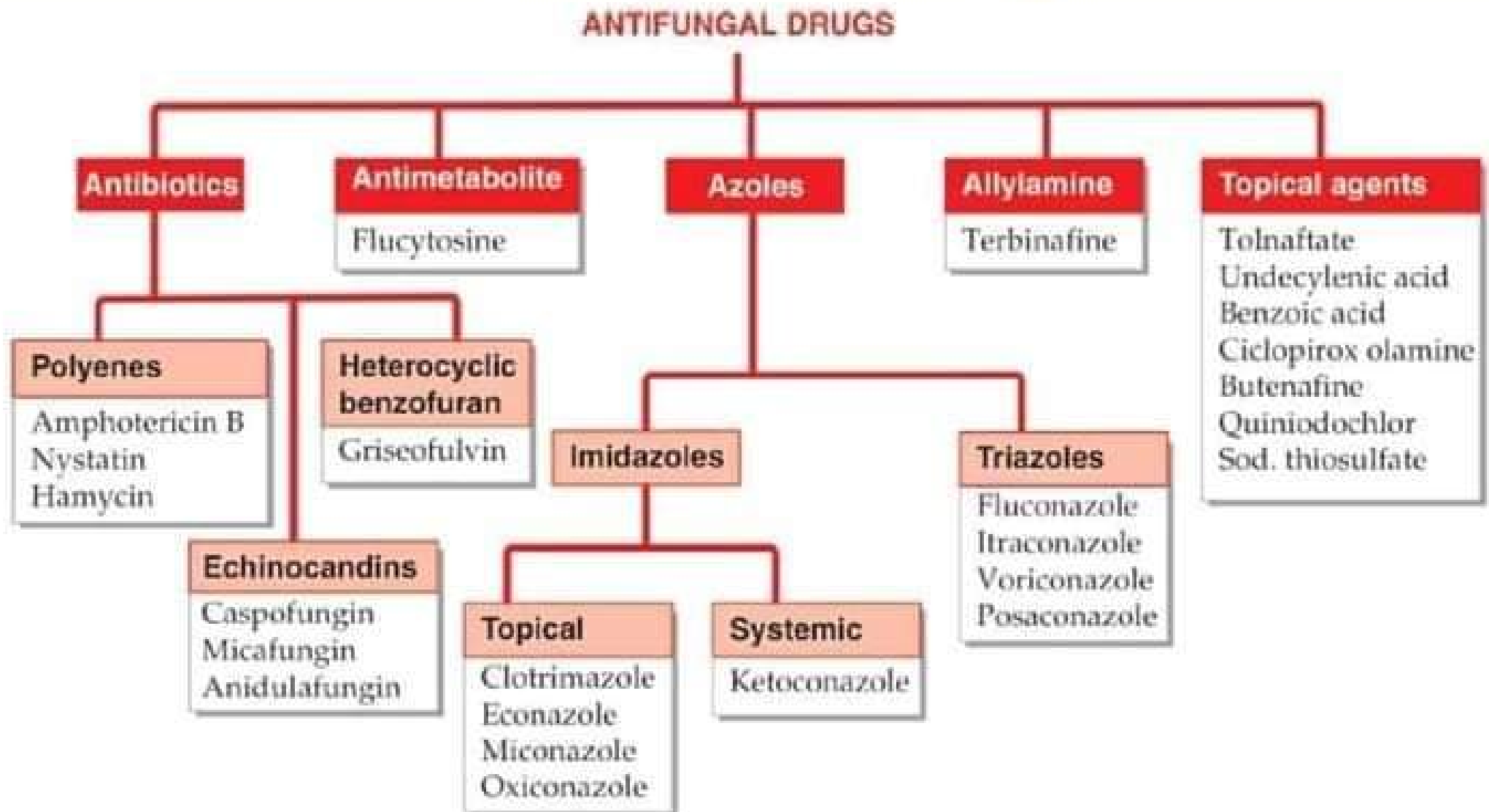
ANTITUBERCULAR DRUGS



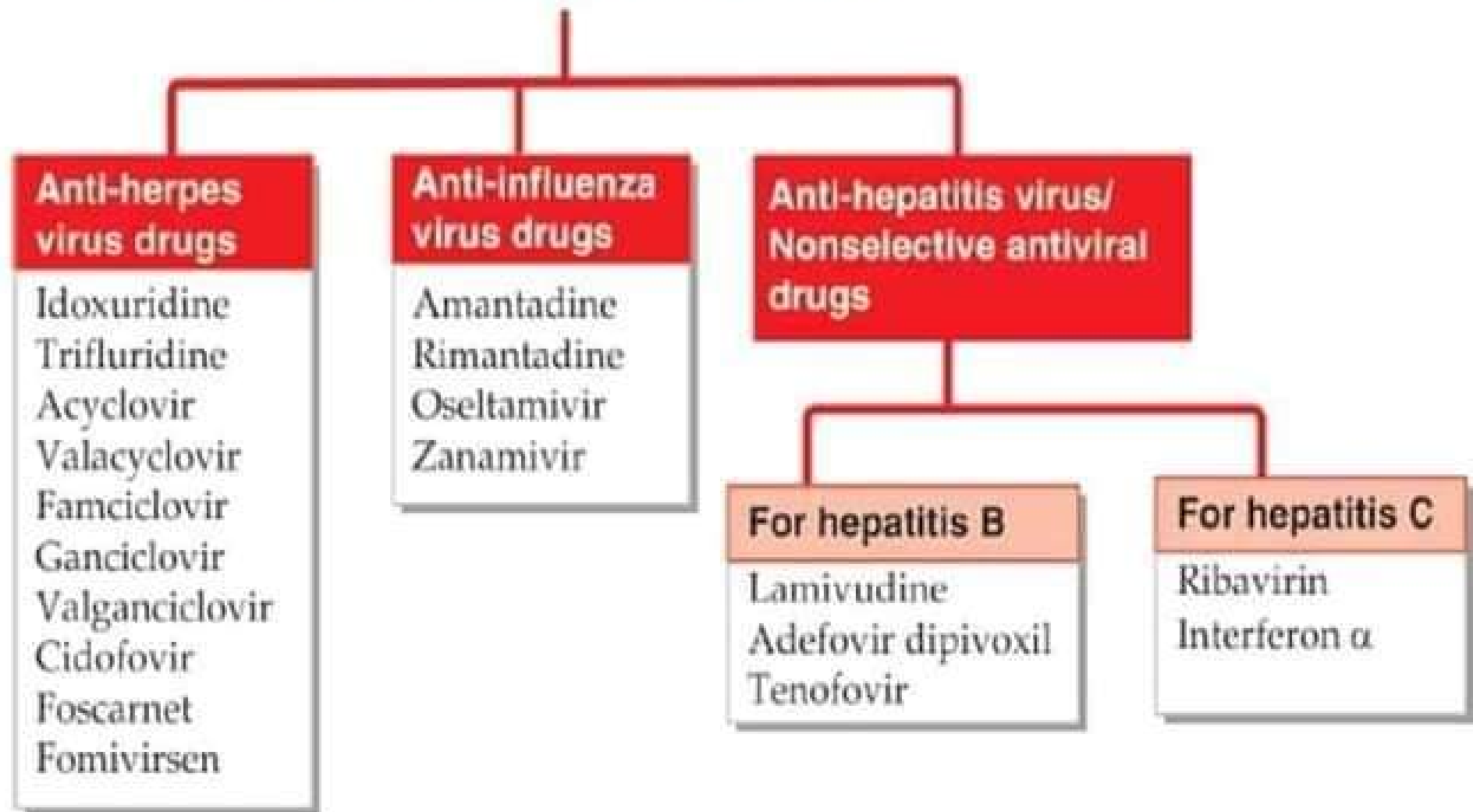
ANTILEPROTIC DRUGS



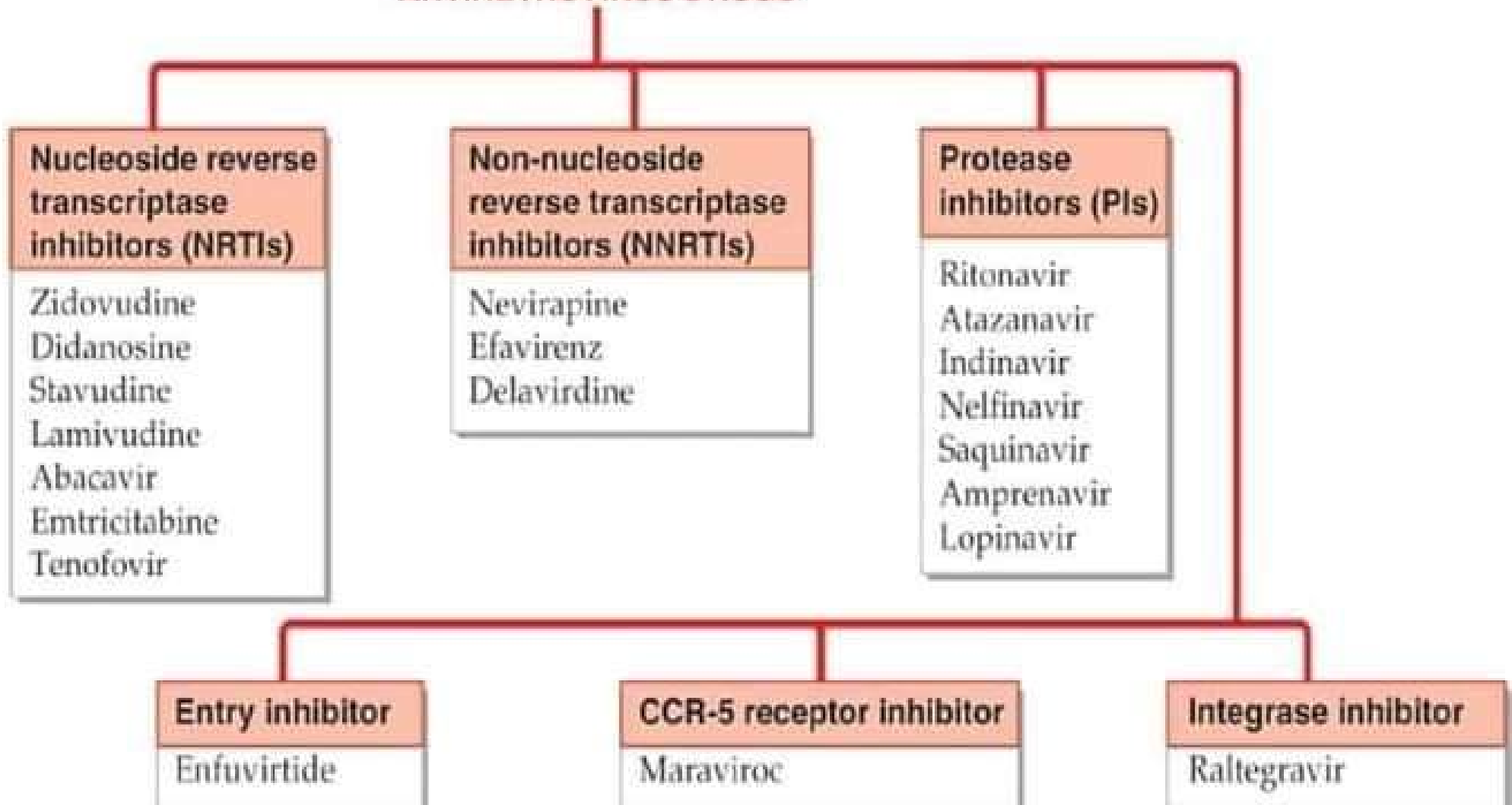
Antifungal, Antiviral, Antiprotozoal and Anthelmintic Drugs



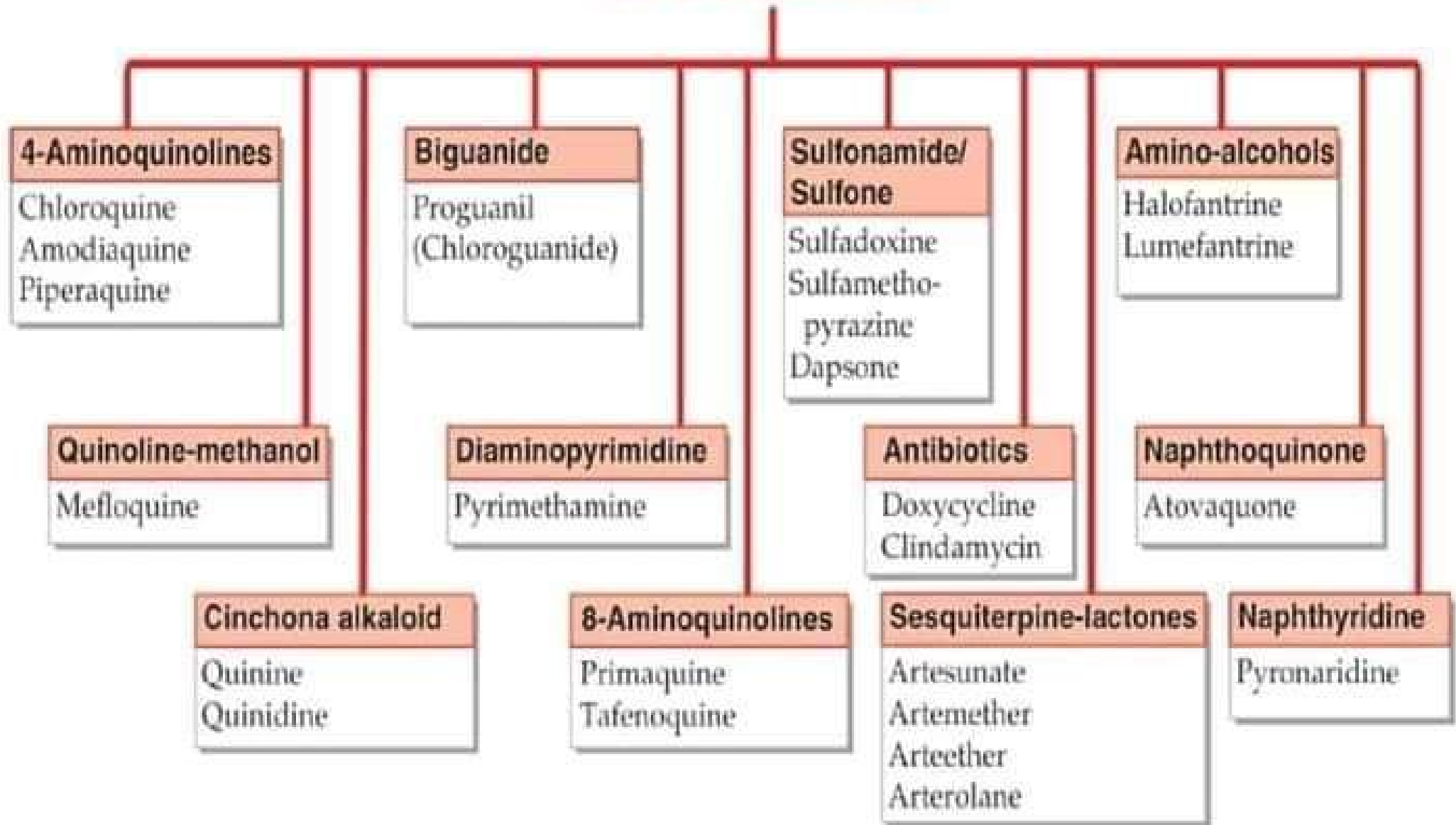
ANTIVIRAL DRUGS (Non-retroviral)



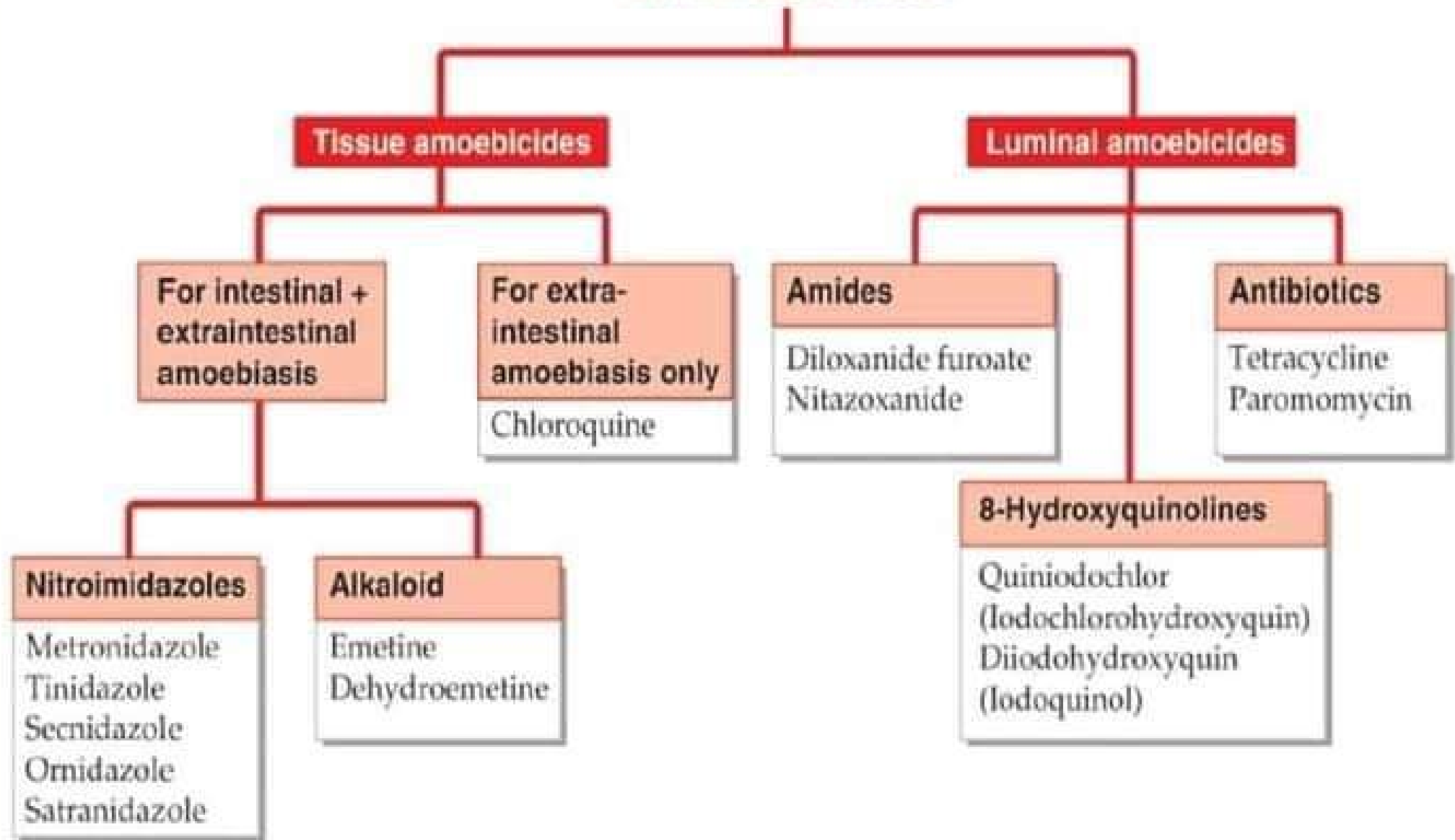
ANTIRETROVIRUS DRUGS



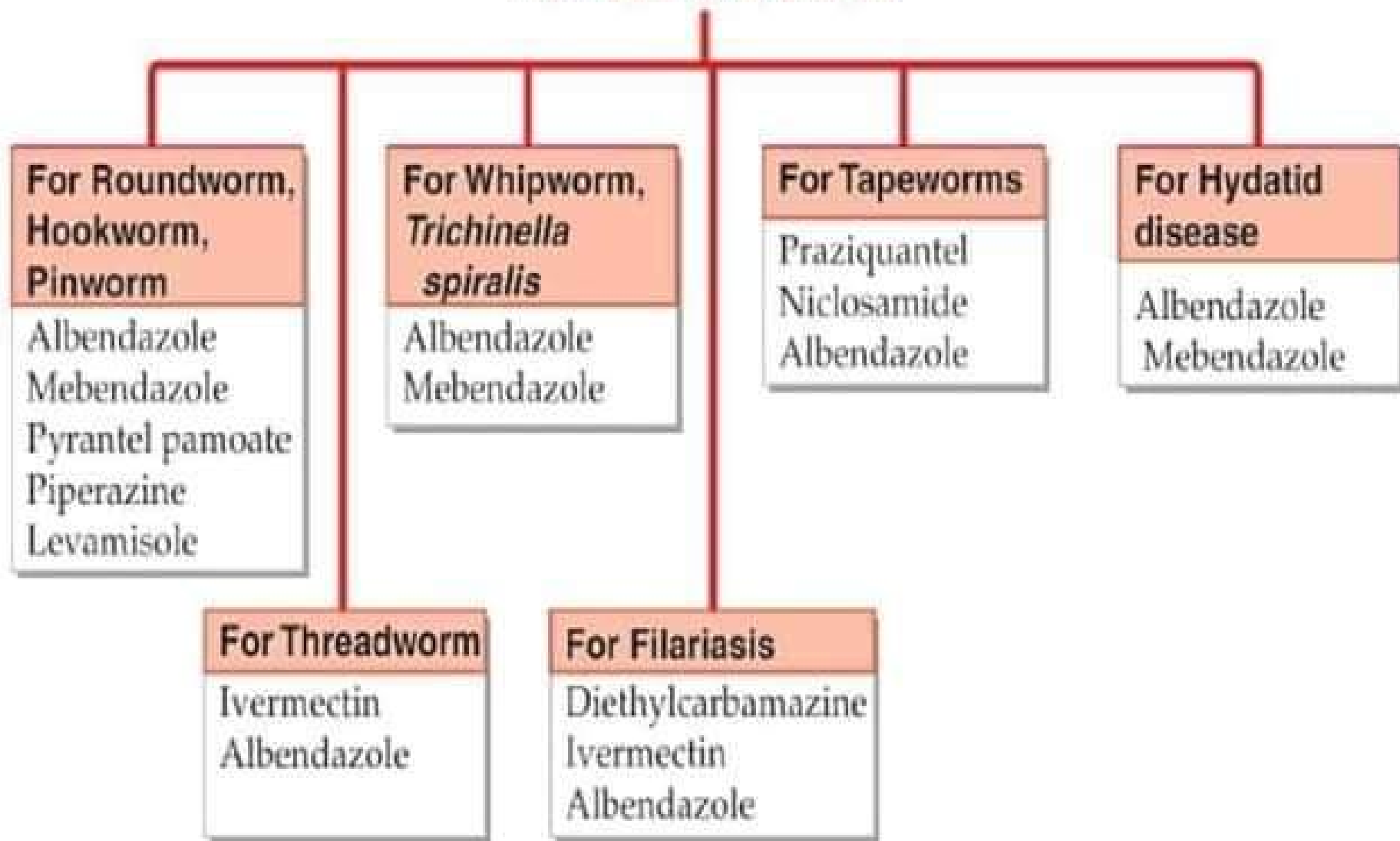
ANTIMALARIAL DRUGS



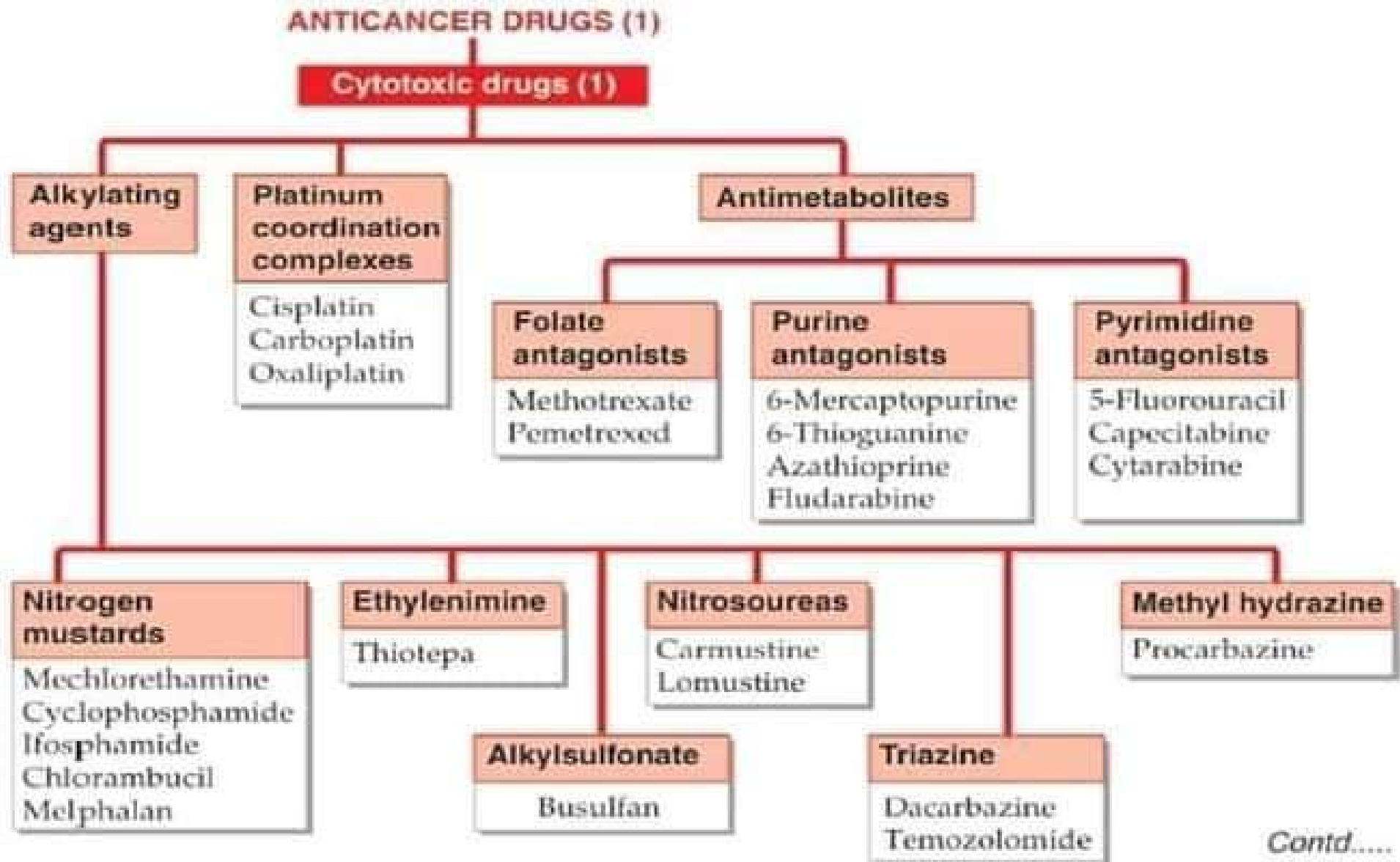
ANTIAMOEBIIC DRUGS



ANTHELMINTIC DRUGS



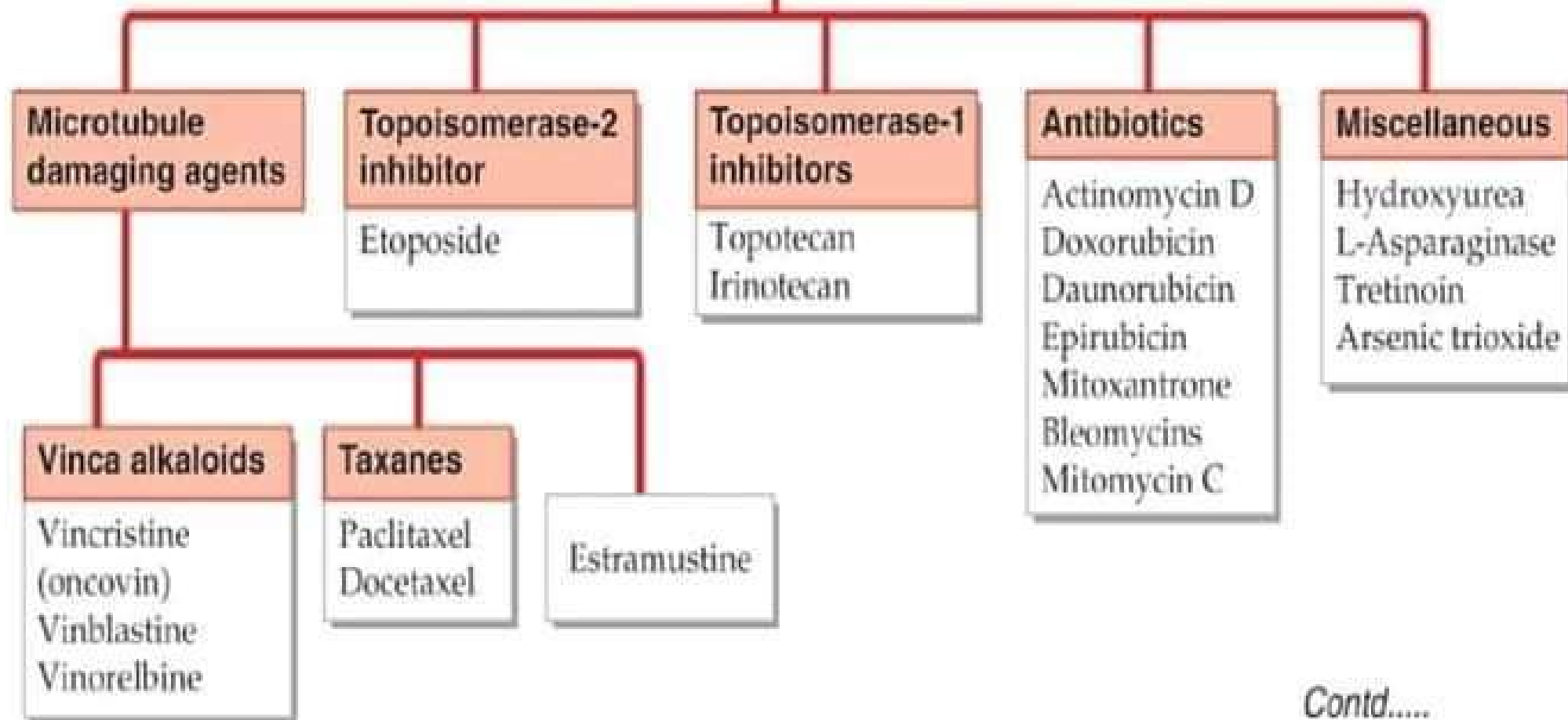
Anticancer Drugs (Antineoplastic Drugs)



Contd.....

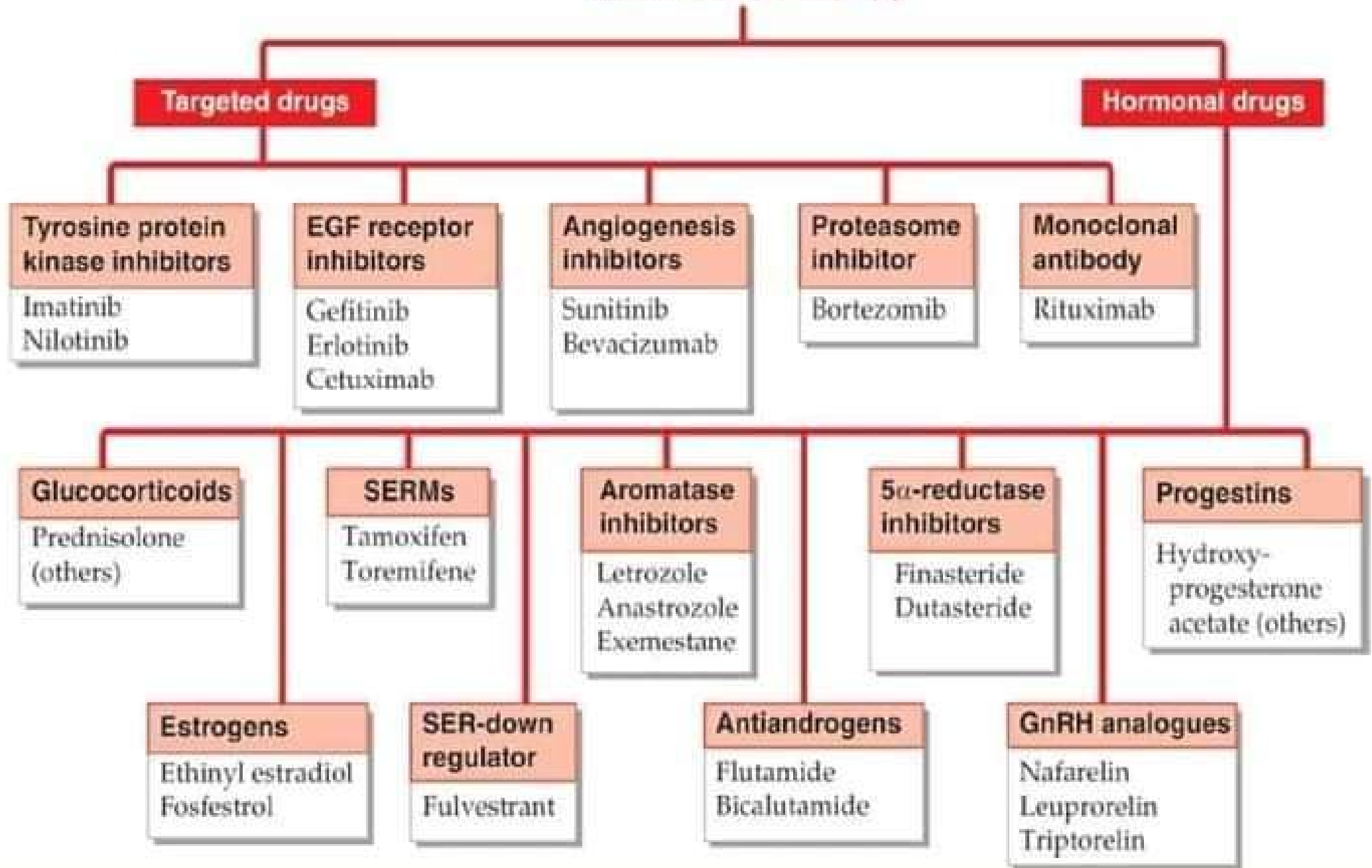
ANTICANCER DRUGS (2)

Cytotoxic drugs (2)



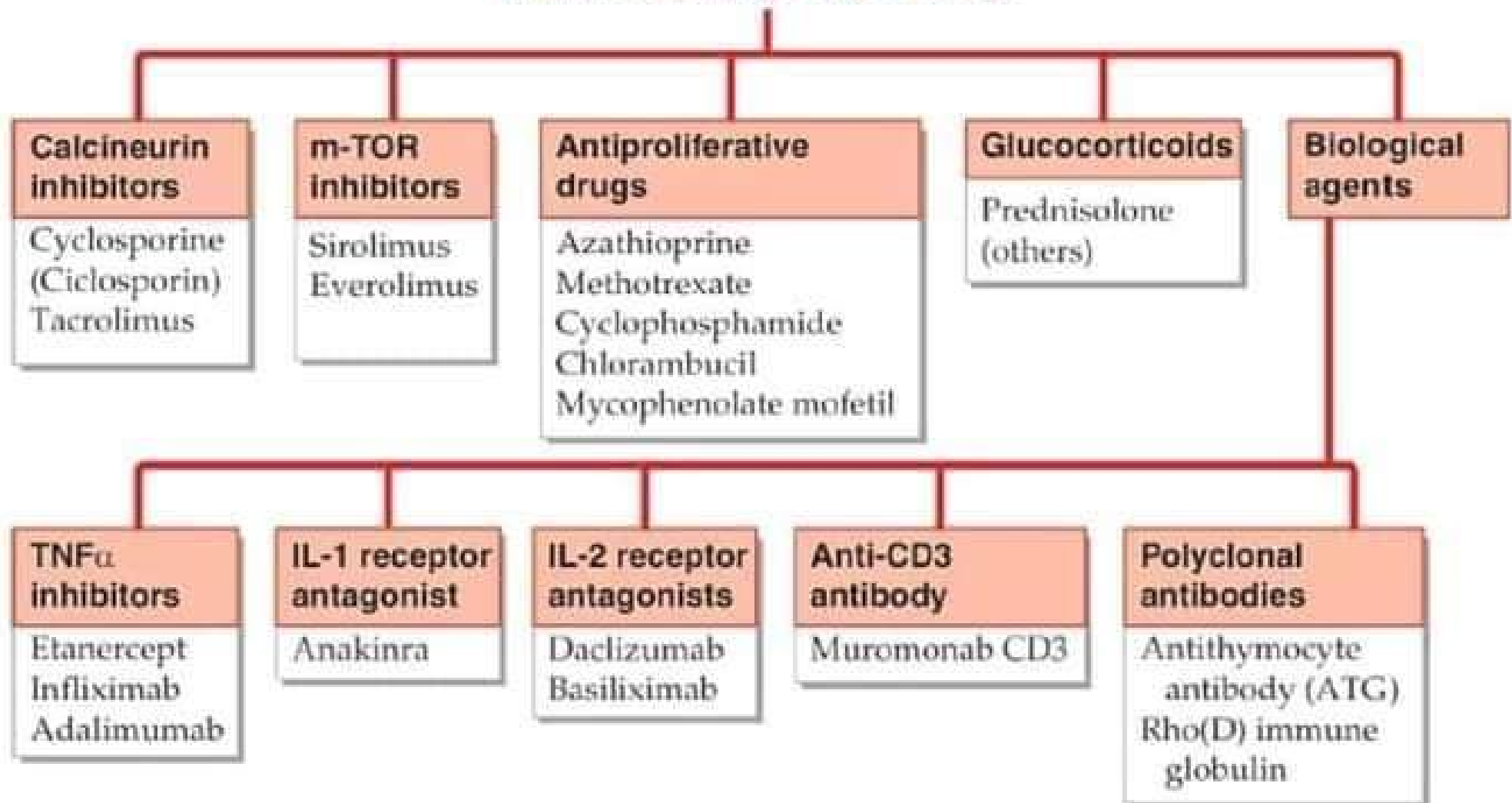
Contd.....

ANTICANCER DRUGS (3)

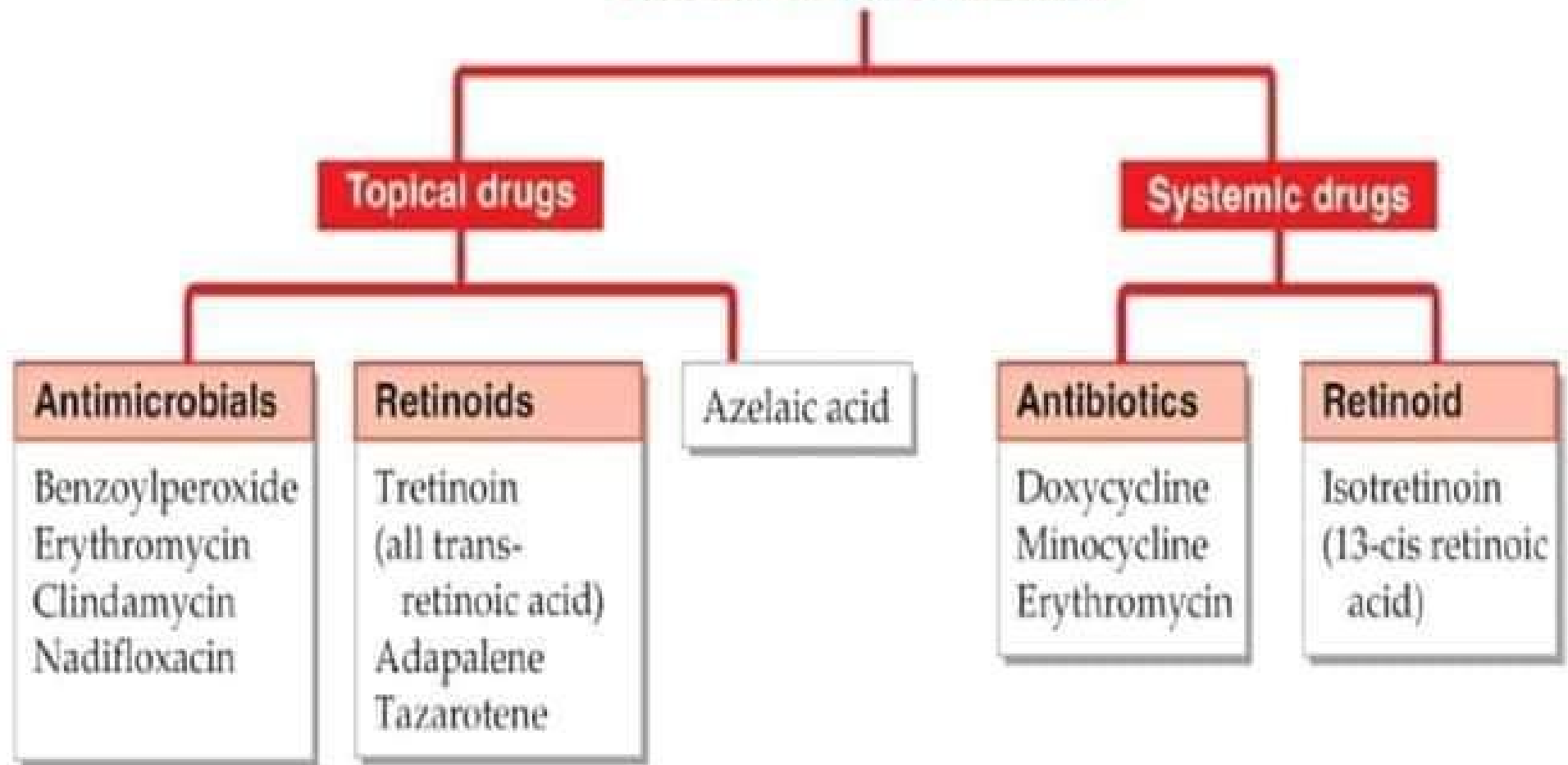


Miscellaneous Drugs

IMMUNOSUPPRESSANT DRUGS



DRUGS FOR ACNE VULGARIS



VACCINES

Killed (inactivated) vaccines

Bacterial

Typhoid - paratyphoid (TAB)
Vi Typhoid polysaccharide
Cholera
Whooping cough (Pertussis)
Meningococcal
Haemophilus influenzae
type B
Plague

Viral

Poliomyelitis inactivated
(IPV; Salk vaccine)
Rabies (Neural tissue)
Rabies (chick embryo cell;
PCEV)
Rabies (Human diploid
cell; HDCV)
Rabies (Vero cell; PVRV)
Influenza
Hepatitis B
Hepatitis A

Combined vaccines

Double antigen: Diphtheria toxoid + tetanus toxoid (DT-DA)
Triple antigen: Diphtheria toxoid + tetanus toxoid + pertussis vaccine (DPT)
Pentavalent vaccine: DPT + Hepatitis B + *H. influenzae* type B vaccines
Measles + mumps + rubella vaccine (MMR)

Live attenuated vaccines

Bacterial

Bacillus
Calmette-
Guerin (BCG)
Typhoid Ty 21a
(oral)

Viral

Poliomyelitis oral
vaccine (OPV, Sabin)
Mumps (Live atte.)
Measles (Live atte.)
Rubella (Live atte.)
Varicella (Live atte.)

Toxoids

Tetanus (fluid / adsorbed)
Diphtheria (adsorbed)